

# SUSPENSION AND AXLE

REFER TO 1990 CELICA REPAIR MANUAL (Pub. No. RM149U)

NOTE: The following pages contain only the points which differ from the above listed manual.

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## WHEEL ALIGNMENT

### 1. MAKE FOLLOWING CHECKS AND CORRECT ANY PROBLEMS

(a) Check the tires for wear proper inflation pressure.

**Cold tire inflation pressure:**

kg/cm<sup>2</sup> (psi, kPa)

| Tire size     | Front         | Rear          |
|---------------|---------------|---------------|
| 215/50R15 88V | 2.2 (32, 220) | 2.1 (30, 210) |

(b) Check the wheel runout.

**Lateral runout: Less than 1.0 mm (0.039 in.)**

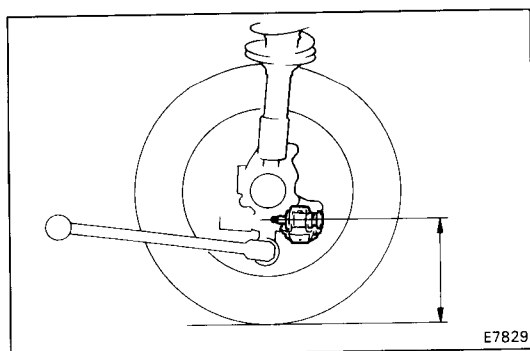
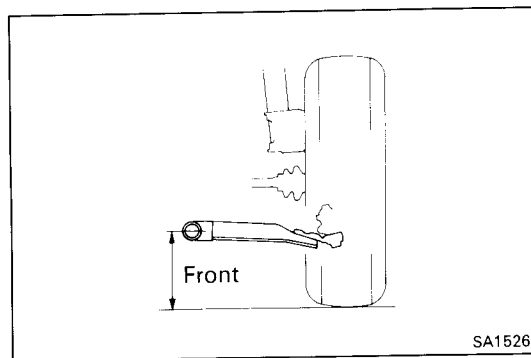
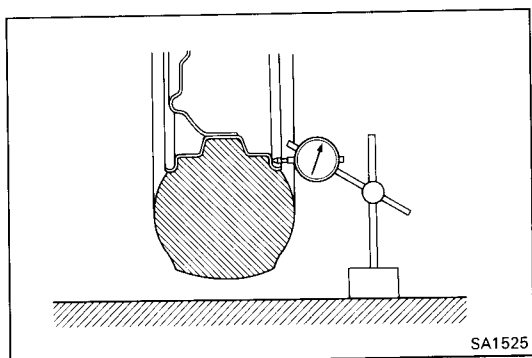
(c) Check the front wheel bearings for looseness.

(d) Check the front suspension for looseness.

(e) Check the steering linkage for looseness.

(f) Check the ball joint for excessive looseness.

(g) Check that the front shock absorber work properly by using the standard bounce test.



### 2. MEASURE VEHICLE HEIGHT

**Vehicle height:**

**Front 191.0 mm (7.520 in.)**

**Rear 233.5 mm (9.193 in.)**

**HINT:**

- **Measuring point**
  - Front – Measure from the ground to the center of the lower suspension arm mounting bolt.
  - Rear – Measure from the ground to the center of the body side No.2 suspension arm mounting bolt.
- Before inspecting the wheel alignment, adjust the chassis ground clearance to specification.

If the clearance of the vehicle is not standard, try to adjust it by pushing down in the body or by lifting the body. If still not correct, check for bad springs or suspension parts.

## Front Wheel Alignment

### 1. INSPECT TOE-IN

Measure toe-in with a toe-in gauge in the following procedure.

- Bounce the vehicle up and down to stabilize the suspension.
- Move the vehicle forward about 5 m (16.4 ft) with the front wheel in the straight-ahead position on a level place.
- Mark the center of each rear tread and measure the distance between the marks of the right and left tires.
- Advance the vehicle until the marks on the rear sides of the tires come to the measuring heights of the gauge on the front side.

**HINT:** If the tire rolls too far, repeat from step (b).

- Measure the distance between the marks on the front of the tires.

**Inspection standard:**  $0 \pm 2 \text{ mm}$  ( $0 \pm 0.08 \text{ in.}$ )

If toe-in is not within specification, adjust by the tie rod end.

### 2. ADJUST TOE-IN

- Remove the boot clips.
- Loosen the tie rod end lock nut.

- Turn the left and right tie rod ends an equal amount to adjust the toe-in.

**Adjustment standard:**  $0 \pm 1 \text{ mm}$  ( $0 \pm 0.04 \text{ in.}$ )

**HINT:** Insure that the lengths of the left and right tie rod ends length are the same.

**Tie rod end length left-right error:**  
 $1.0 \text{ mm}$  ( $0.039 \text{ in.}$ ) or less

- Torque the tie rod end lock nuts.

**Torque:**  $570 \text{ kg-cm}$  ( $41 \text{ ft-lb}$ ,  $56 \text{ N-m}$ )

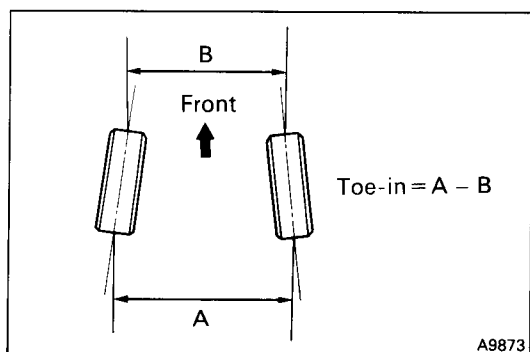
- Place the boot on the seat and clamp it.

**HINT:** Insure that the boots are not twisted.

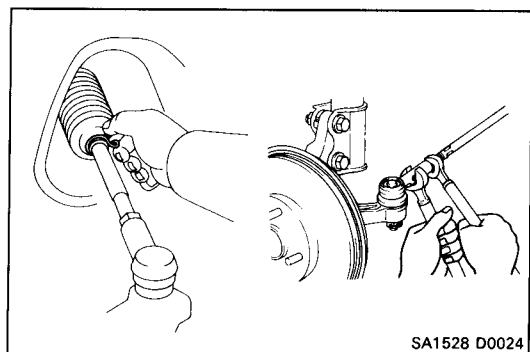
### 3. CHECK WHEEL ANGLE

**Wheel Angle (Maximum):**

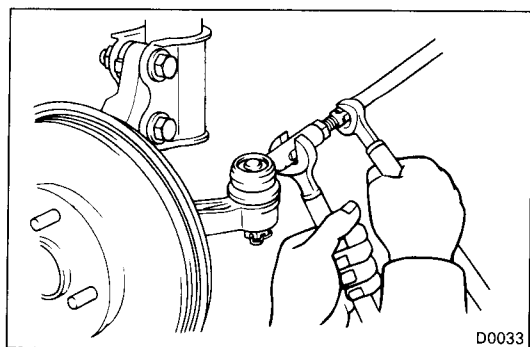
|                      |                                   |
|----------------------|-----------------------------------|
| <b>Inside wheel</b>  | <b><math>33^{\circ}30'</math></b> |
| <b>Outside wheel</b> | <b><math>29^{\circ}30'</math></b> |



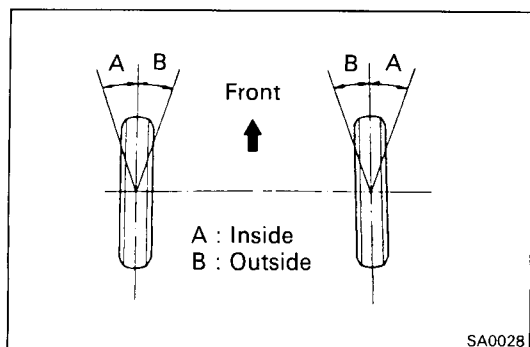
A9873



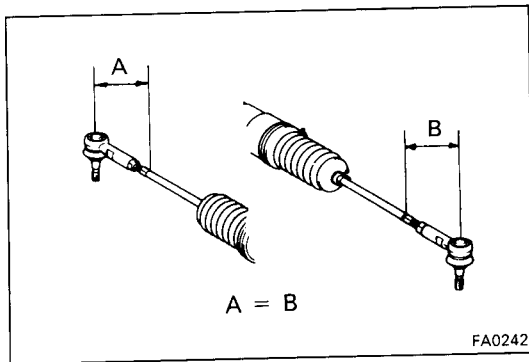
SA1528 D0024



D0033



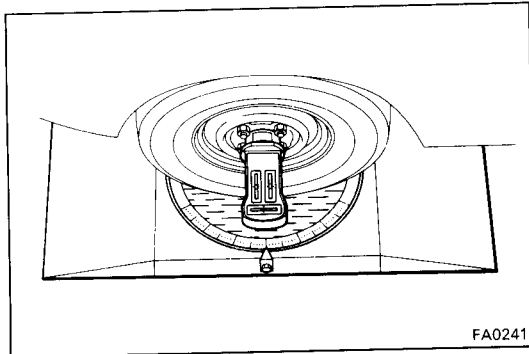
SA0028



If wheel angles differ from the standard specifications, check to see if the lengths of the left and right tie rods are the same.

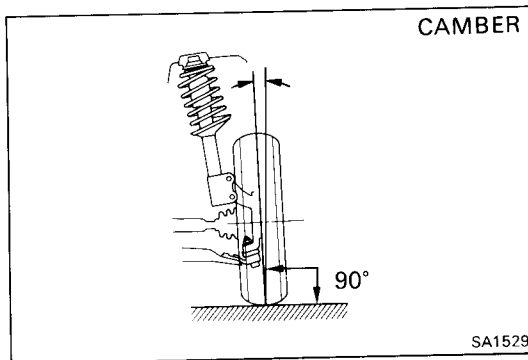
HINT: If the tie rod lengths are not equal, the wheel angle cannot be adjusted properly.

If the tie rod lengths were changed to adjust the wheel angle, inspect the toe-in.



#### 4. INSTALL WHEEL ALIGNMENT EQUIPMENT

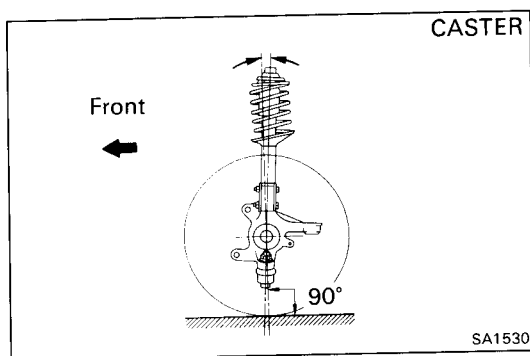
Follow the specific instructions of the equipment manufacturer.



#### 5. CHECK CAMBER

**Camber:** Inspection standard  $-10' \pm 45'$   
Left-right error 30' or less

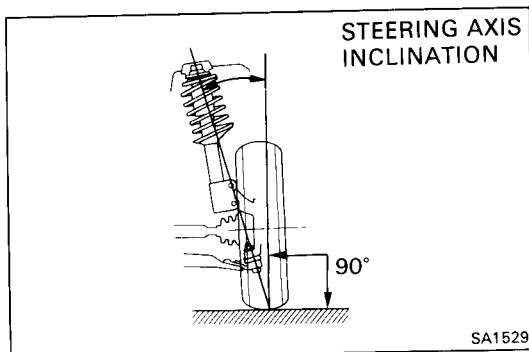
HINT: Camber is not adjustable, if measurement is not within specification, inspect and replace the suspension parts as necessary.



#### 6. CHECK CASTER

**Caster:** Inspection standard  $1^{\circ}00' \pm 45'$   
Left-right error 30' or less

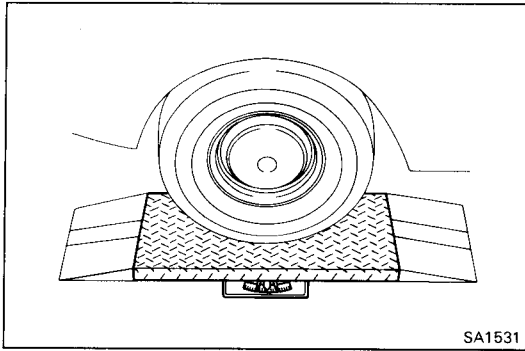
HINT: Caster is not adjustable, if measurement is not within specification, inspect and replace the suspension parts as necessary.



#### 7. CHECK STEERING AXIS INCLINATION

**Steering axis inclination:**  
Inspection standard  $14^{\circ}10' \pm 45'$   
Left-right error 30' or less

HINT: Steering axis inclination is not adjustable, if measurement is not within specification, inspect and replace the suspension parts as necessary.



## 8. CHECK SIDE SLIP (REFERENCE ONLY)

Side slip: 3.0 mm/m (0.118 in./3.3 ft) or less

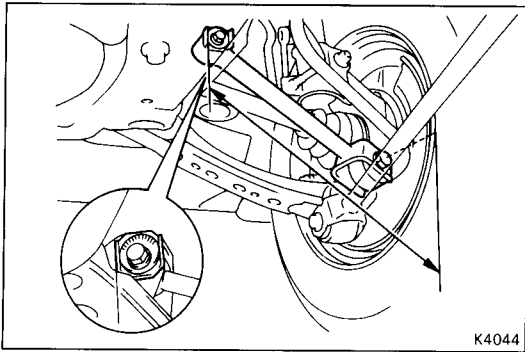
## Rear Wheel Alignment

### 1. INSPECT TOE-IN

(See step 1 on page SA-3)

Inspection standard:  $5 \pm 2$  mm (0.20  $\pm$  0.08 in.)

If toe-in is not within specification, adjust by the cam.

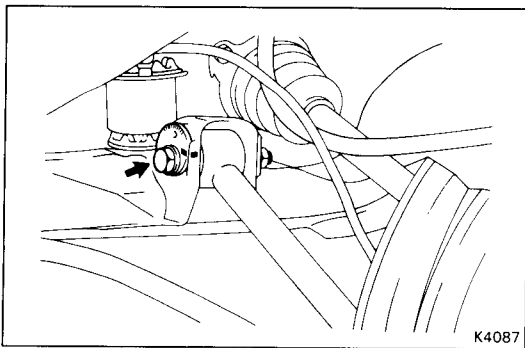


### 2. ADJUST TOE-IN

- (a) Measure the distance between each wheel disc and corner of the cam bracket, then confirm that both are the same.

Left-right error: Less than 3 mm (0.12 in.)

If the left-right error is greater than 3 mm (0.12 in.), adjust following the procedures below.



- (b) Loosen the bolt.

- (c) If the toe-in is out of the standard toward toe-out side, lengthen the longer arm by the cam.

- (d) If the toe-in is out of the standard toward toe-in side, lengthen the shorter arm by the cam.

- (e) Measure the toe-in.

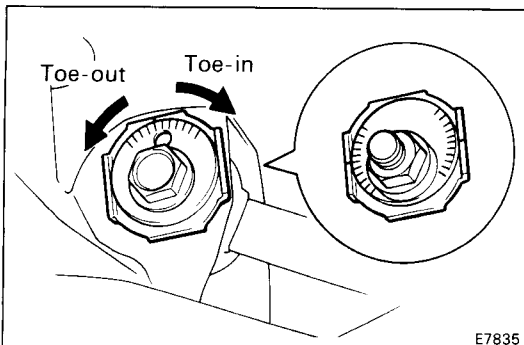
Adjustment standard:  $5 \pm 1$  mm (0.20  $\pm$  0.04 in.)

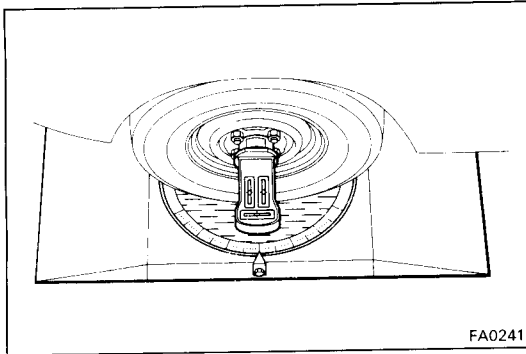
If the left-right error is within specifications but the overall toe-in is not, lengthen or shorten both arms an equal amount by turning the two cams in the opposite direction, until the adjustment standard is obtained.

HINT: The toe-in will change about 4.5 mm (0.177 in.) with each graduation of the cam (one side).

- (f) Tighten the bolt.

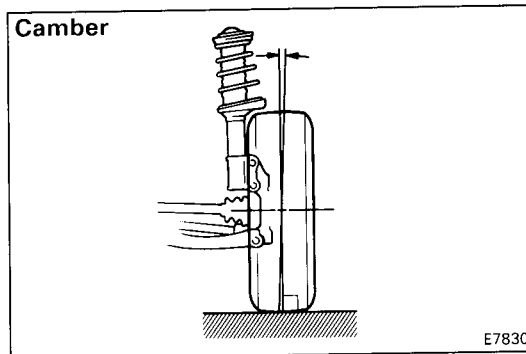
Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)





### 3. INSTALL WHEEL ALIGNMENT EQUIPMENT

Follow the specific instructions of the equipment manufacturer.

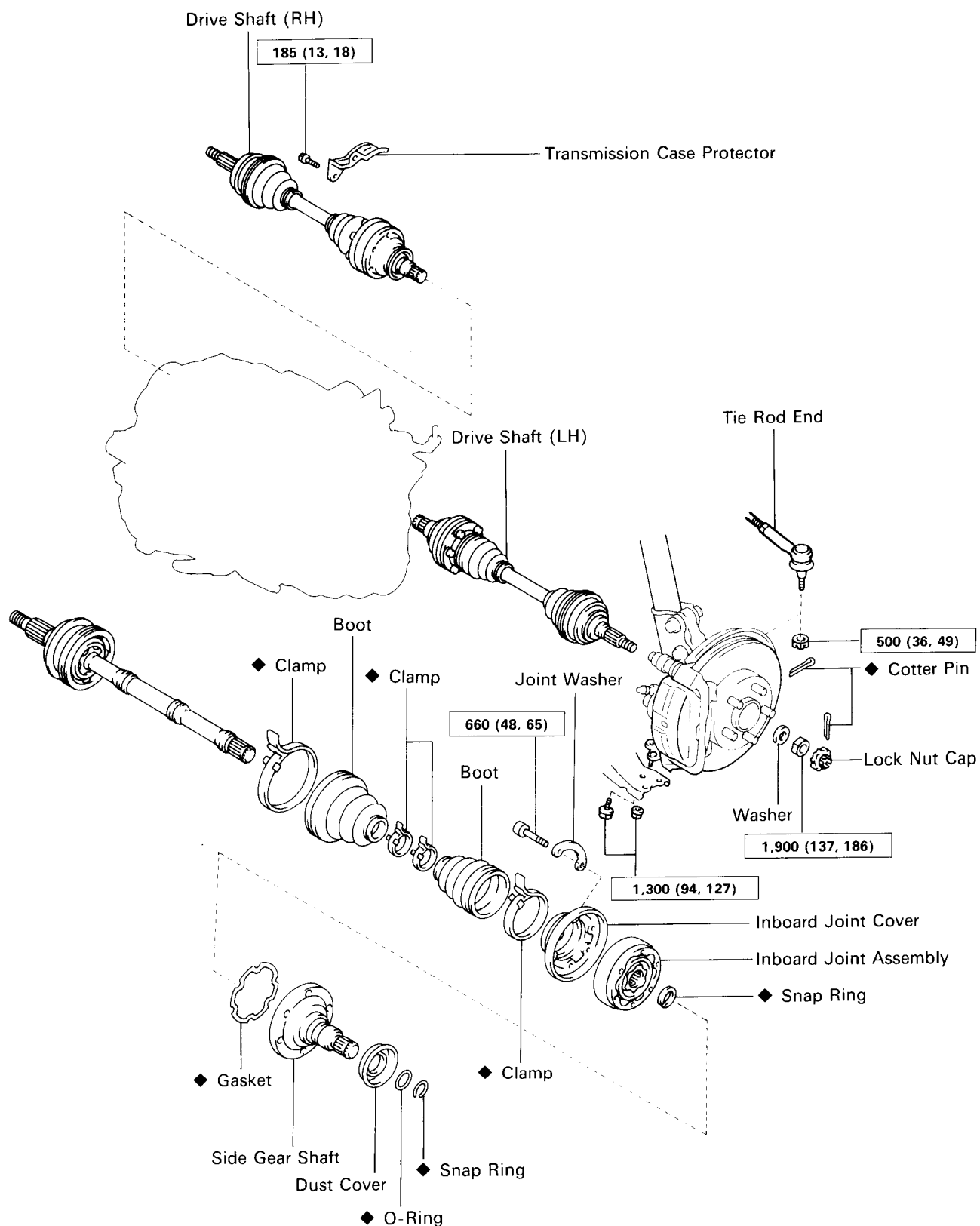


### 4. CHECK CAMBER

**Camber:** Inspection standard  $-45' \pm 30'$   
Left-right error 30' or less

**HINT:** Camber is not adjustable, if measurement is not within specification, inspect and replace the suspension parts as necessary.

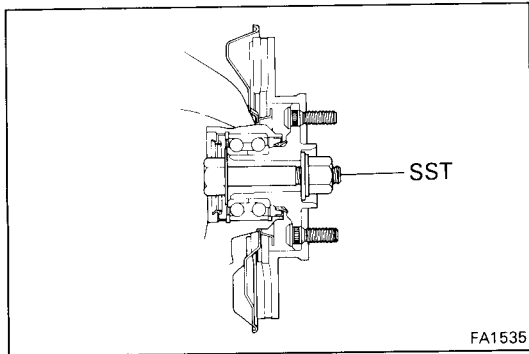
## FRONT DRIVE SHAFT COMPONENTS



kg-cm (ft-lb, N·m) : Specified torque

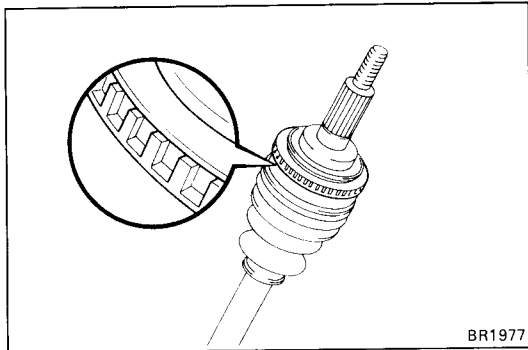
◆ Non-reusable part

SA1648

**NOTICE:**

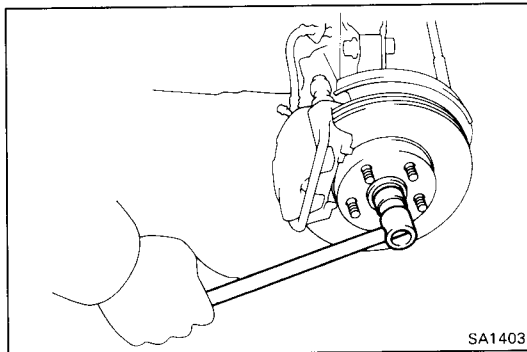
- The hub bearing could be damaged if it is subjected to the vehicle weight, such as when moving the vehicle with the drive shaft removed. Therefore, if it is absolutely necessary to place the vehicle weight on the hub bearing, first support it with SST.

SST 09608-16041 (09608-02020, 09608-02040)



- (w/ ABS)

After disconnecting the drive shaft from the axle hub, work carefully so as not to damage the sensor rotor serrations on the drive shaft.

**REMOVAL OF FRONT DRIVE SHAFT**

(See page SA-7)

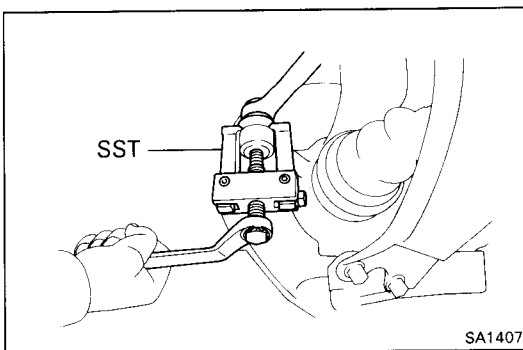
**1. REMOVE FRONT WHEELS****2. REMOVE COTTER PIN, LOCK NUT CAP AND LOCK NUT**

- Remove the cotter pin and lock nut cap.
- Loosen the bearing lock nut while depressing the brake pedal.
- Remove the bearing lock nut and plate washer.

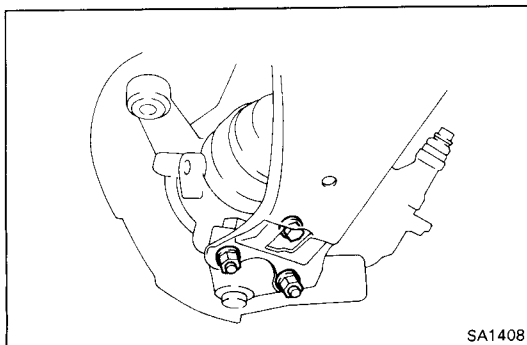
**3. DISCONNECT TIE ROD END**

- Remove the cotter pin and nut from the tie rod end.
- Using SST, disconnect the tie rod end from the steering knuckle.

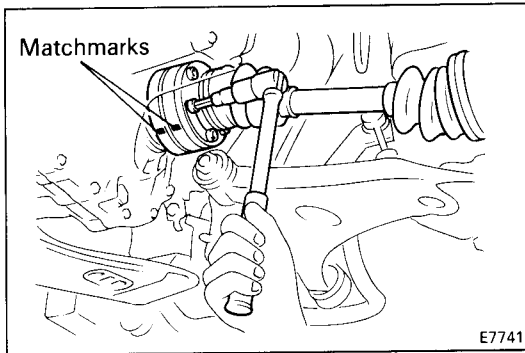
SST 09628-62011

**4. DISCONNECT STEERING KNUCKLE FROM LOWER ARM**

Remove the bolt and two nuts and disconnect the steering knuckle from the lower arm.

**5. DRAIN TRANSAXLE OIL**





## 6. REMOVE DRIVE SHAFT

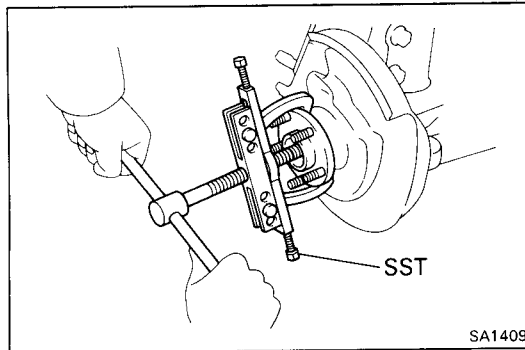
- (a) Place matchmarks on the drive shaft and side gear shaft.

**NOTICE:** Do not use a punch to mark the matchmarks.

Use paint, etc.

- (b) Using SST, loosen the six hexagon bolts while depressing the brake pedal.

SST 09043-88010

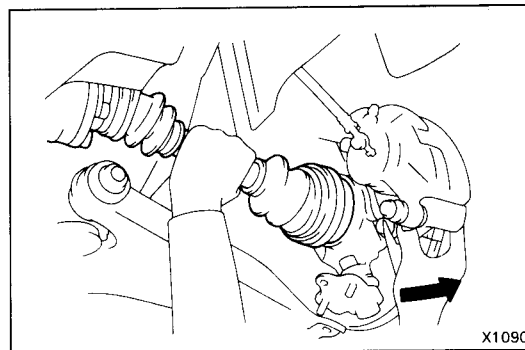


- (c) Using SST, disconnect the drive shaft from the steering knuckle.

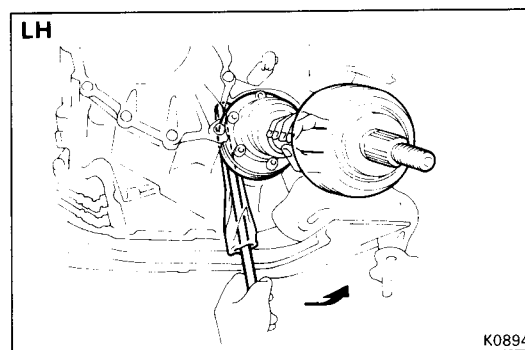
SST 09950-20017

**NOTICE:**

- Before removing the drive shaft, wrap vinyl tape around the threads of the drive shaft to prevent damaging the oil seal.
- Cover the drive shaft boot with cloth to protect it from damage.



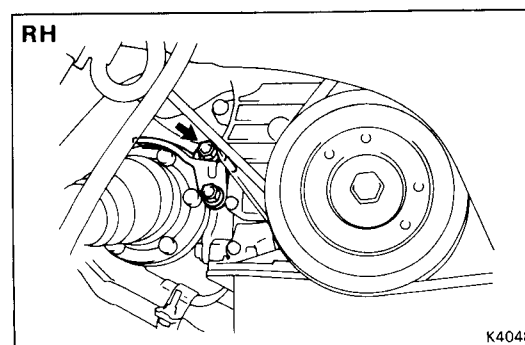
- (d) Push the front axle carrier towards the outside of the vehicle, and separate the drive shaft from the steering knuckle.



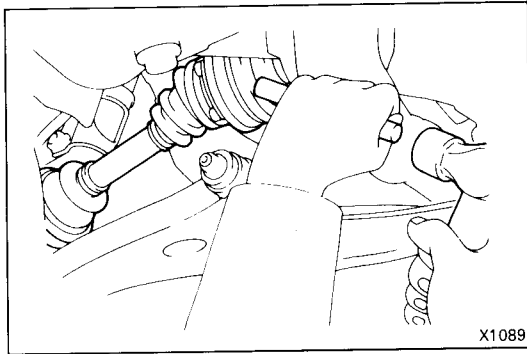
- (e) (LH drive shaft)  
Using a hub nut wrench or equivalent, pry out the LH drive shaft.

**NOTICE:**

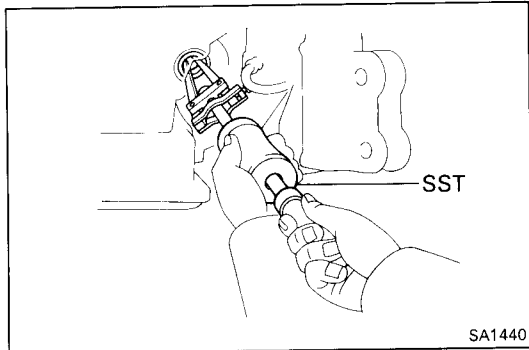
- Be careful not to damage the dust cover.
- Cover the hub nut wrench or an equivalent with cloth so as not to damage the transaxle body.



- (f) Remove the two bolts and transmission case protector.



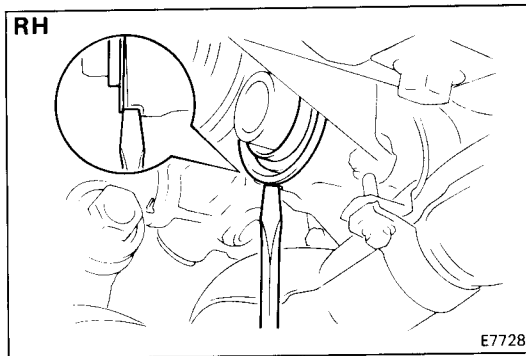
- (g) Using a brass bar and hammer, drive out the drive shaft.



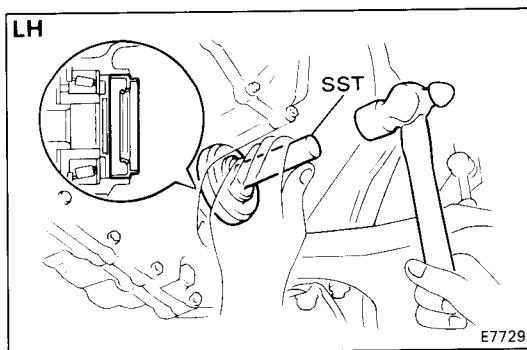
## ON-VEHICLE REPLACEMENT OF SIDE GEAR SHAFT OIL SEAL

1. **REMOVE DRIVE SHAFT**  
(See page SA-7)

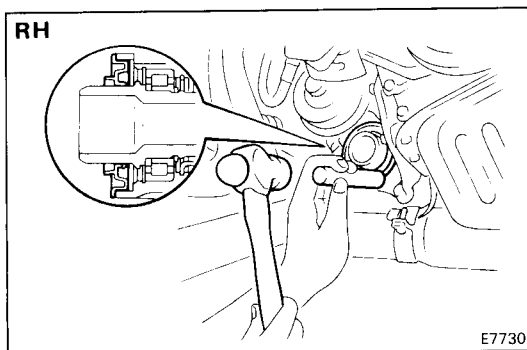
2. **REMOVE LH OIL SEAL**  
Using SST, drive out the oil seal from the case.  
SST 09308-00010



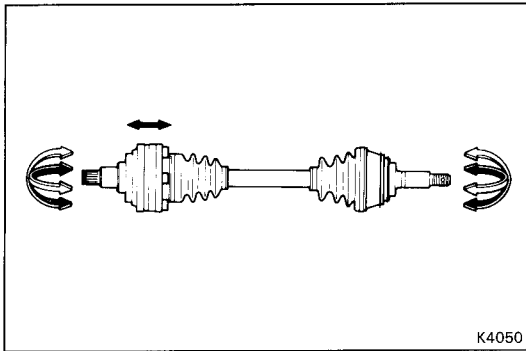
3. **REMOVE RH OIL SEAL**  
Using a screwdriver, remove the oil seal as shown.



4. **INSTALL LH OIL SEAL**  
Using SST and hammer, tap in a new oil seal.  
SST 09223-15010  
HINT: Coat the oil seal lip with MP grease.



5. **INSTALL RH OIL SEAL**  
Using a brass bar and hammer, tap in a new oil seal.  
HINT: Coat the oil seal lip with MP grease.
6. **INSTALL DRIVE SHAFT**  
(See page SA-18)

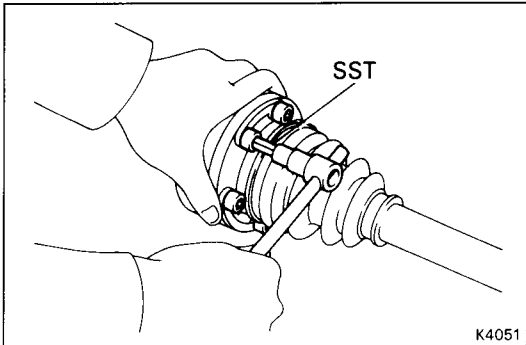


## DISASSEMBLY OF FRONT DRIVE SHAFT

(See page SA-7)

### 1. CHECK DRIVE SHAFT

- Check to see that there is no play in the outboard joint.
- Check to see that the inboard joint slides smoothly in the thrust direction.
- Check to see that there is no remarkable play in the radial direction of the inboard joint.
- Check for damage to boots.

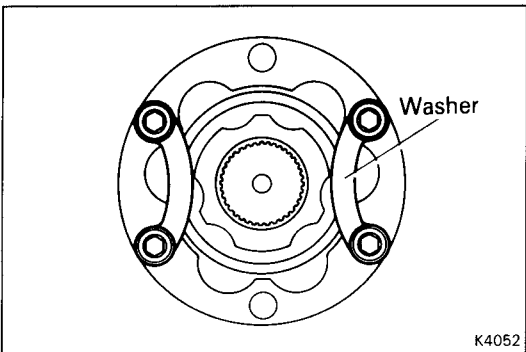


### 2. DISCONNECT SIDE GEAR SHAFT

- Using SST, remove the six hexagon bolts and the three washers.

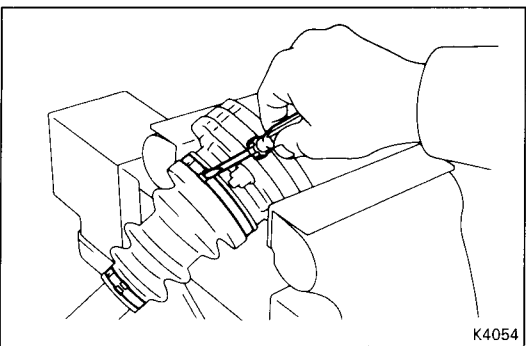
SST 09043-88010

- Disconnect the side gear shaft from the drive shaft.



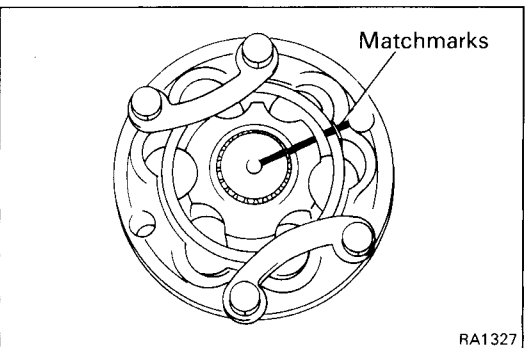
- Use bolts, nuts and washers to keep the inboard joint together.

**NOTICE:** Tighten the bolts by hand to avoid scratching the flange surface.



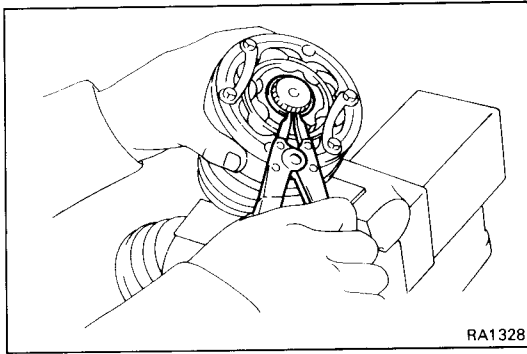
### 3. REMOVE BOOT CLAMPS

- Using a screwdriver, remove the four boot clamps from the inboard joint and outboard joint.
- Remove the inboard joint boot from the inboard joint cover.

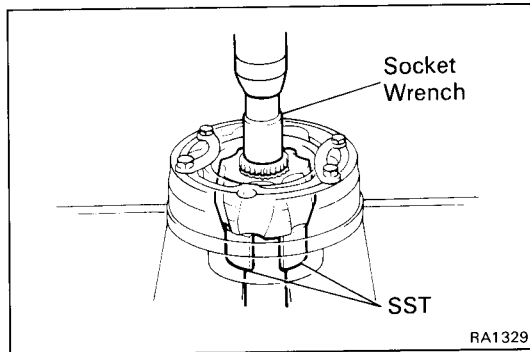


### 4. DISASSEMBLE INBOARD JOINT

- Place matchmarks on the inboard joint and drive shaft.



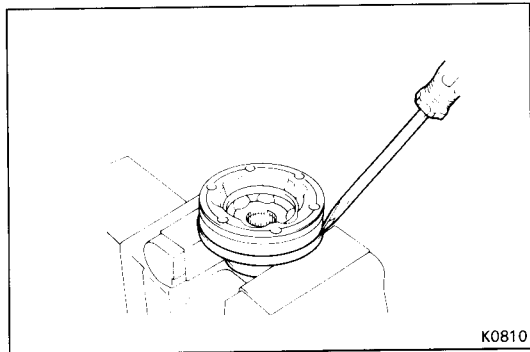
(b) Using snap ring pliers, remove the snap ring.



(c) Using SST, a socket wrench and a press, remove the inboard joint from the drive shaft.

SST 09726-10010 (09726-00030)

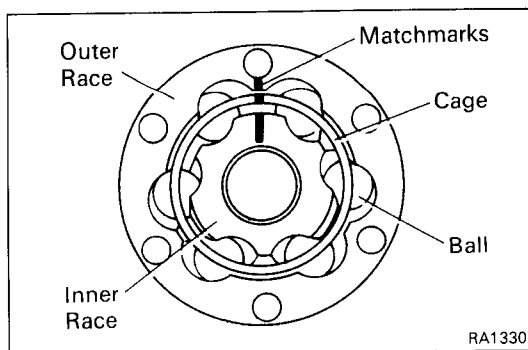
(d) Remove the four bolts and two washers from the inboard joint.



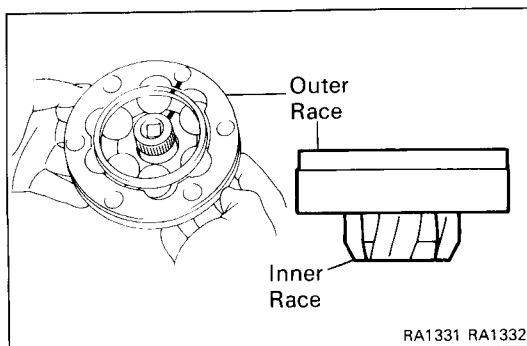
(e) Using a screwdriver, unstake the inboard joint cover.

(f) Using a screwdriver, pry out the inboard joint from the inboard joint cover.

**NOTICE:** When lifting the inboard joint, hold onto the inner race and outer race.

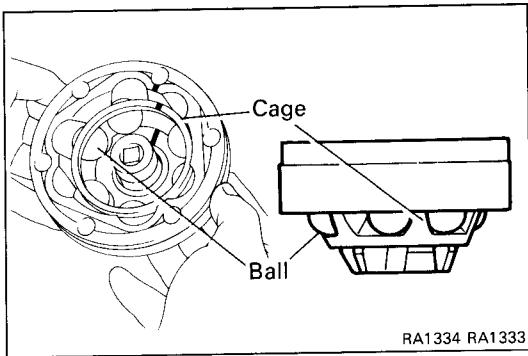


**HINT:** Should the joint become disassembled, re-assemble it in the way shown.

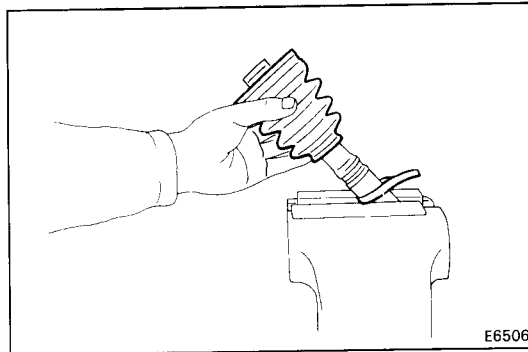


## SERVICE HINT

- (a) Align the matchmarks placed before disassembly.
- (b) Insert the spark plug wrench into the inner race.
- (c) Lift the outer race and cage, and insert the six balls.

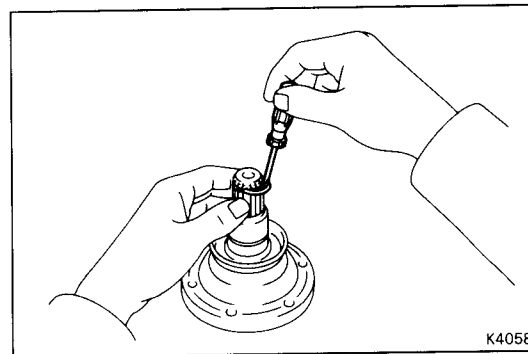


- (d) Jiggle the outer race and cage as shown to place the balls in their respective grooves.
- (e) Lower the outer race and cage so that they fit tightly with the inner race.



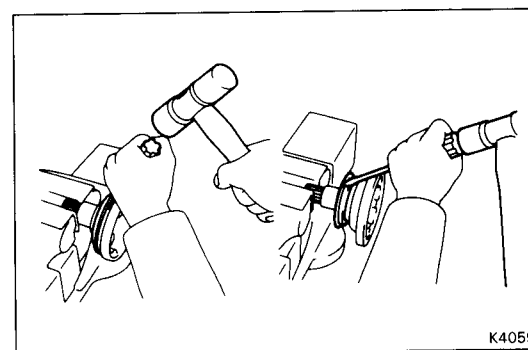
## 5. REMOVE BOOTS

Remove boots of the inboard joint and outboard joint.



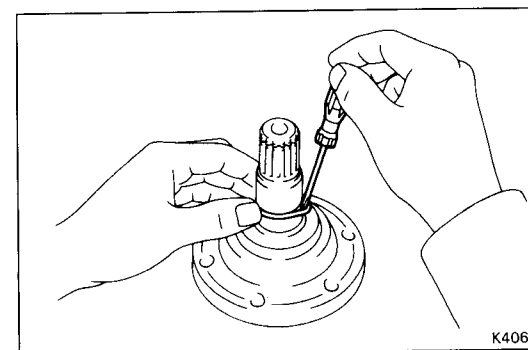
## 6. REPLACE SIDE GEAR SHAFT SNAP RING

- (a) Using a screwdriver, pry out the snap ring.
- (b) Using snap ring pliers, install the new snap ring.



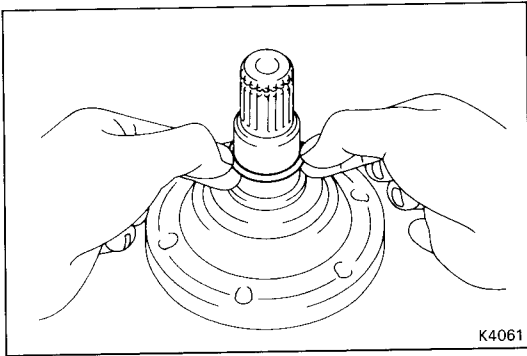
## 7. REMOVE DUST COVER FROM SIDE GEAR SHAFT

Using a screwdriver and hammer, tap out the dust cover.

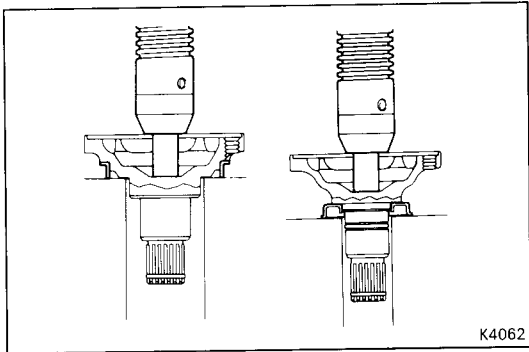


## 8. REPLACE SIDE GEAR SHAFT O-RING

- (a) Using a screwdriver, remove the O-ring.



- (b) Coat O-ring with MP grease.
- (c) Install a new O-ring.



- 9. INSTALL DUST COVER TO SIDE GEAR SHAFT**  
Using a press, install a new dust cover.

## ASSEMBLY OF FRONT DRIVE SHAFT

(See page SA-7)

### 1. TEMPORARILY INSTALL BOOTS AND NEW BOOT CLAMPS

**HINT:** Before installing the boot, wrap vinyl tape around the spline of the shaft to prevent damaging the boot.

**NOTICE:** The boot and clamp of the outboard joint are smaller than those of the inboard joint.

Temporarily install the two boots and four new clamps to the outboard joint and inboard joint.

### 2. INSTALL BOOT TO OUTBOARD JOINT

(a) Before assembling the boots, pack in grease.

**HINT:** Use the grease supplied in the boot kit.

**Grease capacity:** 120 g (0.26 lb)

(b) Install the two boot clamps to the outboard joint boot.

(c) Bend the boot clamp and lock it as shown.

### 3. INSTALL INBOARD JOINT COVER

(a) Clean contacting surfaces of any residual packing material using gasoline or alcohol.

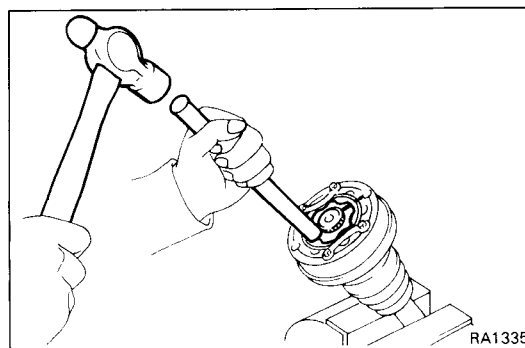
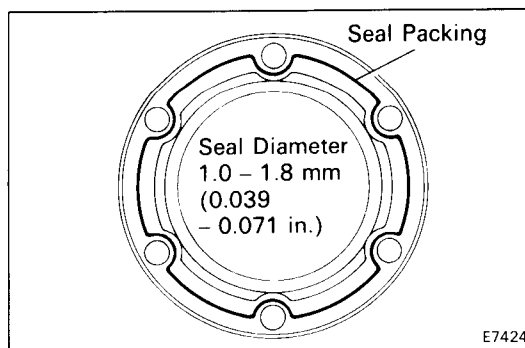
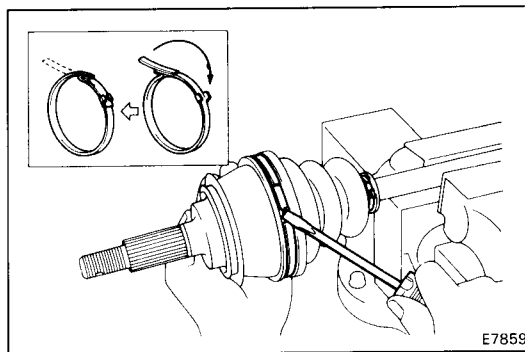
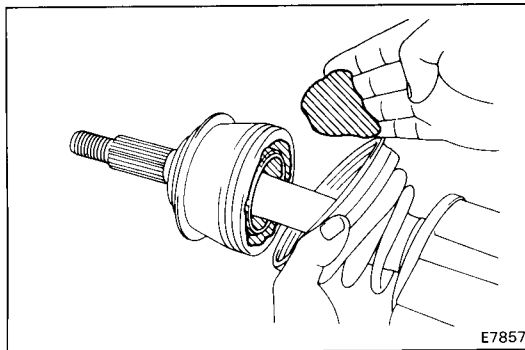
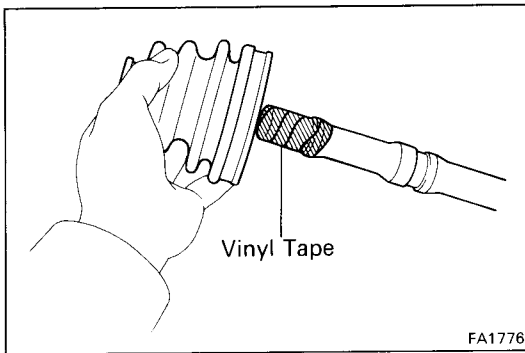
(b) Apply seal packing to the inboard joint cover as shown.

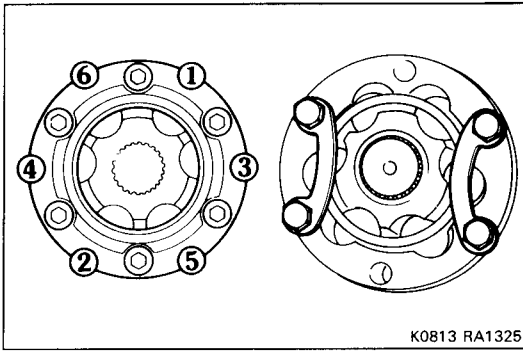
**Seal packing:** Part No.08826-00801, THREE BOND 1121 or equivalent

**HINT:** Avoid applying an excess amount to the surface.

(c) Align the bolt holes of the cover with those of the inboard joint, then insert the hexagon bolts.

(d) Using a hammer and brass bar, tap the rim of the inboard joint cover into place. Do this in the order shown, and repeat several times.



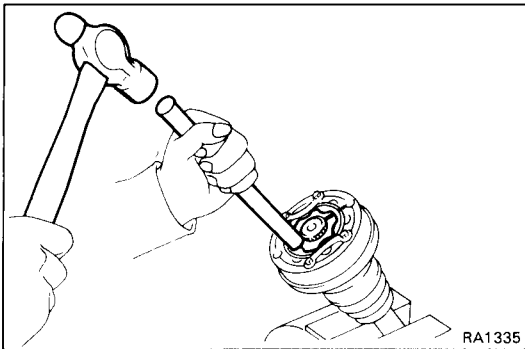
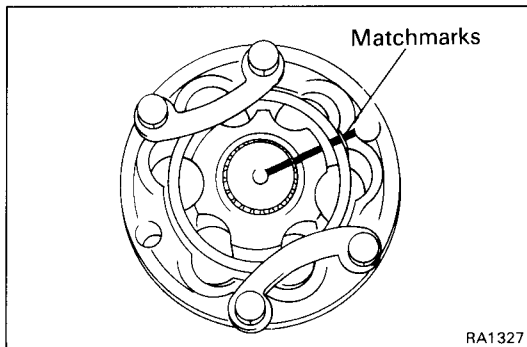


- (e) Use bolts, nuts and washers to keep the inboard joint together.

**NOTICE:** Tighten the bolts by hand to avoid scratching the flange surface.

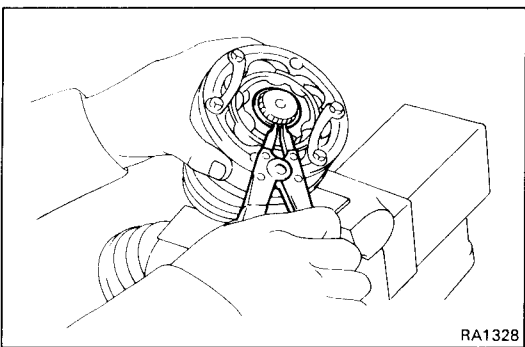
#### 4. ASSEMBLE INBOARD JOINT

- (a) Align the matchmarks placed before disassembly.

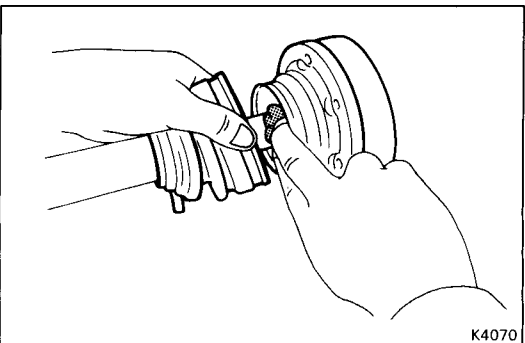


- (b) Using a brass bar and hammer, tap the inboard joint onto the drive shaft.

**NOTICE:** Make sure that the brass bar is touching the inner race, and not the cage.



- (c) Using snap ring pliers, install a new snap ring.

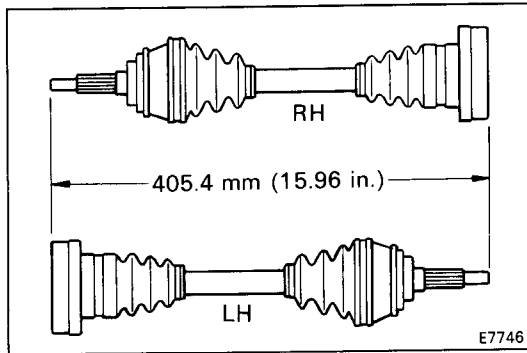


- (d) Pack in grease to the inboard tulip and boot.

**HINT:** Use the grease supplied in the boot kit.

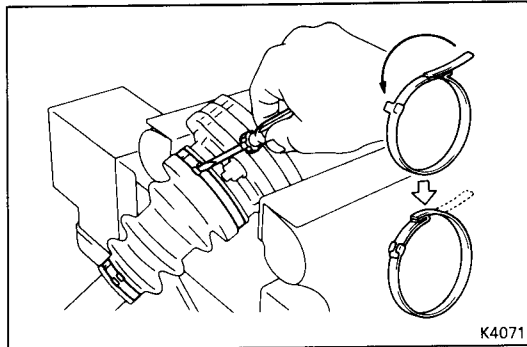
**Grease capacity:** 90 g (0.20 lb)



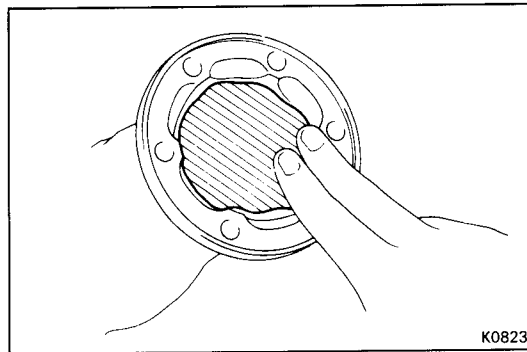


- (e) Be sure the boot is on the shaft groove.
- (f) Insure that the boot is not stretched or contracted when the drive shaft is at standard length.

**Drive shaft length: 405.4 mm (15.96 in.)**



- (g) Bend the boot clamp and lock it as shown.



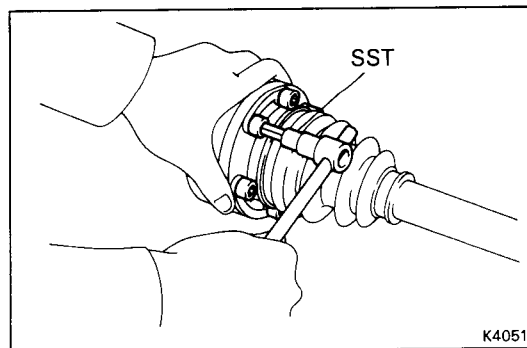
## 5. INSTALL SIDE GEAR SHAFT

- (a) Pack in grease to the side gear shaft.

**HINT:** Use the grease supplied in the boot kit.

**Grease capacity: 43 g (0.09 lb)**

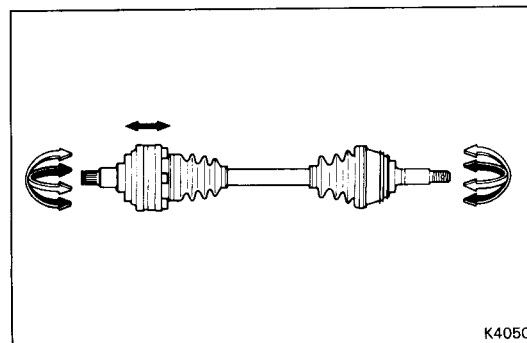
- (b) Remove the two washers and four bolts from the drive shaft.



- (c) Align matchmarks and install the side gear shaft with a new gasket to the drive shaft.

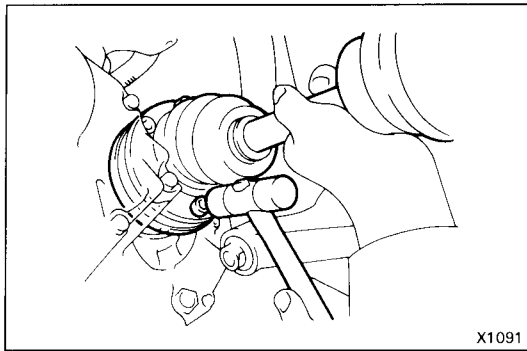
- (d) Using SST, finger tighten the six hexagon bolts with three washers.

**SST 09043-88010**

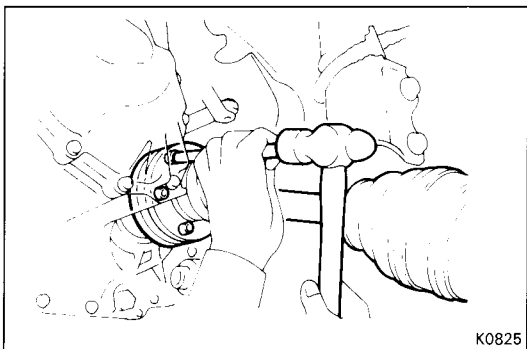


## 6. CHECK DRIVE SHAFT

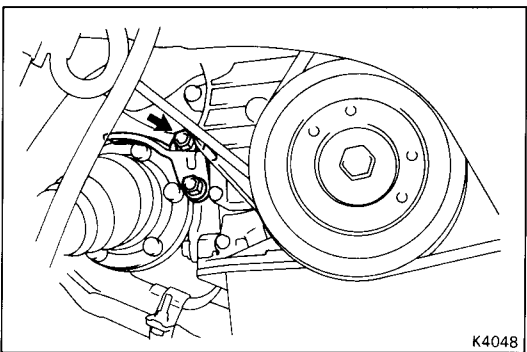
- (a) Check to see that there is no play in the inboard joint and outboard joint.
- (b) Check to see that the inboard joint slide smoothly in the thrust direction.



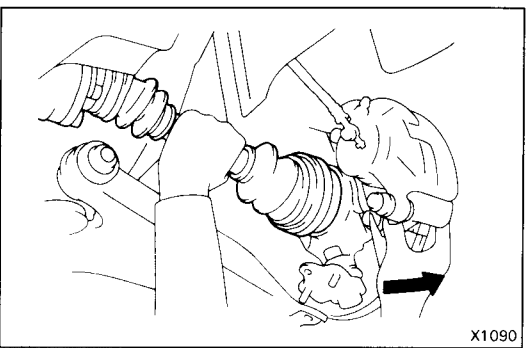
X1091



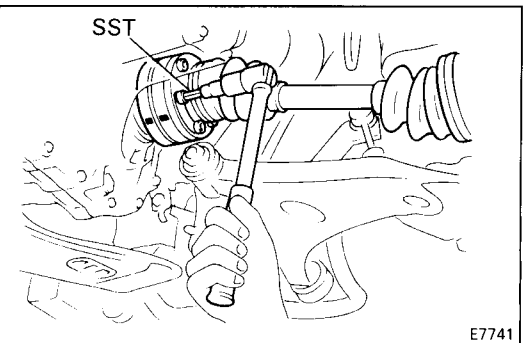
K0825



K4048



X1090



E7741

## INSTALLATION OF FRONT DRIVE SHAFT

(See page SA-7)

### 1. INSTALL DRIVE SHAFT

- (a) Temporarily install the drive shaft with a plastic hammer.

**HINT:** Before installing the drive shaft, set the snap ring opening side facing downward.

**NOTICE:** Be careful not to damage the boot, oil seal and deflector.

- (b) Using a brass bar and hammer, tap in the hexagon bolt head of the drive shaft until it makes contact with the pinion shaft.

**HINT:** Whether or not the drive shaft is making contact with the pinion shaft can be known by sound or feeling when driving it in.

- (c) Install the transmission case protector with two bolts.

**Torque:** 185 kg-cm (13 ft-lb, 18 N·m)

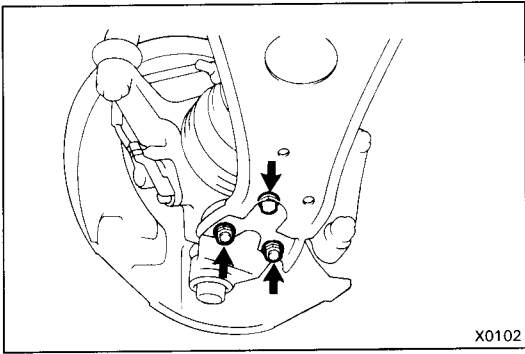
- (d) Install the outboard joint side of the drive shaft to the axle hub.

**NOTICE:** Be careful not to damage the oil seal and boot.

- (e) Using SST, torque the six hexagon bolts while depressing the brake pedal.

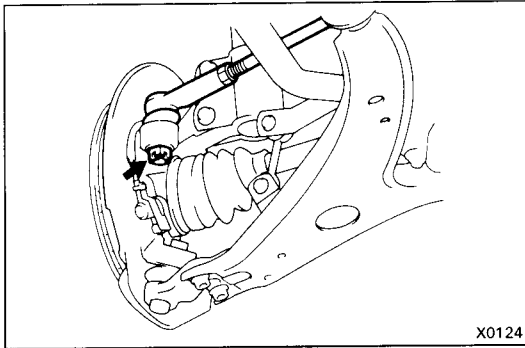
SST 09043-88010

**Torque:** 660 kg-cm (48 ft-lb, 65 N·m)

**2. CONNECT STEERING KNUCKLE TO LOWER ARM**

- (a) Install the steering knuckle to the lower arm.
- (b) Install and torque the bolt and two nuts.

**Torque: 1,300 kg-cm (94 ft-lb, 127 N·m)**

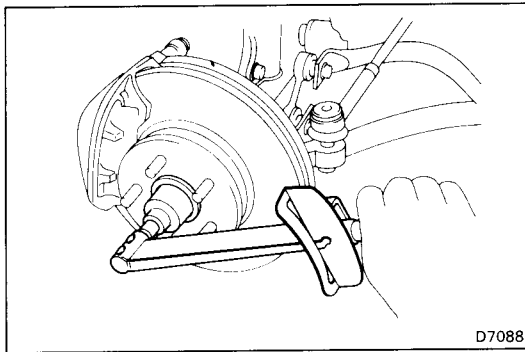
**3. CONNECT TIE ROD END TO STEERING KNUCKLE**

- (a) Install the tie rod end to the steering knuckle with a nut.
- (b) Torque the nut.

**Torque: 500 kg-cm (36 ft-lb, 49 N·m)**

- (c) Install a new cotter pin.

**HINT:** If the cotter pin hole does not line up, correct by tightening the nut by the smallest amount possible.

**4. INSTALL BEARING LOCK NUT, LOCK NUT CAP AND NEW COTTER PIN**

- (a) Install the washer and lock nut.
- (b) Torque the lock nut while depressing the brake pedal.

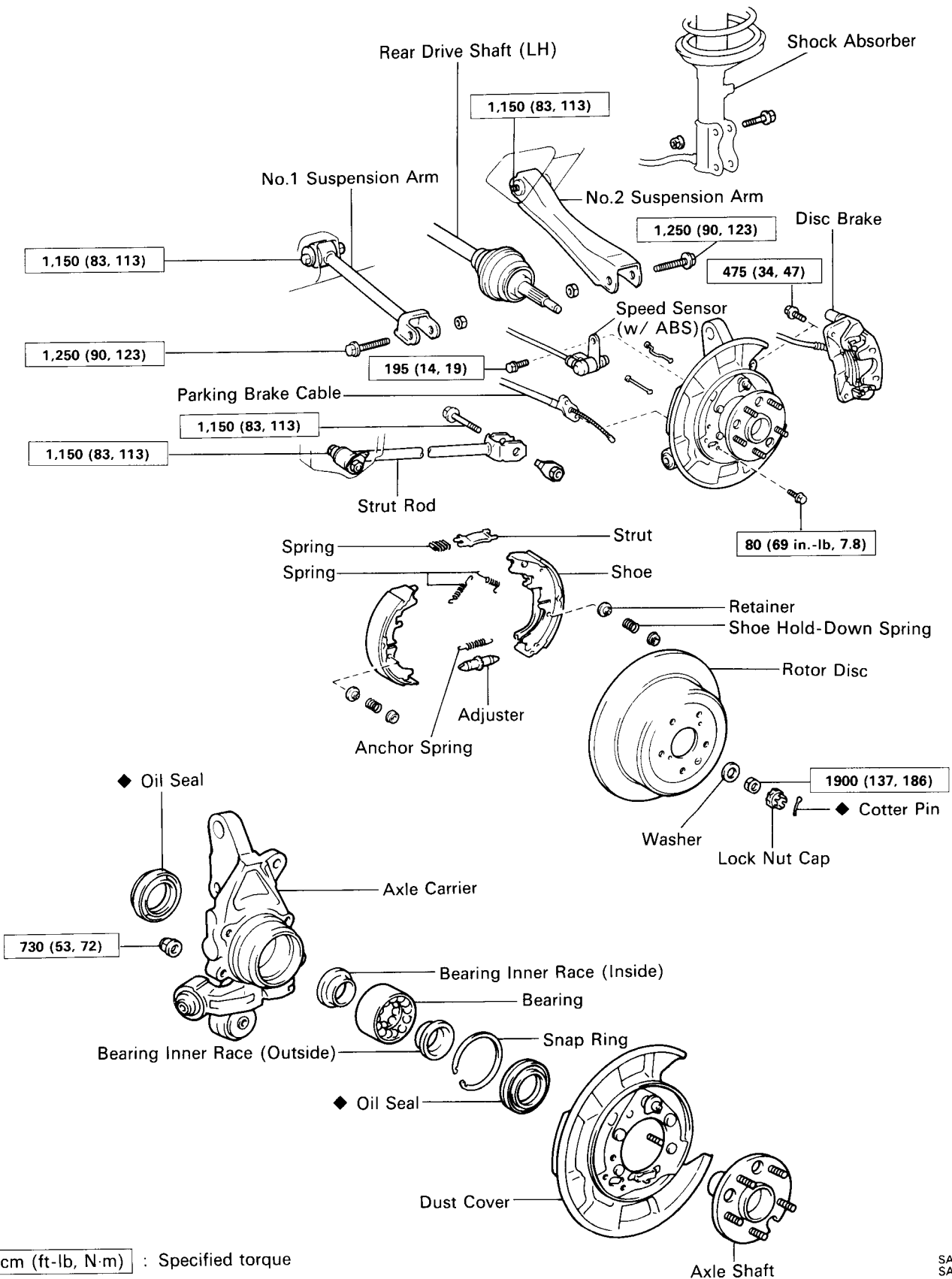
**Torque: 1,900 kg-cm (137 ft-lb, 186 N·m)**

- (c) Install the lock nut cap, and using pliers, install a new cotter pin.

**5. INSTALL FRONT WHEELS****6. FILL TRANSAXLE WITH GEAR OIL****7. CHECK FRONT WHEEL ALIGNMENT**

(See page SA-3)

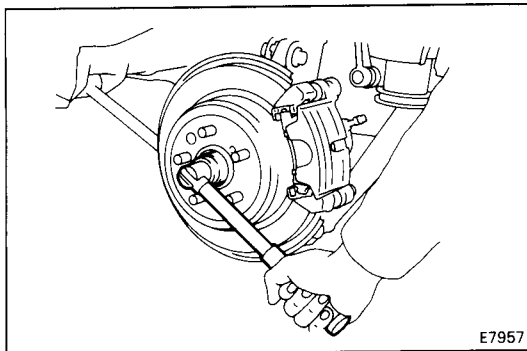
## REAR AXLE HUB AND CARRIER COMPONENTS



kg-cm (ft-lb, N·m) : Specified torque

◆ Non-reusable part

SA1645  
SA1646  
RA1412

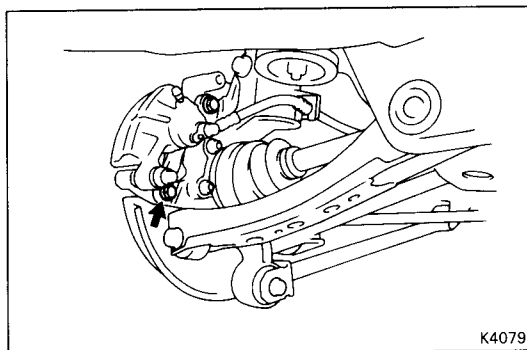


## REMOVAL OF REAR AXLE HUB AND CARRIER

(See page SA-20)

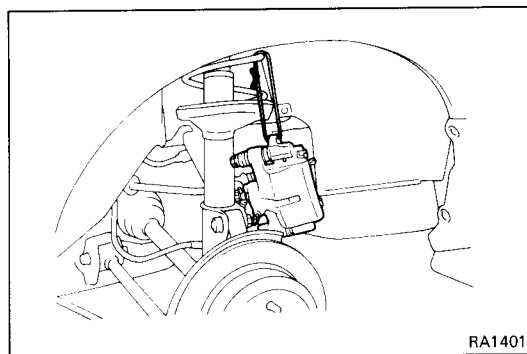
### 1. REMOVE COTTER PIN, BEARING LOCK NUT CAP AND BEARING LOCK NUT

- Remove the cotter pin and bearing lock nut cap.
- With the parking brake engaged, remove the bearing lock nut with plate washer.

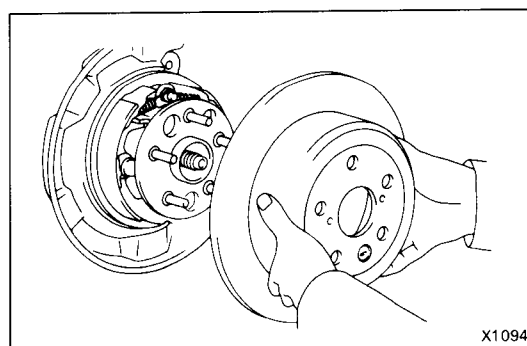


### 2. REMOVE BRAKE CALIPER

- Remove the two bolts.

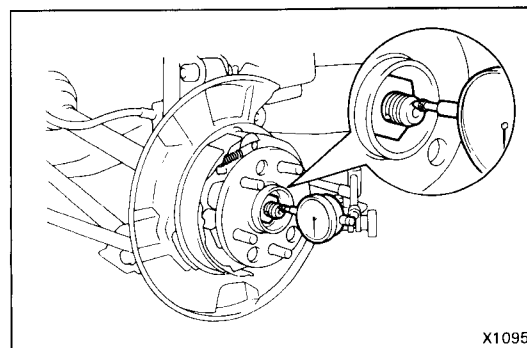


- Remove the brake caliper from the rear axle carrier and suspend it with wire.



### 3. REMOVE ROTOR DISC

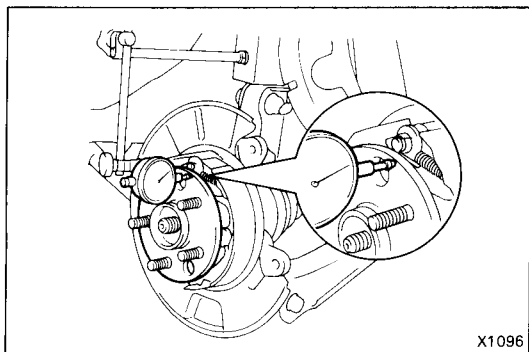
- Place matchmarks on the rotor disc and axle hub.
- Disengage the parking brake, and remove the rotor disc.



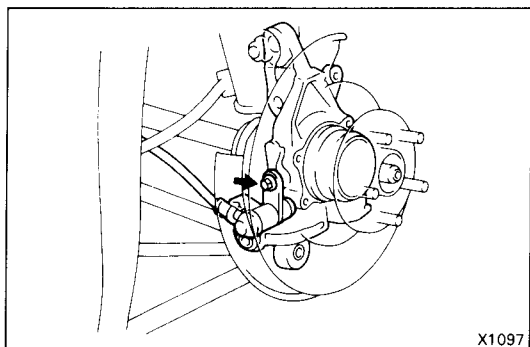
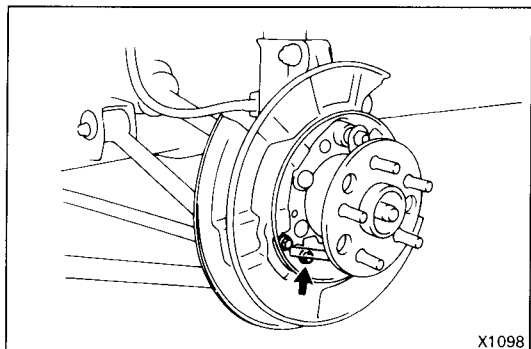
### 4. CHECK BEARING PLAY IN AXIAL DIRECTION

**Bearing play: 0.05 mm (0.0020 in.) or less**

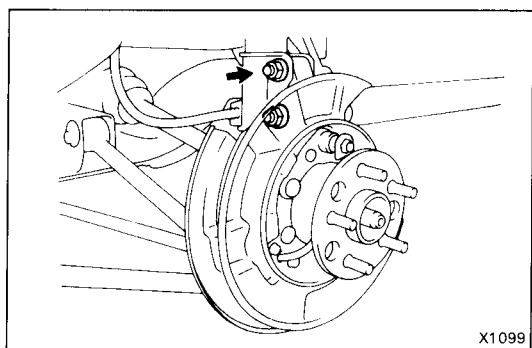
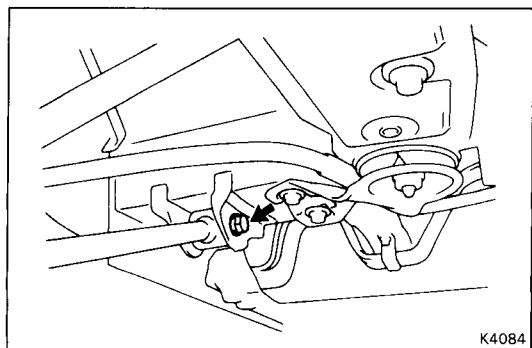
If the bearing play is greater than the maximum, replace the bearing.

**5. CHECK AXLE SHAFT FLANGE RUNOUT**

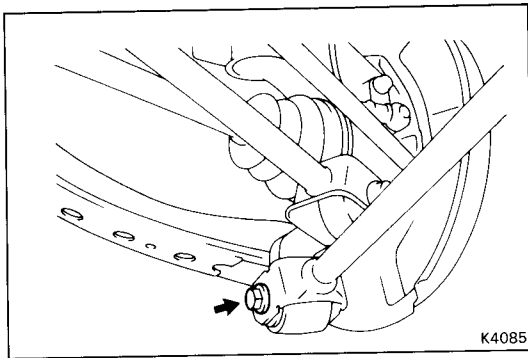
Maximum flange runout: 0.07 mm (0.0028 in.) or less

**6. REMOVE PARKING BRAKE ASSEMBLY****7. (w/ ABS)  
REMOVE REAR SPEED SENSOR****8. REMOVE PARKING BRAKE CABLE**

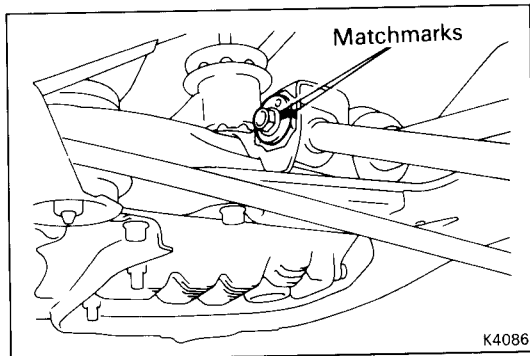
Remove the two bolts and pull out the parking brake cable.

**9. LOOSEN TWO AXLE CARRIER MOUNTING NUTS****10. DISCONNECT STRUT ROD**

(a) Loosen the strut rod mounting nut (body side).

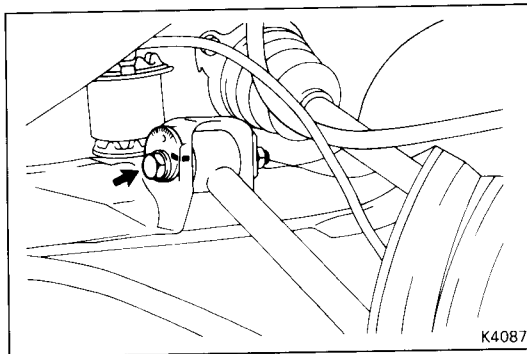


- (b) Remove the strut rod mounting bolt and nut from the axle carrier.

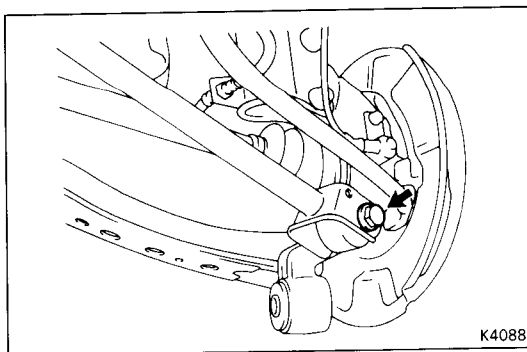


# **11. DISCONNECT NO.1 SUSPENSION ARM**

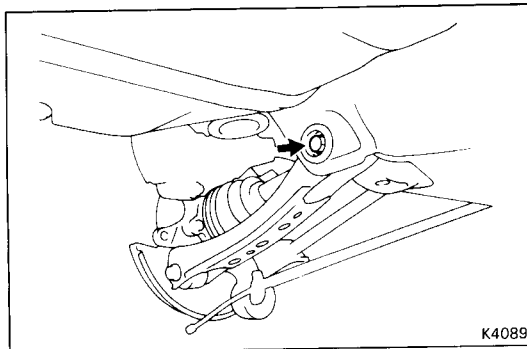
- (a) Place matchmarks on the toe adjusting cam and control arm retainer.



- (b) Loosen the No.1 suspension arm mounting bolt (rear suspension member side).

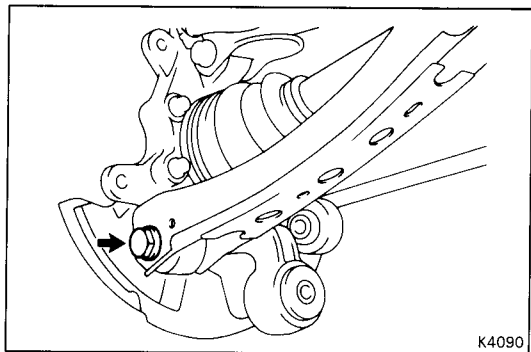


- (c) Remove the No.1 suspension arm mounting bolt and nut from the axle carrier.

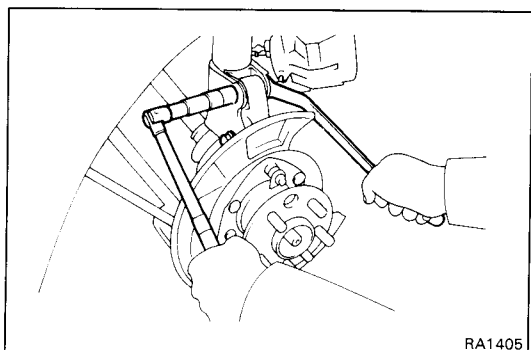


# **12. DISCONNECT NO.2 SUSPENSION ARM**

- (a) Loosen the No.2 suspension arm mounting bolt (rear suspension member side).



(b) Remove the bolt and nut from the axle carrier.

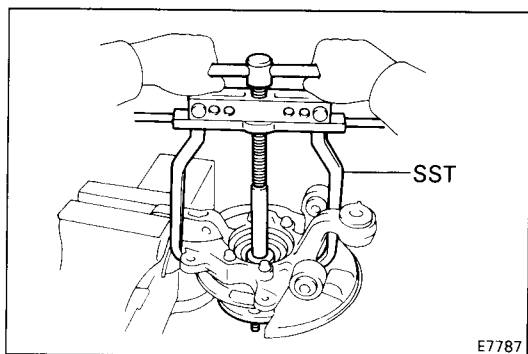


### 13. REMOVE AXLE CARRIER WITH AXLE HUB

(a) Remove the two axle carrier mounting bolts and nuts.

(b) Remove the axle carrier with axle hub.

**NOTICE:** Cover the drive shaft boot with cloth to protect it from damage.

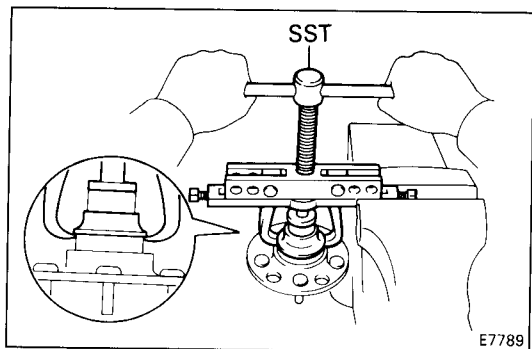


## REPLACEMENT OF AXLE HUB AND BEARING

### 1. REMOVE AXLE SHAFT FROM AXLE HUB

Using SST, push the axle shaft off the axle hub.

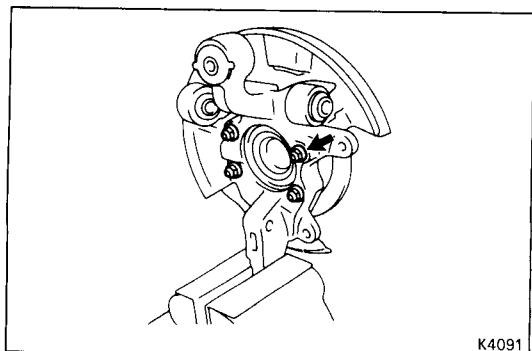
SST 09950-20017



### 2. REMOVE BEARING INNER RACE (OUTSIDE) FROM AXLE SHAFT

Using SST, pull off the bearing inner race (outside) from the axle shaft.

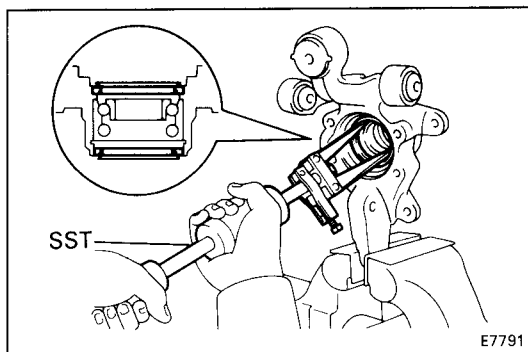
SST 09950-20017



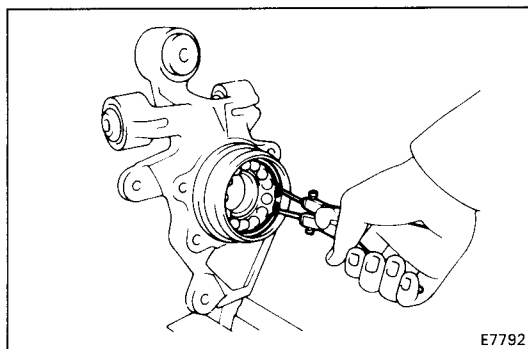
### 3. REMOVE BACKING PLATE

Remove the four nuts and backing plate.

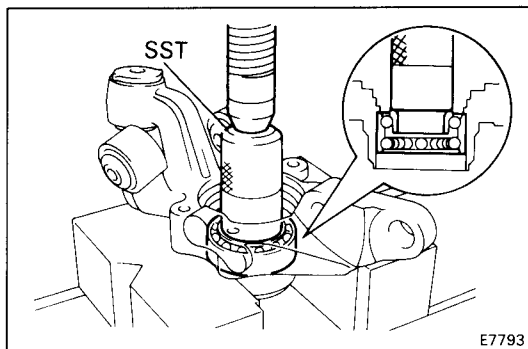


**4. REMOVE INNER AND OUTER OIL SEAL**

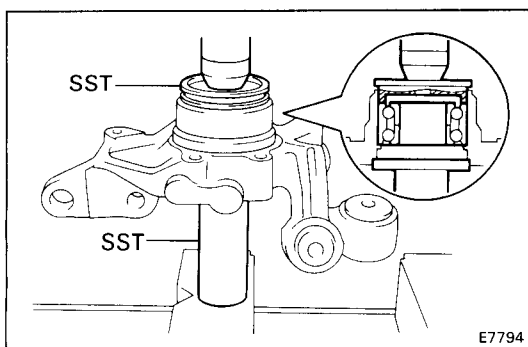
Using SST, remove the oil seal from the axle carrier.  
SST 09308-00010

**5. REMOVE SNAP RING**

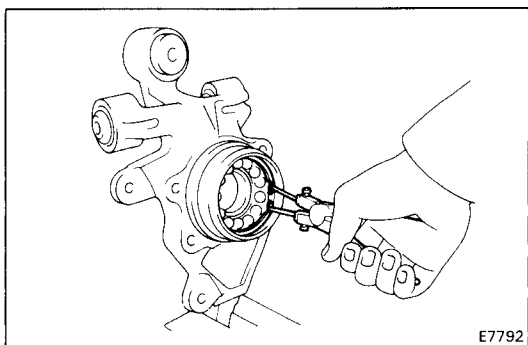
Using snap ring pliers, remove the snap ring from the axle carrier.

**6. REMOVE BEARING**

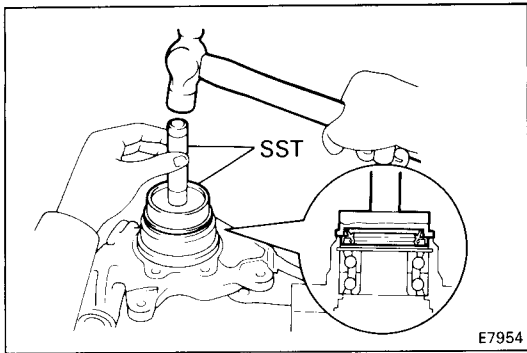
Using SST, press out the bearing from the axle carrier.  
SST 09636-20010

**7. INSTALL BEARING**

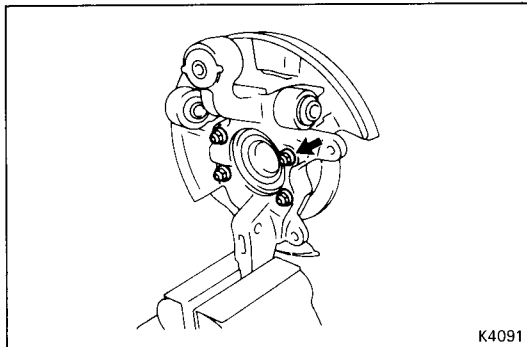
Using SST, press the bearing into the axle hub.  
SST 09309-36010, 09608-32010

**8. INSTALL SNAP RING**

Using snap ring pliers, install a snap ring into the axle carrier.

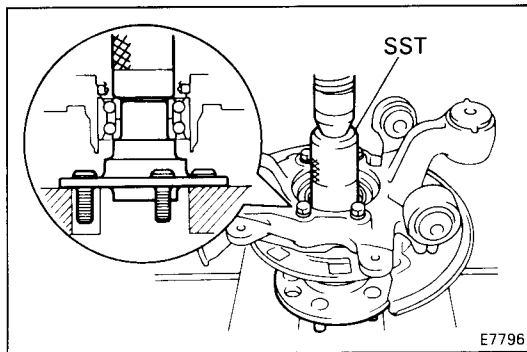
**9. INSTALL OUTER OIL SEAL**

- (a) Using SST, drive in a new oil seal to the axle carrier.  
SST 09608-30012 (09608-04020), 09608-32010
- (b) Apply MP grease to the oil seal lip.

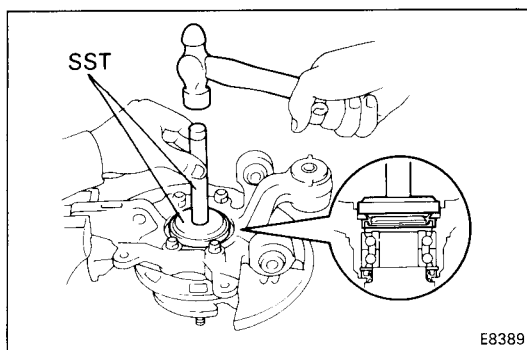
**10. INSTALL BACKING PLATE**

Install the backing plate to the axle carrier with new four nuts.

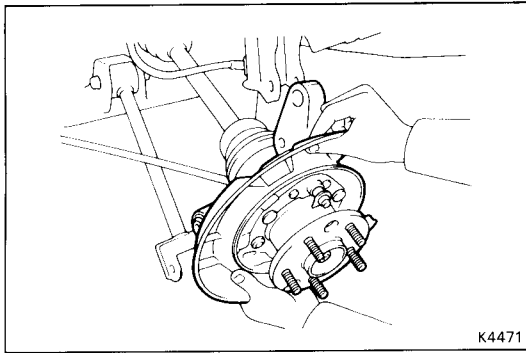
**Torque:** 730 kg-cm (53 ft-lb, 72 N·m)

**11. INSTALL AXLE SHAFT**

Using SST, press in the axle shaft to the axle carrier.  
SST 09636-20010

**12. INSTALL INNER OIL SEAL**

- (a) Using SST, drive in a new oil seal to the axle carrier.  
SST 09608-30012 (09608-04020, 09608-04110)
- (b) Apply MP grease to the oil seal lip.



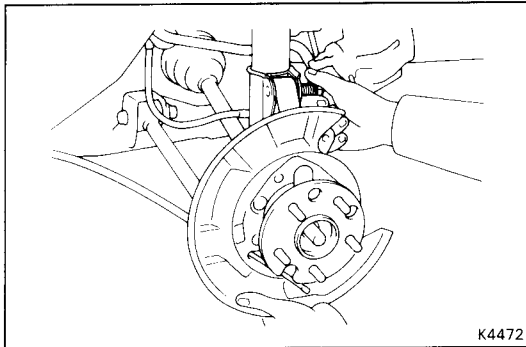
## INSTALLATION OF REAR AXLE HUB AND CARRIER

### 1. INSTALL AXLE CARRIER WITH AXLE HUB

- (a) Install the axle carrier to the drive shaft.

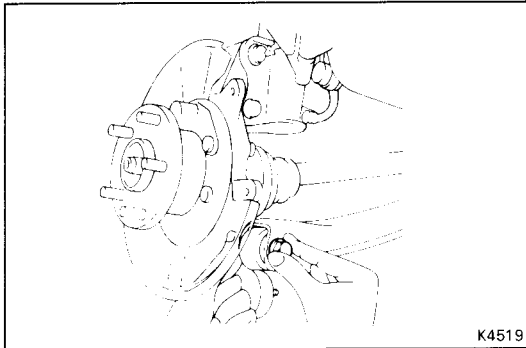
**NOTICE:** Be careful not to damage the oil seal.

- (b) Connect the No.2 suspension arm with a bolt to the axle carrier.



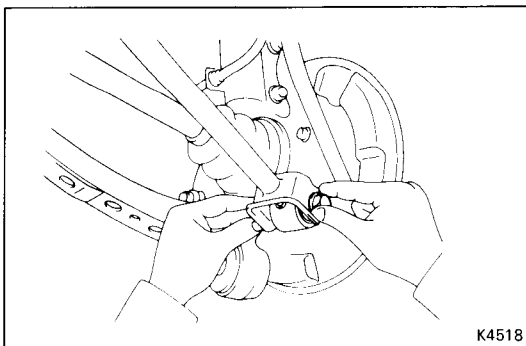
- (c) Install the axle carrier to the shock absorber with two bolts.

- (d) Install and finger tighten the two nuts.



### 2. INSTALL NO.2 SUSPENSION ARM NUT

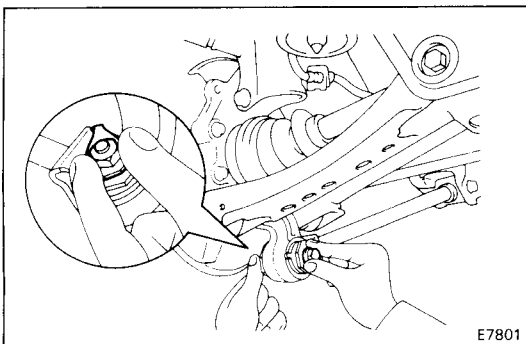
Install and finger tighten the nut.



### 3. CONNECT NO.1 SUSPENSION ARM

- (a) Connect No.1 suspension arm with a bolt to the axle carrier.

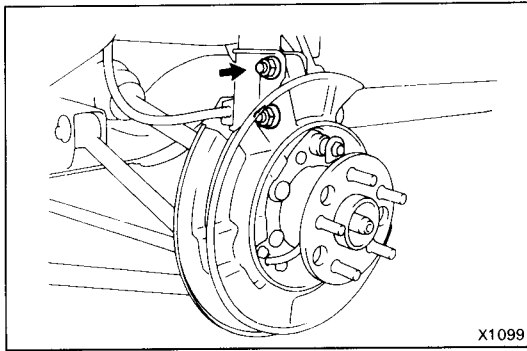
- (b) Install and finger tighten the nut.



### 4. CONNECT STRUT ROD

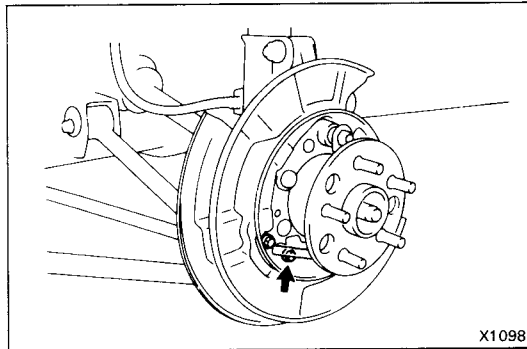
- (a) Connect strut rod with a bolt to the axle carrier.

- (b) Install and finger tighten the nut.

**5. TORQUE AXLE CARRIER NUTS**

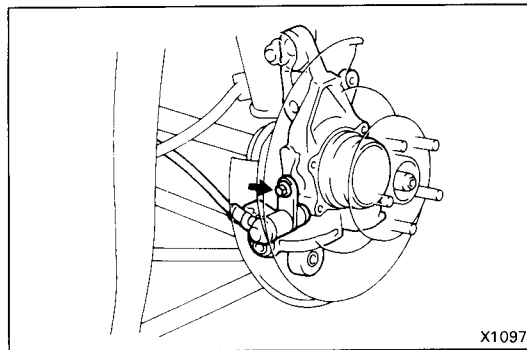
Torque the two bolts and nuts.

**Torque: 2,600 kg-cm (188 ft-lb, 255 N·m)**

**6. INSTALL PARKING BRAKE CABLE**

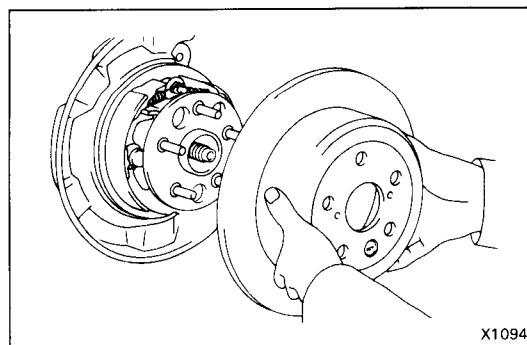
Install the parking brake cable with two bolts to the backing plate.

**Torque: 80 kg-cm (69 in.-lb, 7.8 N·m)**

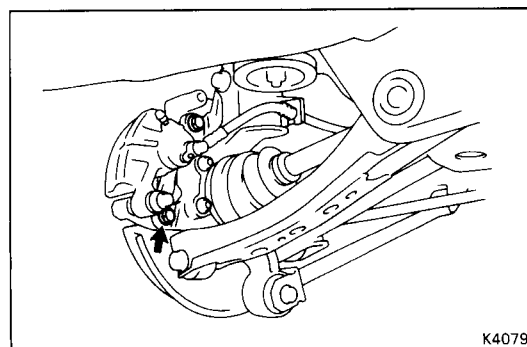
**7. (w/ ABS)****INSTALL REAR SPEED SENSOR**

Install the rear speed sensor with a bolt.

**Torque: 195 kg-cm (14 ft-lb, 19 N·m)**

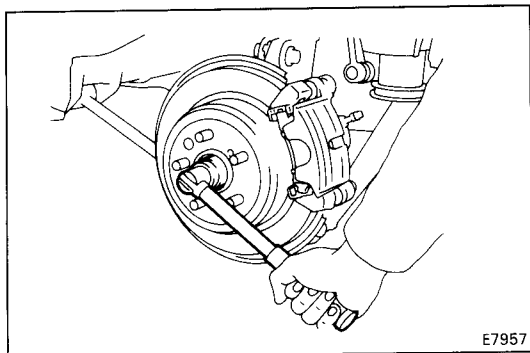
**8. INSTALL PARKING BRAKE ASSEMBLY****9. INSTALL ROTOR DISC**

Align matchmarks on the axle hub and rotor disc, and install the rotor disc.

**10. INSTALL BRAKE CALIPER**

Install the brake caliper with two bolts and torque the two bolts.

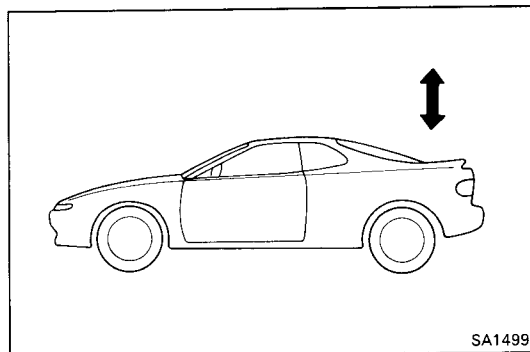
**Torque: 475 kg-cm (34 ft-lb, 47 N·m)**

**11. INSTALL PLATE WASHER, BEARING LOCK NUT, BEARING LOCK NUT CAP AND NEW COTTER PIN**

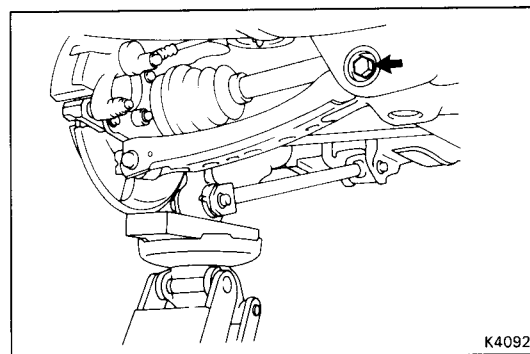
- (a) Install the plate washer and lock nut.
- (b) With the parking brake engaged, and torque the nut.

**Torque: 1,900 kg-cm (137 ft-lb, 186 N·m)**

- (c) Install the lock nut cap and a new cotter pin.

**12. STABILIZE SUSPENSION**

- (a) Temporarily install wheels.
- (b) Remove stands and bounce the vehicle up and down to stabilize the suspension.

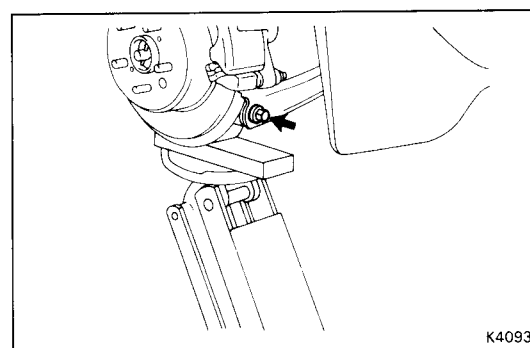
**13. JACK UP VEHICLE AND REMOVE WHEELS****14. JACK UP AXLE CARRIER**

Place wooden block on the jack and jack up axle carrier.

**15. TORQUE NO.2 SUSPENSION ARM MOUNTING BOLT**

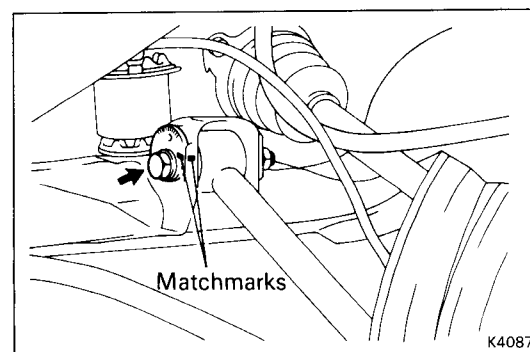
- (a) Torque the No.2 suspension arm body side bolt.

**Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)**



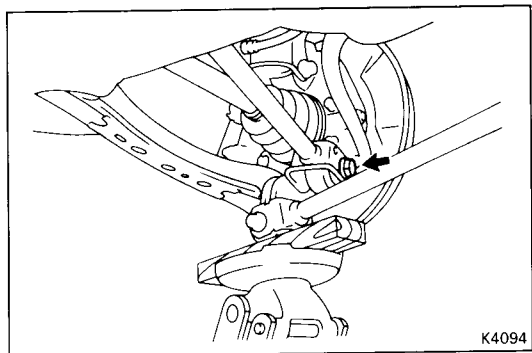
- (b) Torque the No.2 suspension arm axle carrier side bolt.

**Torque: 1,250 kg-cm (90 ft-lb, 123 N·m)**

**16. TORQUE NO.1 SUSPENSION ARM MOUNTING BOLT**

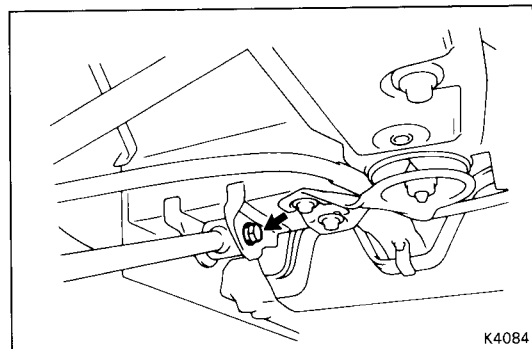
- (a) Align matchmarks on the adjusting cam and control arm retainer.
- (b) Torque the No. 1 suspension arm mounting bolt.

**Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)**



(c) Torque the No.1 suspension arm axle carrier side bolt.

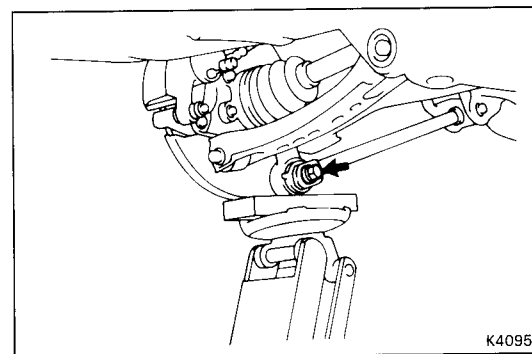
**Torque: 1,250 kg-cm (90 ft-lb, 123 N·m)**



#### **17. TORQUE STRUT ROD MOUNTING BOLT**

(a) Torque the strut rod body side bolt.

**Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)**



(b) Torque the strut rod axle carrier side bolt.

**Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)**

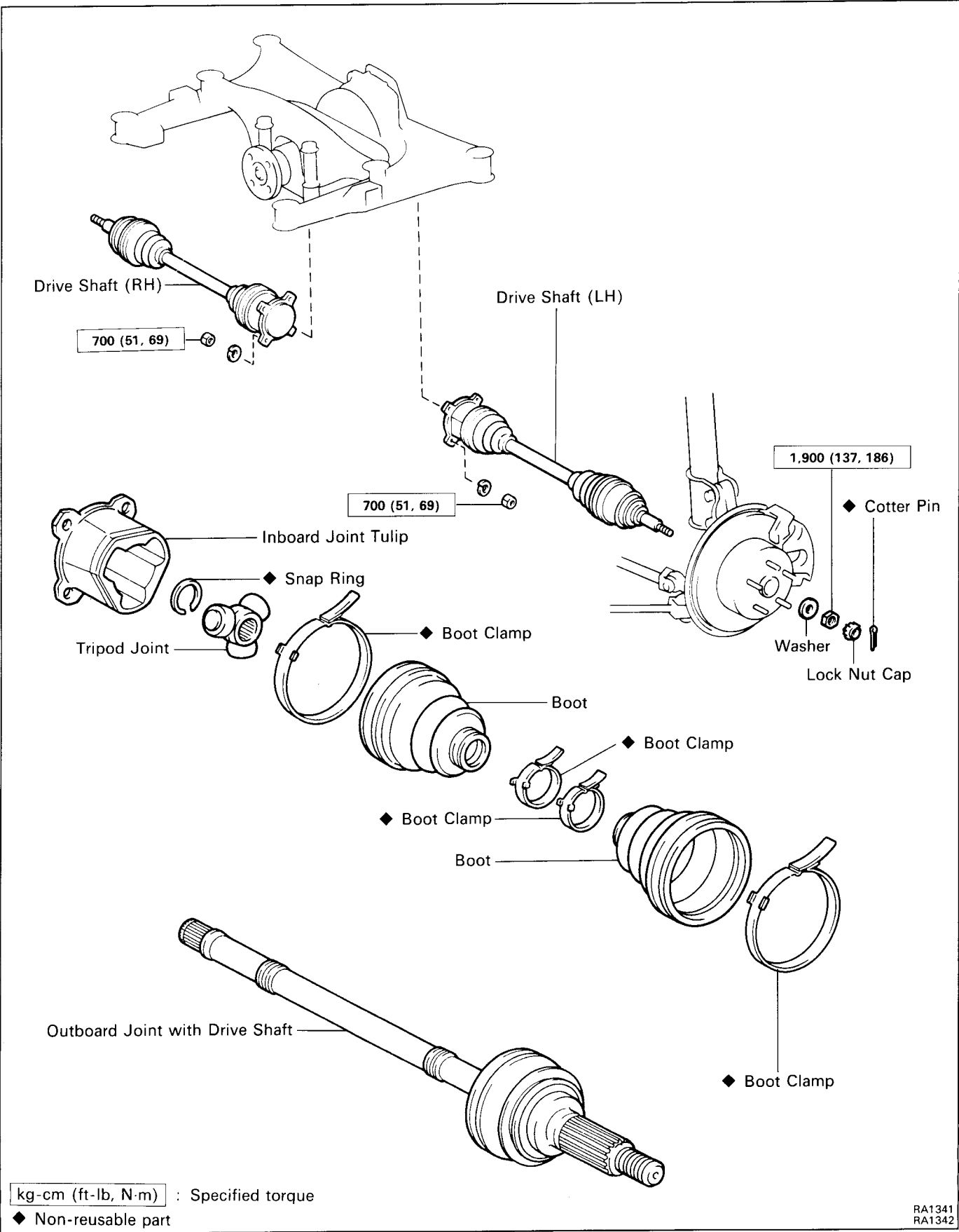
#### **18. INSTALL AND TORQUE WHEELS**

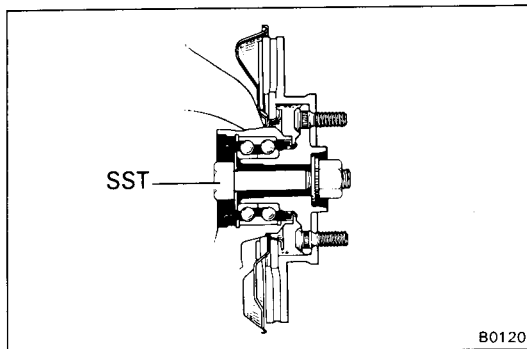
**Torque: 1,050 kg-cm (76 ft-lb, 103 N·m)**

#### **19. BLEED BRAKE SYSTEM**

#### **20. CHECK REAR WHEEL ALIGNMENT** (See page SA-5)

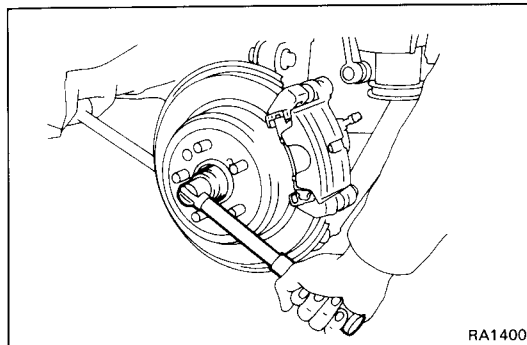
REAR DRIVE SHAFT  
COMPONENTS





**NOTICE:** The hub bearing could be damaged if it is subjected to the vehicle weight, such as when moving the vehicle with the drive shaft removed. Therefore, if it is absolutely necessary to place the vehicle weight on the hub bearing, first support it with SST.

SST 09608-16041 (09608-02020, 09608-02040)



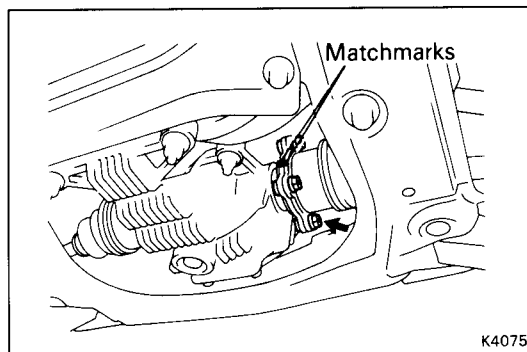
## REMOVAL OF REAR DRIVE SHAFT

(See page SA-31)

### 1. REMOVE REAR WHEELS

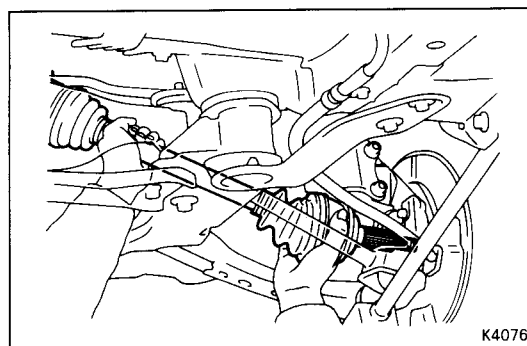
### 2. REMOVE COTTER PIN, LOCK NUT CAP AND LOCK NUT

- (a) Remove the cotter pin and lock nut cap.
- (b) With the parking brake engaged, loosen the bearing lock nut.



### 3. REMOVE DRIVE SHAFT

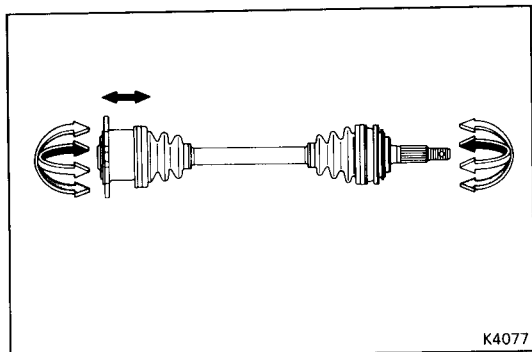
- (a) Place matchmarks on the inboard joint tulip and the side gear shaft flange.
- (b) With the parking brake engaged, remove the four nuts and washers.
- (c) Disconnect the drive shaft from the side gear shaft.



- (d) Remove the drive shaft from the axle carrier.

**HINT:** Push the axle carrier towards the outside of vehicle, and separate the drive shaft from the axle carrier.



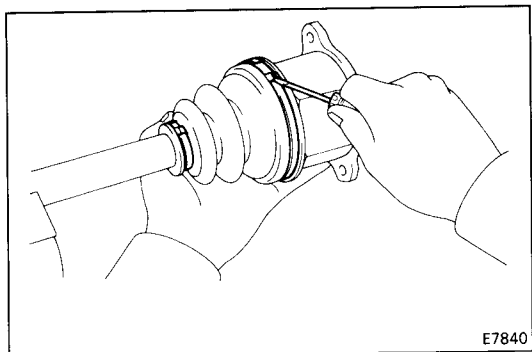


## DISASSEMBLY OF REAR DRIVE SHAFT

(See page SA-31)

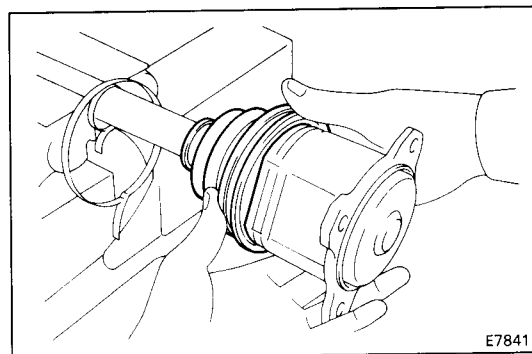
### 1. CHECK DRIVE SHAFT

- Check to see that there is no play in the inboard and outboard joints.
- Check to see that the inboard joint slide smoothly in the thrust direction.
- Check to see that there is no remarkable play in the radial direction of the inboard joint.
- Check the damage of boot.

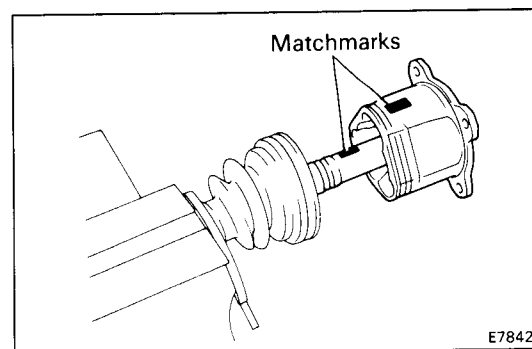


### 2. REMOVE INBOARD JOINT BOOT

- Using a screwdriver, remove the two boot clamps.



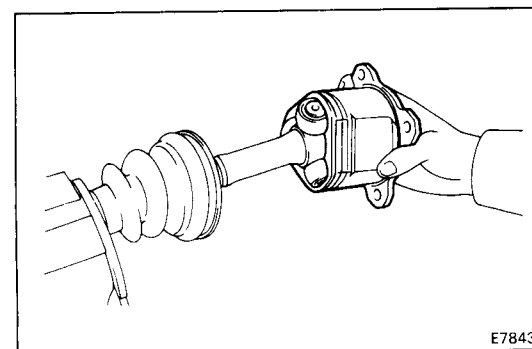
- Slide the inboard joint boot toward the outboard joint.



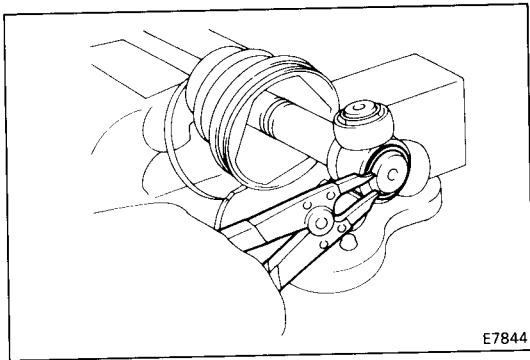
### 3. REMOVE INBOARD JOINT OUTER RACE

- Place matchmarks on the inboard joint tulip and drive shaft.

**NOTICE:** Do not punch the marks.

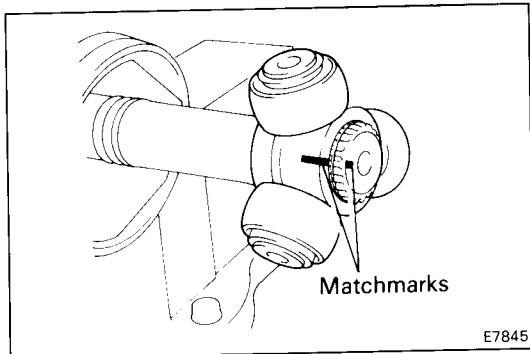


- Remove the inboard joint tulip from the drive shaft.



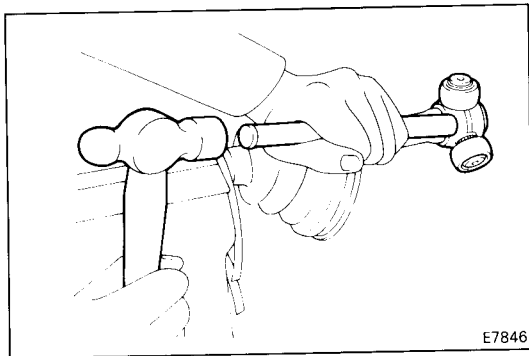
#### 4. REMOVE TRIPOD JOINT

- (a) Using snap ring pliers, remove the snap ring.



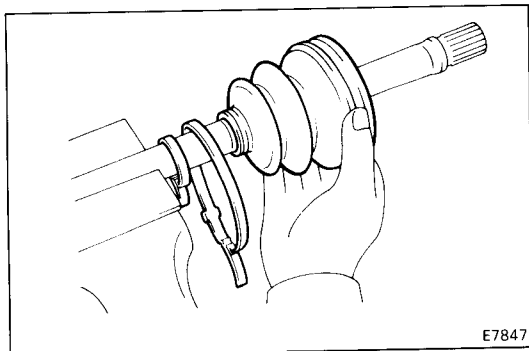
- (b) Place matchmarks on the shaft and the tripod joint.

**NOTICE:** Do not use a punch to mark the matchmarks. Use paint, etc.

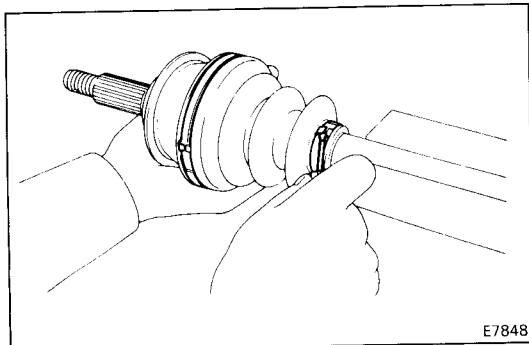


- (c) Using a hammer and brass bar, drive out the tripod joint from the drive shaft.

**NOTICE:** Do not tap the roller.



#### 5. REMOVE INBOARD JOINT BOOT AND CLAMPS

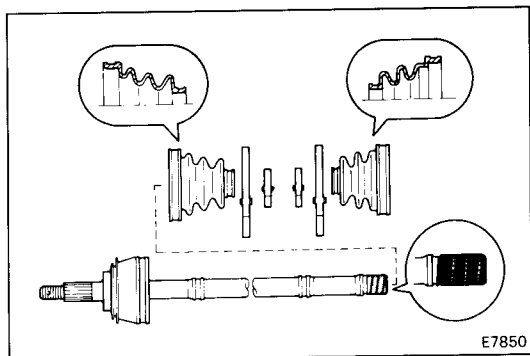


#### 6. REMOVE OUTBOARD JOINT BOOT

- (a) Using a screwdriver, remove the two boot clamps of the outboard joint boot.

- (b) Remove the boot from the outboard joint.

**NOTICE:** Do not disassemble the outboard joint.



## ASSEMBLY OF REAR DRIVE SHAFT

(See page SA-31)

### 1. TEMPORARILY INSTALL BOOTS AND NEW BOOT CLAMPS

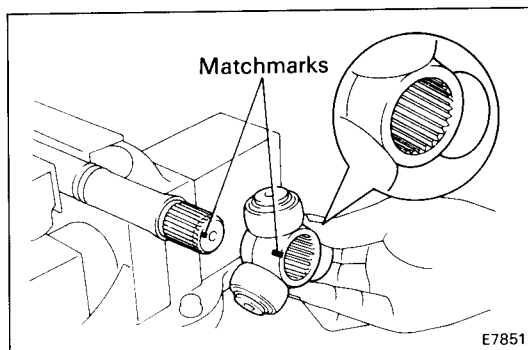
**NOTICE:** The boot and clamp of the outboard joint are smaller than those of the inboard joint.

**HINT:** Before installing the boot, wrap vinyl tape around the spline of the shaft to prevent damaging the boot.

Temporarily install the boots and new clamps to the drive shaft.

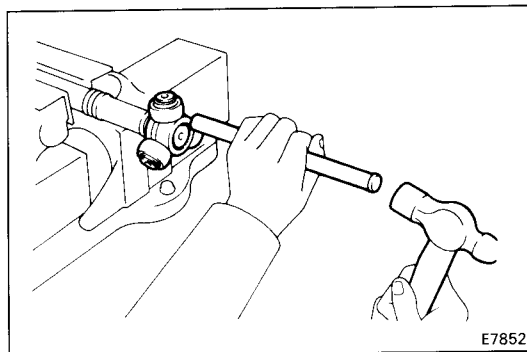
### 2. INSTALL TRIPOD JOINT

(a) Align the matchmarks placed before remove.

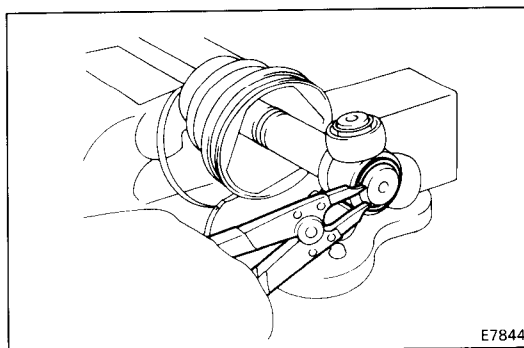


(b) Using a brass bar and hammer, tap in the tripod joint to the drive shaft.

**NOTICE:** Do not tap the roller.



(c) Using snap ring pliers, install a new snap ring.

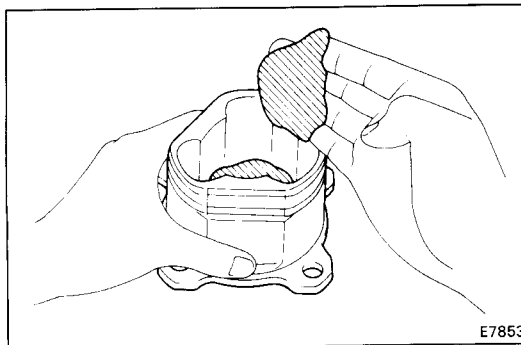


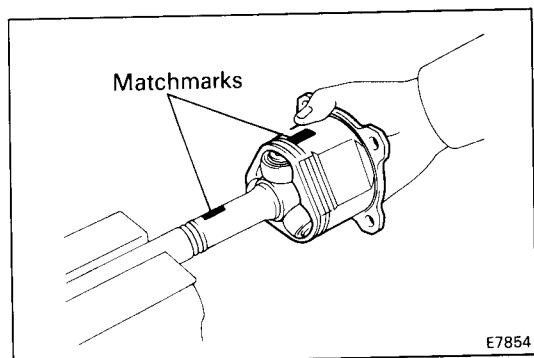
### 3. INSTALL INBOARD JOINT TULIP TO DRIVE SHAFT

(a) Pack in the grease to the inboard tulip.

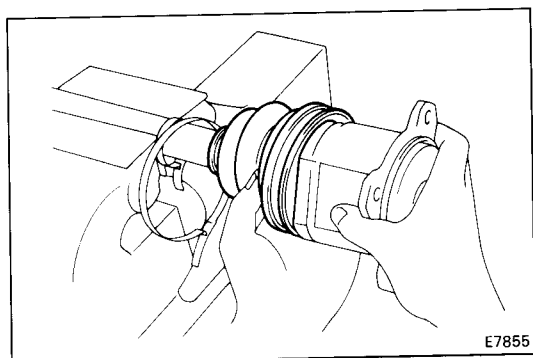
**HINT:** Use the grease supplied in the boot kit.

**Grease capacity: 180 g (0.40 lb)**





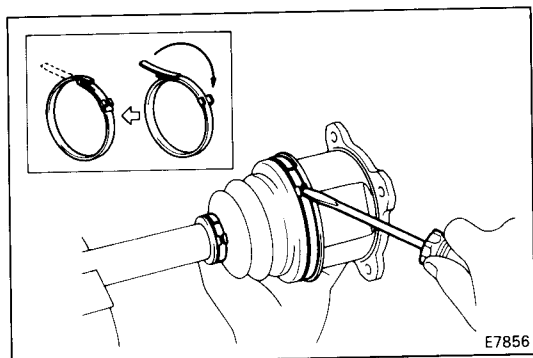
- (b) Align the matchmarks placed before remove, and install the inboard joint tulip to the drive shaft.



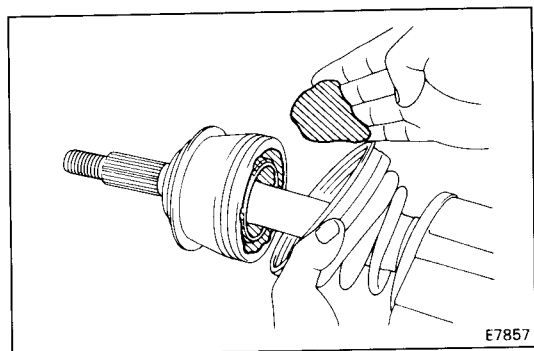
#### 4. INSTALL INBOARD JOINT BOOT

**NOTICE:** The clamps of the inboard joint are smaller than those of the outboard joint.

- (a) Be sure the boot is on the shaft groove.



- (b) Using a screwdriver, bend the boot clamp and lock it as shown.



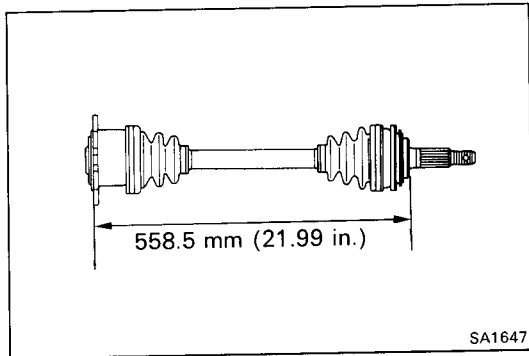
#### 5. INSTALL OUTBOARD JOINT BOOT

- (a) Before installing the boot, pack in grease.

**HINT:** Use the grease supplied in the boot kit.

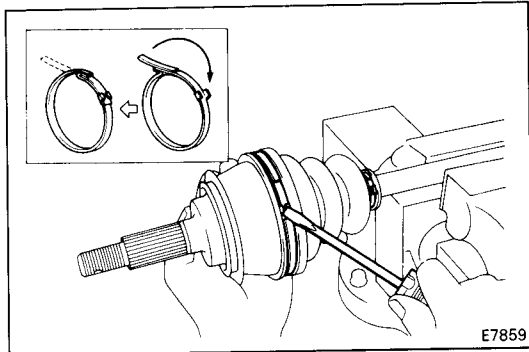
**Grease capacity:** 120 g (0.26 lb)

- (b) Be sure the boot is on the shaft groove.

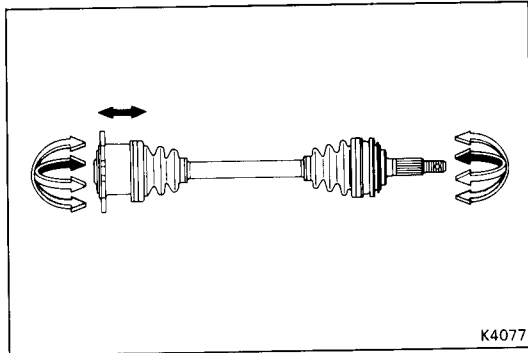


- (c) Insure that the boot is not stretched or contracted when drive shaft is at standard length.

**Drive shaft length: 558.5 mm (21.99 in.)**

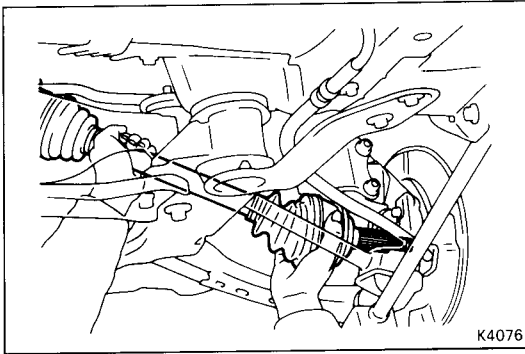


- (d) Using a screwdriver, bend the boot clamp and lock it as shown.



## 6. CHECK DRIVE SHAFT

- (a) Check to see that there is no play in the inboard joint and outboard joint.
- (b) Check to see that the inboard joint side smoothly in the thrust direction.



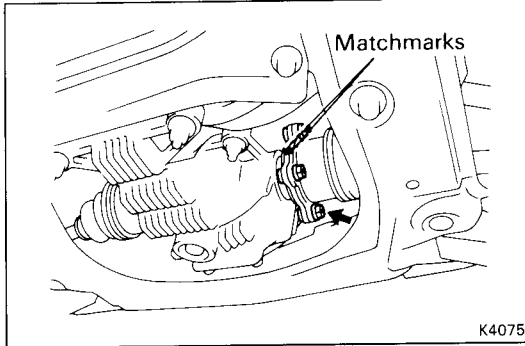
## INSTALLATION OF REAR DRIVE SHAFT

(See page SA-31)

### 1. INSTALL DRIVE SHAFT

- (a) Install the drive shaft to the axle carrier.

HINT: Be careful not to damage the boot.

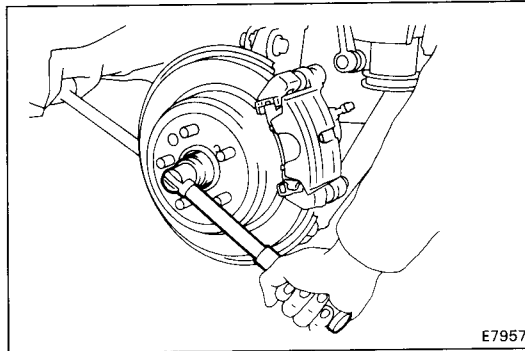


- (b) Align the matchmarks on the inboard joint tulip and the side gear shaft flange.

- (c) Connect the drive shaft with the four bolts and washer to the side gear shaft.

- (d) With the parking brake engaged, torque the four bolts.

**Torque: 700 kg-cm (51 ft-lb, 69 N·m)**



### 2. INSTALL BEARING LOCK NUT, LOCK NUT CAP AND COTTER PIN

- (a) Install the washer and lock nut.

- (b) With the parking brake engaged, torque the lock nut.

**Torque: 1,900 kg-cm (137 ft-lb, 186 N·m)**

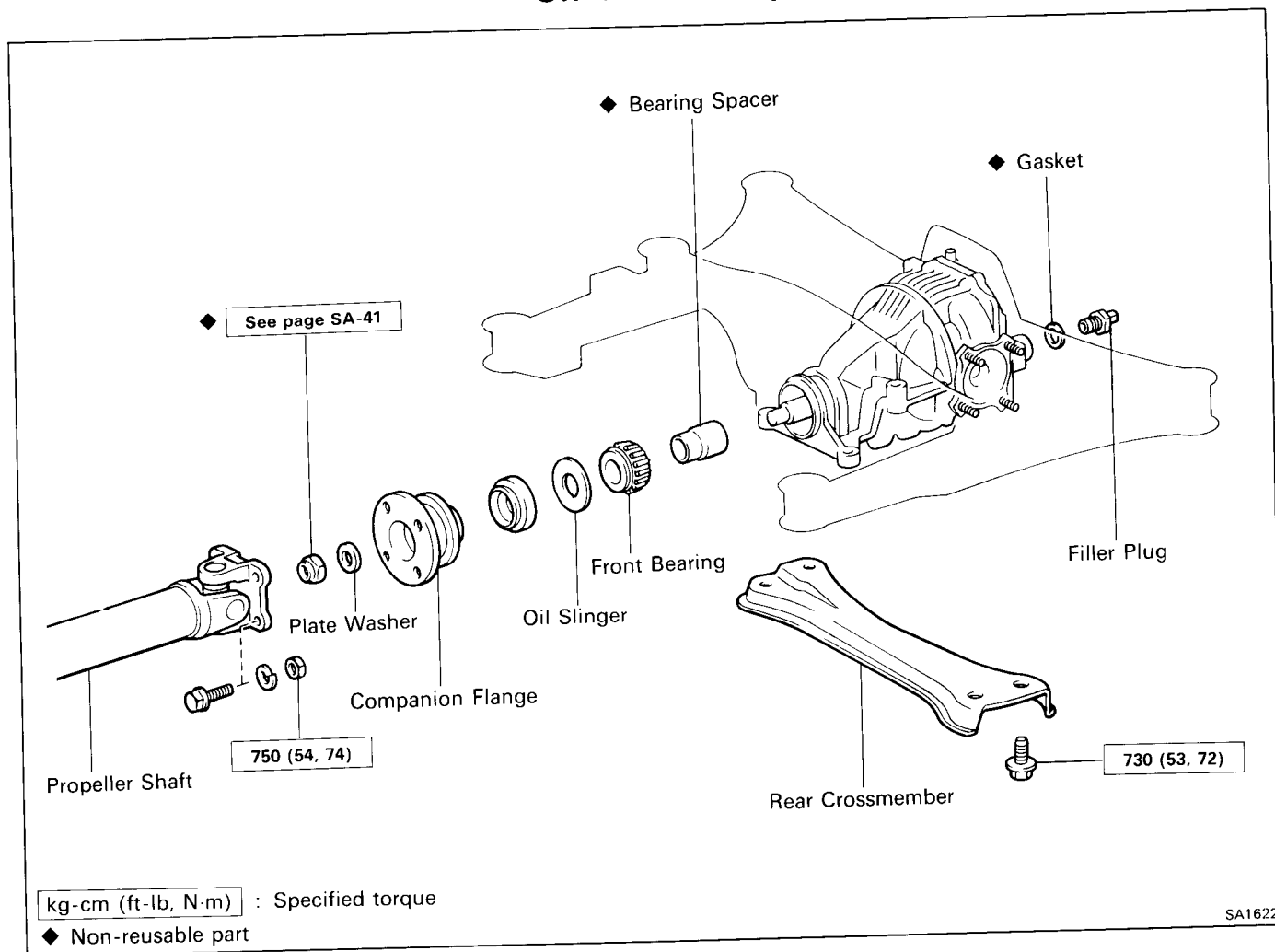
- (c) Install the lock nut cap, and using pliers, install a new cotter pin.

### 3. INSTALL REAR WHEELS

### 4. CHECK REAR WHEEL ALIGNMENT (See page SA-5)

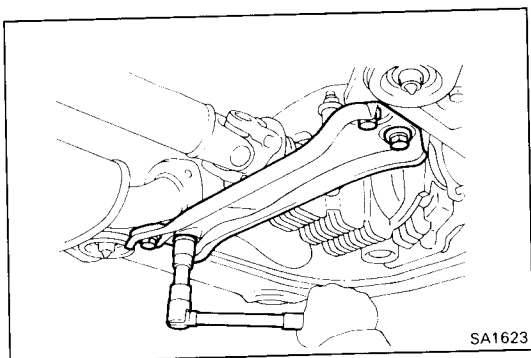
# DIFFERENTIAL

## On-Vehicle Repair



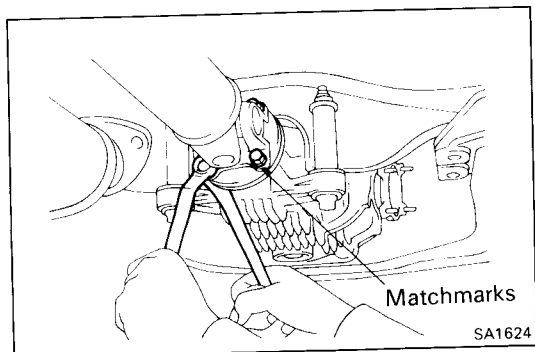
## REPLACEMENT OF FRONT OIL SEAL

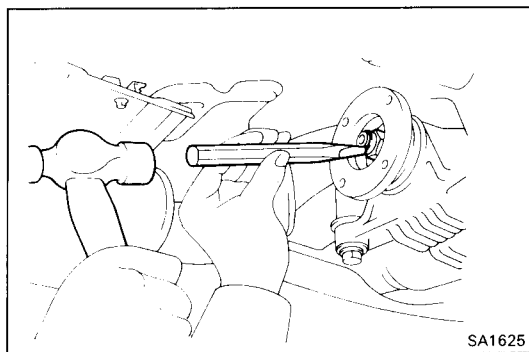
### 1. REMOVE REAR CROSSMEMBER



### 2. DISCONNECT PROPELLER SHAFT

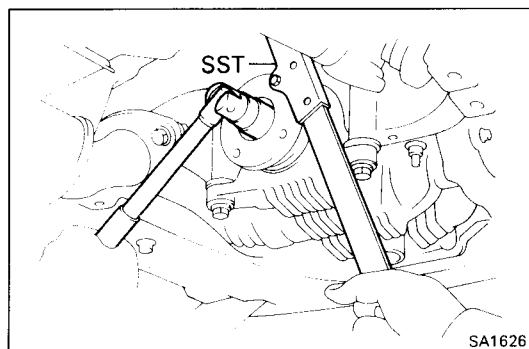
- Place the matchmarks on the both flanges.
- Remove the four bolts, washers and nuts.
- Disconnect the propeller shaft from the differential.





### 3. REMOVE COMPANION FLANGE

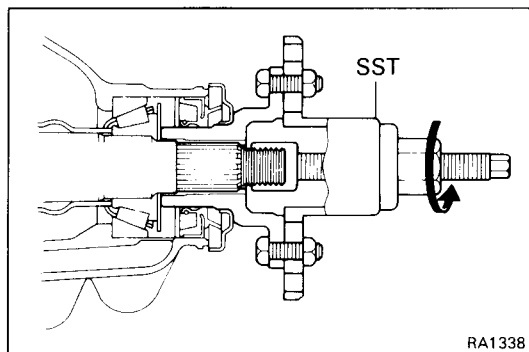
- (a) Using a hammer and chisel, loosen the staked part of the nut.



- (b) Using SST to hold the flange, remove the nut.

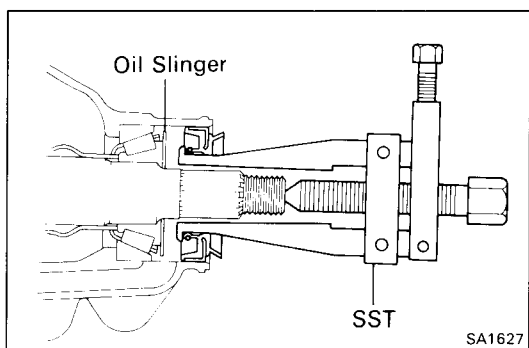
SST 09330-00021

- (c) Remove the plate washer.



- (d) Using SST, remove the companion flange.

SST 09557-22022

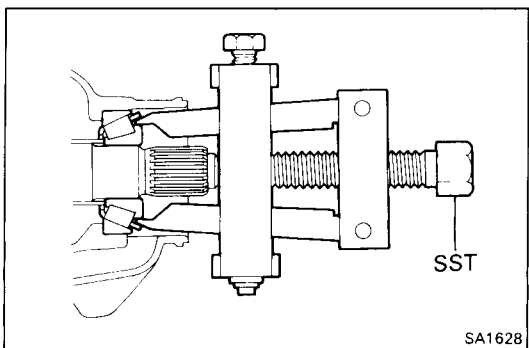


### 4. REMOVE FRONT OIL SEAL AND OIL Slinger

- (a) Using SST, remove the front oil seal.

SST 09308-10010

- (b) Remove the oil slinger.

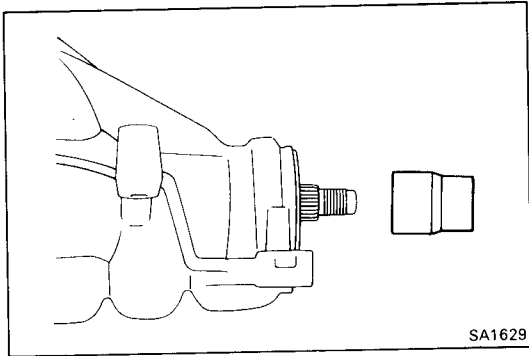


### 5. REMOVE FRONT BEARING AND BEARING SPACER

- (a) Using SST, remove the front bearing.

SST 09556-22010

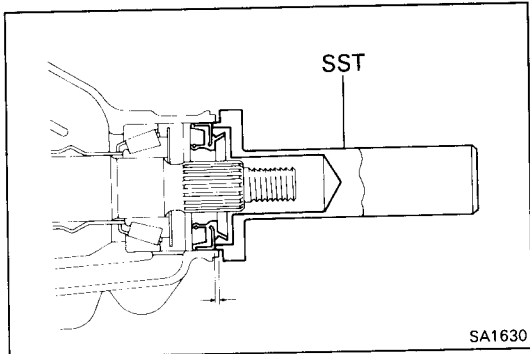




(b) Remove the bearing spacer.

## 6. INSTALL NEW BEARING SPACER AND FRONT BEARING

- (a) Install a new bearing spacer on the shaft.
- (b) Install the front bearing on the shaft.



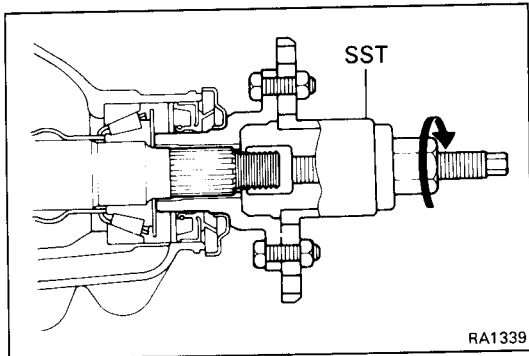
## 7. INSTALL OIL SLINGER AND NEW OIL SEAL

- (a) Install the oil slinger on the shaft.
- (b) Using SST, drive in a new oil seal.

SST 09554-22010

Oil seal drive in depth: 2.0 mm (0.079 in.)

- (c) Apply MP grease to the oil seal lip.



## 8. INSTALL COMPANION FLANGE

- (a) Using SST, install the companion flange.
- SST 09557-22022

- (b) Install the plate washer.
  - (c) Coat the threads of a new nut with gear oil.
  - (d) Using SST to hold the flange, tighten the nut.
- SST 09330-00021

Torque: 1,100 kg-cm (80 ft-lb, 108 N·m)

## 9. CHECK DRIVE PINION BEARING PRELOAD

Using a torque wrench, measure the preload of the backlash between the drive pinion and ring gear.

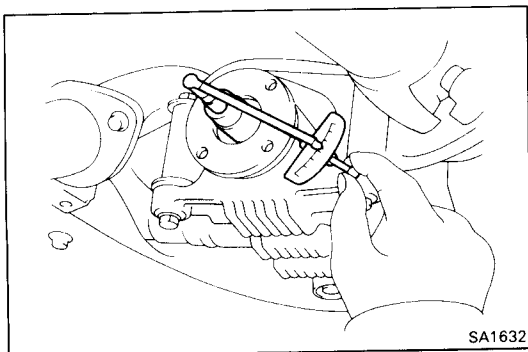
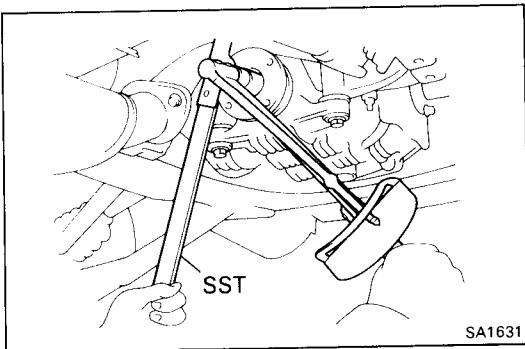
Preload (at starting):

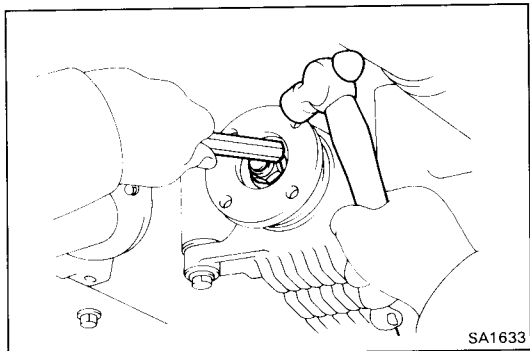
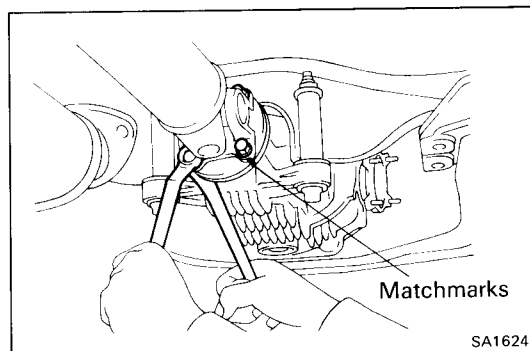
|                |                                                     |
|----------------|-----------------------------------------------------|
| New bearing    | 10 – 16 kg-cm<br>(8.7 – 13.9 in.-lb, 1.0 – 1.6 N·m) |
| Reused bearing | 5 – 8 kg-cm<br>(4.3 – 6.9 in.-lb, 0.5 – 0.8 N·m)    |

- If preload is greater than specification, replace the bearing spacer.
- If preload is less than specification, retighten the nut 130 kg-cm (9 ft-lb, 13 N·m) at a time until the specified preload is reached.

If the maximum torque is exceed while retightening the nut, replace the bearing spacer and repeat the preload procedure. Do not back off the pinion nut to reduce the preload.

Maximum torque: 2,400 kg-cm (174 ft-lb, 235 N·m)

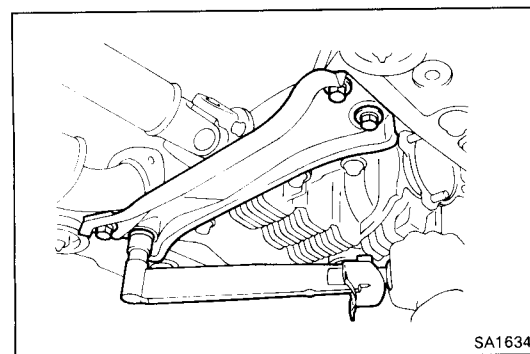


**10. STAKE DRIVE PINION NUT****11. CONNECT PROPELLER SHAFT**

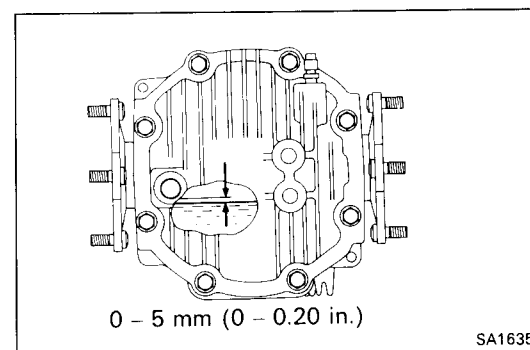
(a) Align the matchmarks on the flanges and connect the propeller shaft with the four bolts, washers and nuts.

(b) Torque the bolts and nuts.

**Torque: 750 kg-cm (54 ft-lb, 74 N·m)**

**12. INSTALL REAR CROSSMEMBER**

**Torque: 730 kg-cm (53 ft-lb, 72 N·m)**

**13. CHECK OIL LEVEL**

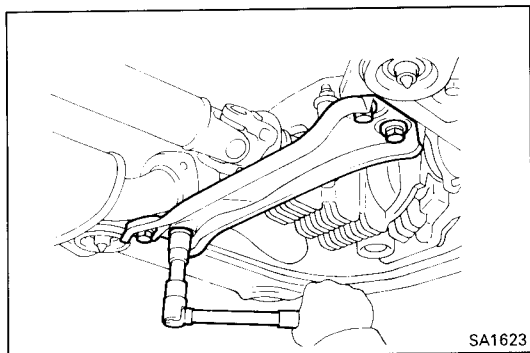
**Oil grade: API GL-5 hypoid gear oil**

**Viscosity: Above -18°C (0°F) SAE 90**

**Below -18°C (0°F) SAE 80W-90**

**Capacity: 1.1 liters (1.2 US qts, 1.0 Imp.qts)**

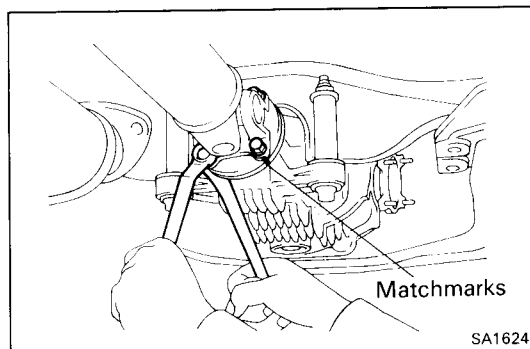
0 – 5 mm (0 – 0.20 in.)



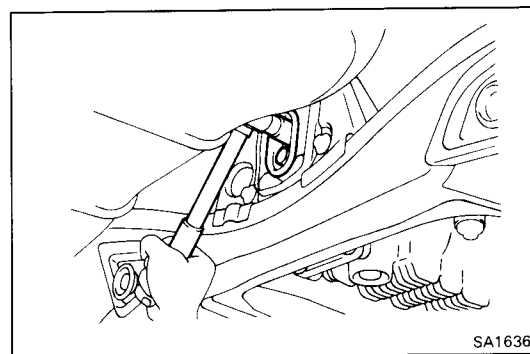
## REMOVAL OF DIFFERENTIAL

(See page SA-39)

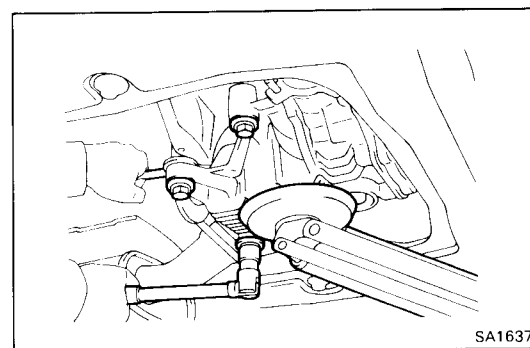
1. DRAIN DIFFERENTIAL OIL
2. REMOVE DRIVE SHAFTS (See page SA-31)
3. REMOVE REAR CROSSMEMBER



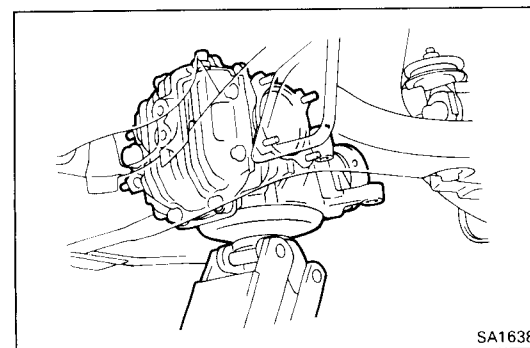
4. DISCONNECT PROPELLER SHAFT
  - (a) Place the matchmarks on the both flanges.
  - (b) Remove the four bolts, washers and nuts.
  - (c) Disconnect the propeller shaft from the differential.



5. REMOVE DIFFERENTIAL
  - (a) Jack up the differential slightly.
  - (b) Remove the two bolts.

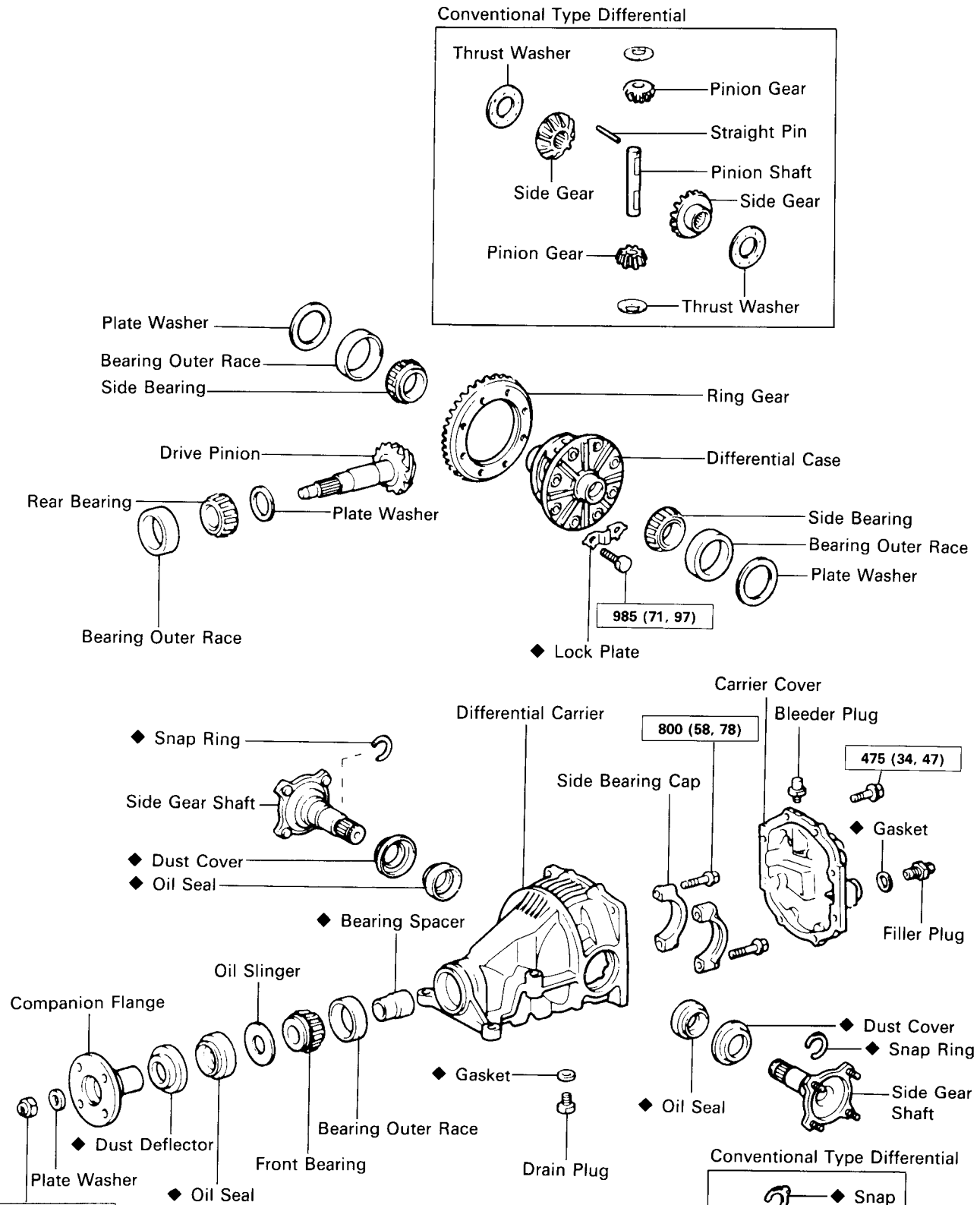


- (c) Remove the four nuts and bolts.



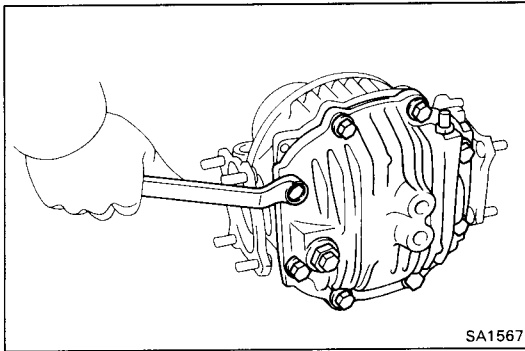
- (d) Remove the differential from the body.

## DIFFERENTIAL CARRIER



Kg-cm (ft-lb, N-m) : Specified torque

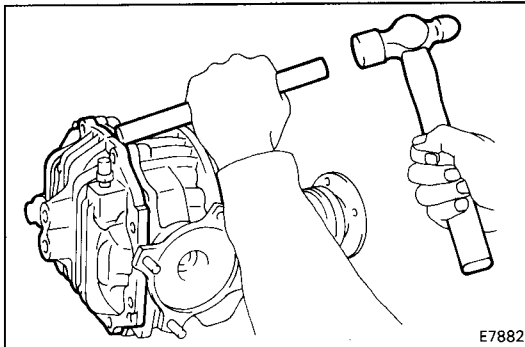
◆ Non-reusable part



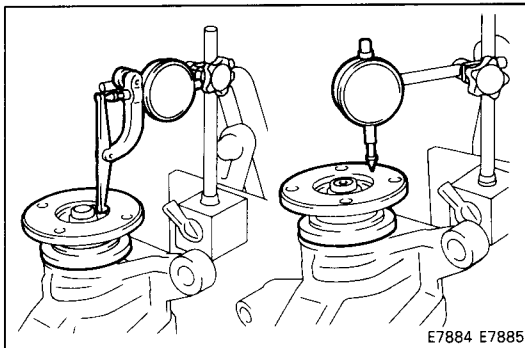
## PRE-INSPECTION OF DIFFERENTIAL CARRIER

### 1. REMOVE DIFFERENTIAL CARRIER COVER

(a) Remove the eight bolts.



(b) Using a brass bar and hammer, separate the cover and carrier.



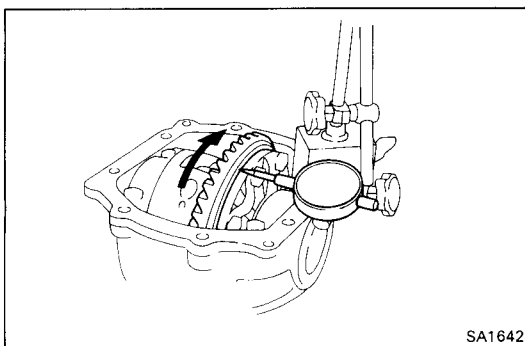
### 2. CHECK COMPANION FLANGE RUNOUT

Using a dial indicator, measure the lateral and radial runout of the companion flange.

**Maximum lateral runout:** 0.10 mm (0.039 in.)

**Maximum radial runout:** 0.10 mm (0.039 in.)

If the runout is greater than the maximum, replace the companion flange.

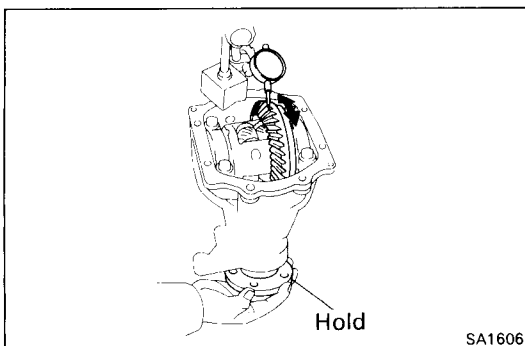


### 3. CHECK RING GEAR RUNOUT

Using a dial indicator, measure the runout of the ring gear.

**Maximum runout:** 0.07 mm (0.0028 in.)

If the runout is greater than the maximum, replace the ring gear.

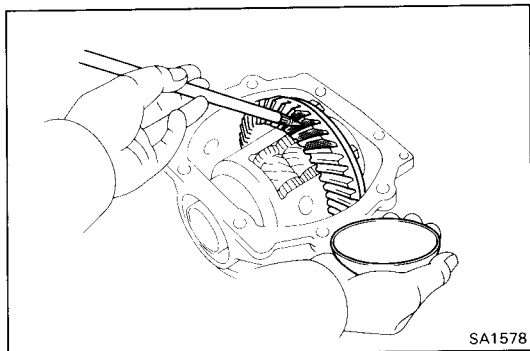


### 4. CHECK RING GEAR BACKLASH

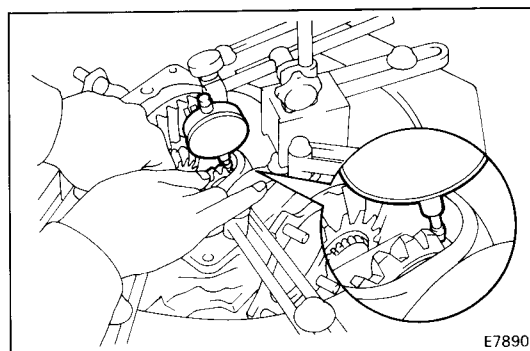
Using a dial indicator, check the backlash of the ring gear.

**Backlash:** 0.13 – 0.18 mm (0.0051 – 0.0071 in.)

If the backlash is not within specification, adjust the side bearing preload.



#### 5. CHECK TOOTH CONTACT (See page SA-58)

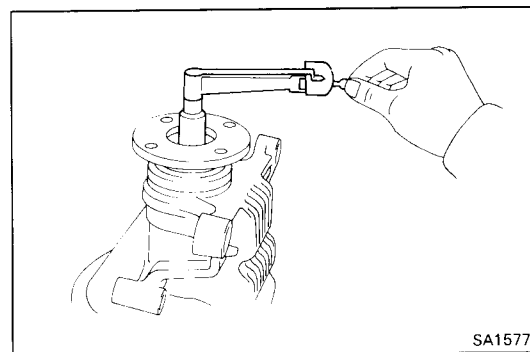


#### 6. (CONVENTIONAL TYPE DIFFERENTIAL) CHECK SIDE GEAR BACKLASH

Using a dial indicator, check the backlash of the side gear while holding one pinion gear toward the differential case.

**Backlash:** 0.05 – 0.20 mm (0.0020 – 0.0079 in.)

If the backlash is not within specification, install the side gear thrust washers of different thickness.



#### 7. MEASURE DRIVE PINION PRELOAD

Using a torque wrench, measure the preload of the backlash between the drive pinion and ring gear.

**Preload (at starting):**

5 – 8 kg-cm (4.3 – 6.9 in.-lb, 0.5 – 0.8 N·m)

#### 8. CHECK TOTAL PRELOAD

Using a torque wrench, measure the total preload.

**Total preload (at starting):**

In addition to drive pinion preload

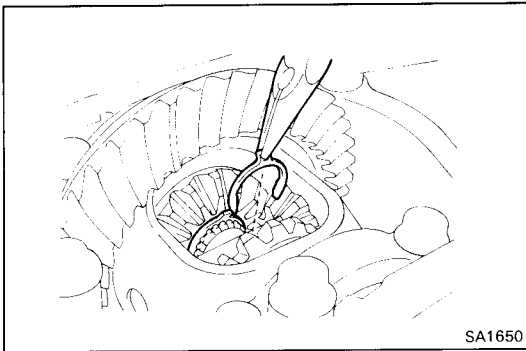
3 – 5 kg-cm (2.6 – 4.3 in.-lb, 0.3 – 0.5 N·m)

If necessary disassemble and inspect a differential.

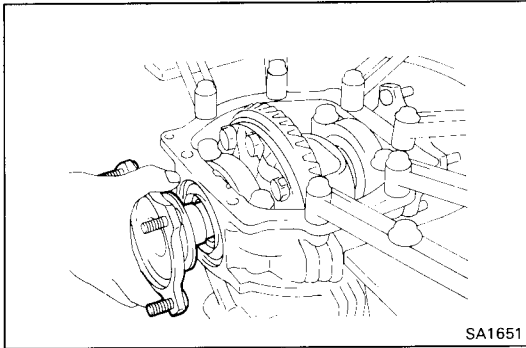
**DISASSEMBLY OF DIFFERENTIAL CARRIER****1. REMOVE SIDE GEAR SHAFTS**

(Conventional Type Differential)

(a) Using needle nose pliers, remove the two snap ring.



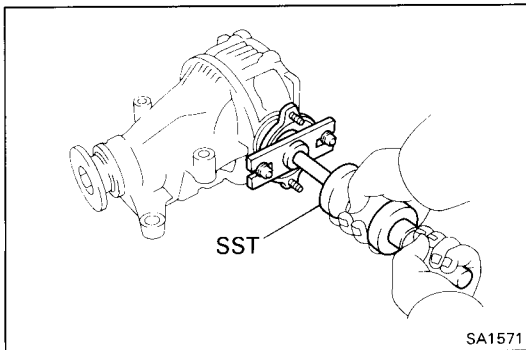
(b) Pull out the two side gear shafts.



(Torque Sensing Limited Slip Differential)

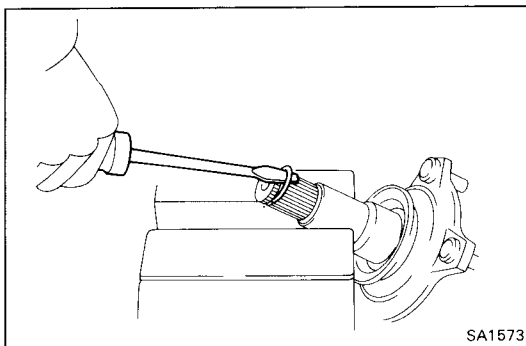
(a) Using SST, drive out the side gear shaft.

SST 09520-24010



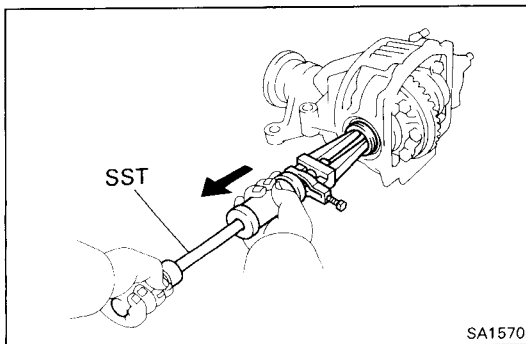
(b) Remove the snap ring from the side gear shaft.

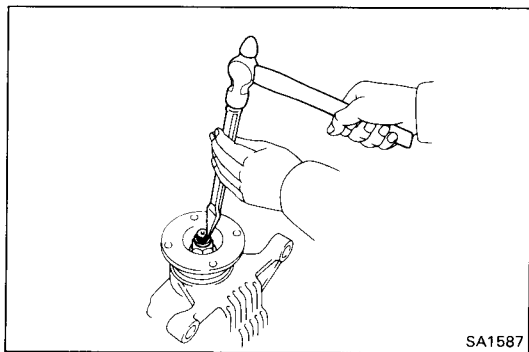
HINT: Use a soft jaw vise.

**2. REMOVE SIDE GEAR SHAFT OIL SEALS**

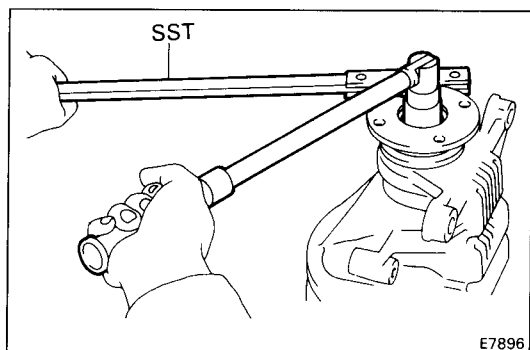
Using SST, remove the two oil seals from the housing.

SST 09308-00010



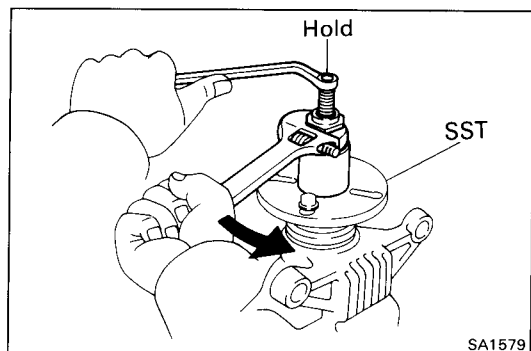
**3. REMOVE COMPANION FLANGE**

- (a) Using a hammer and chisel, loosen the staked part of the nut.

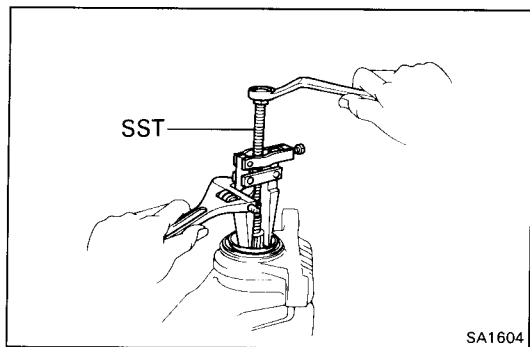


- (b) Using SST to hold the flange, remove the nut.  
SST 09330-00021

- (c) Remove the plate washer.

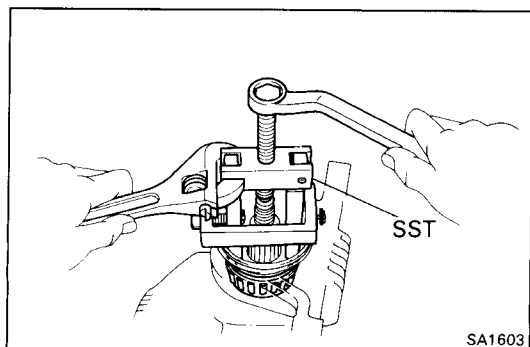


- (d) Using SST, remove the companion flange.  
SST 09557-22022 (09557-22040)

**4. REMOVE FRONT OIL SEAL AND OIL SLINGER**

- (a) Using SST, remove the oil seal from the housing.  
SST 09308-10010

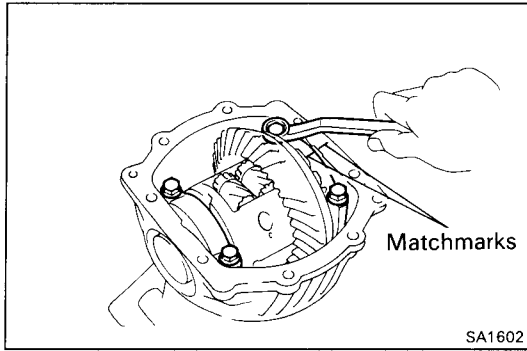
- (b) Remove the oil slinger.

**5. REMOVE FRONT BEARING AND BEARING SPACER**

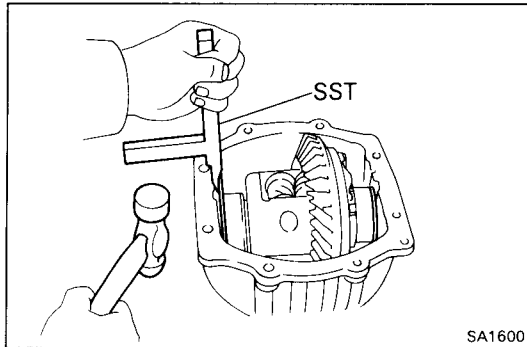
- (a) Using SST, remove the bearing from the housing.  
SST 09556-22010

- (b) Remove the bearing spacer.



**6. REMOVE DIFFERENTIAL CASE**

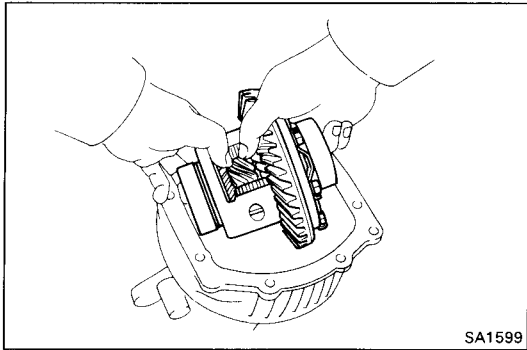
- (a) Place matchmarks on the bearing cap and differential carrier.
- (b) Remove the two bearing caps.



- (c) Using SST, remove the two side bearing preload adjusting plate.

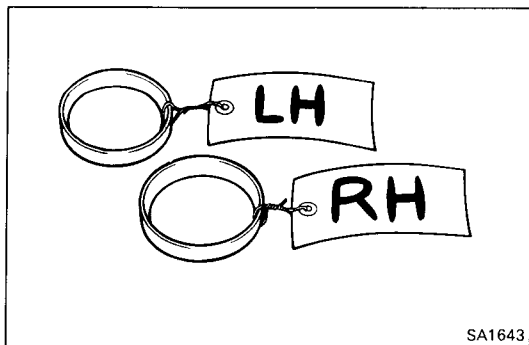
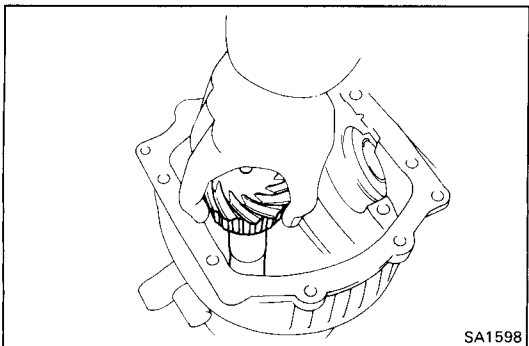
SST 09504-22011

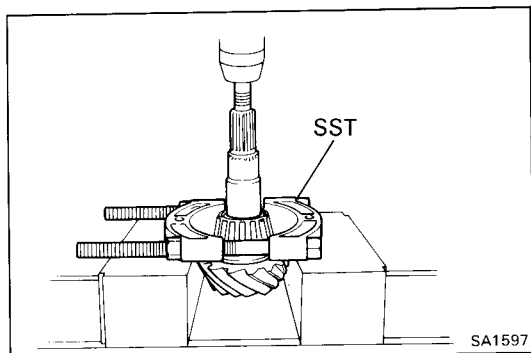
HINT: Measure the adjusting plate washer and note the thickness.



- (d) Remove the differential case and bearing outer race from the carrier.

HINT: Tag the bearing outer races to show the location for reassembly.

**7. REMOVE DRIVE PINION FROM DIFFERENTIAL CARRIER**



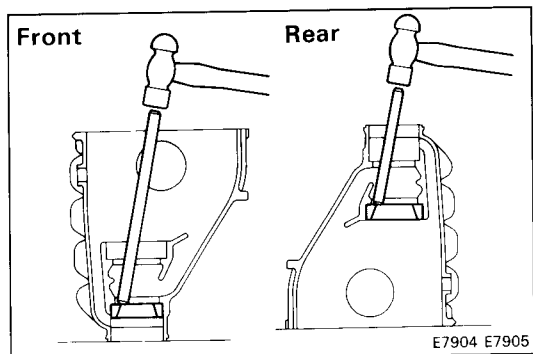
## 8. REMOVE DRIVE PINION REAR BEARING

- (a) Using SST and a press, remove the bearing from the drive pinion.

SST 09950-00020

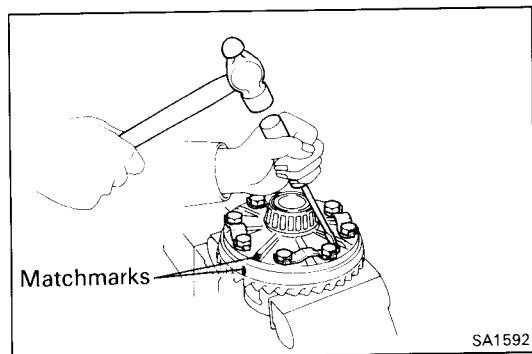
HINT: If the drive pinion or ring gear are damaged replace them a set.

- (b) Remove the plate washer.



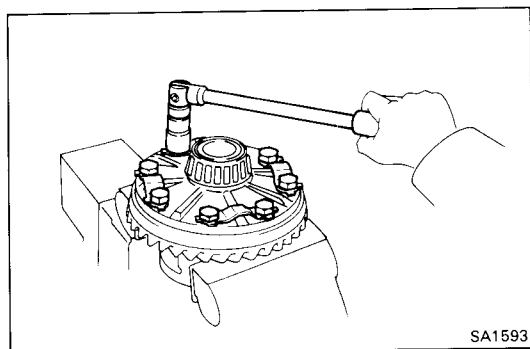
## 9. REMOVE FRONT AND REAR BEARING OUTER RACES

Using a hammer and brass bar, drive out the outer races from the carrier.

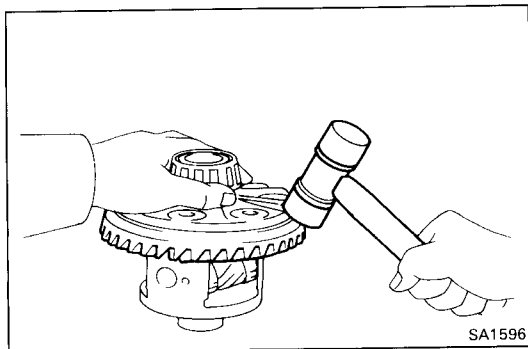


## 10. REMOVE RING GEAR

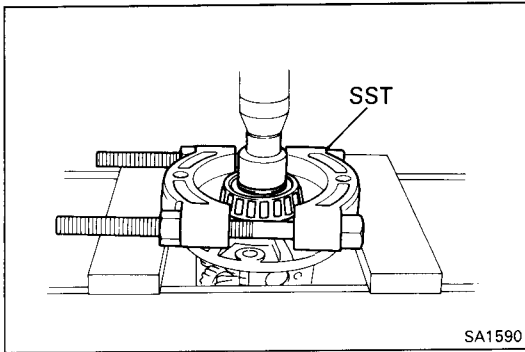
- (a) Place the matchmarks on the ring gear and differential case.
- (b) Unstake the lock plates.



- (c) Remove the eight bolts and four lock plates.

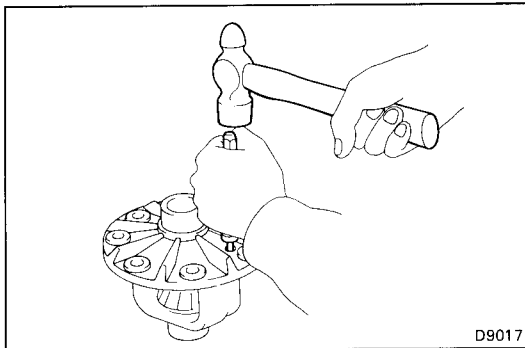


- (d) Using a plastic hammer, tap on the ring gear to separate it from differential case.

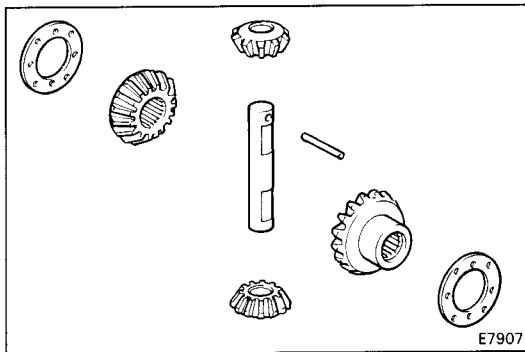
**11. REMOVE SIDE BEARINGS**

Using SST, press out two side bearings from differential case.

SST 09950-00020

**12. (CONVENTIONAL TYPE DIFFERENTIAL)  
DISASSEMBLE DIFFERENTIAL CASE**

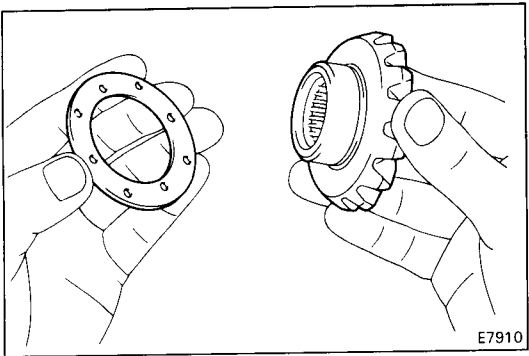
(a) Using a hammer and punch, drive out the straight pin.



(b) Remove the following parts from the differential case:

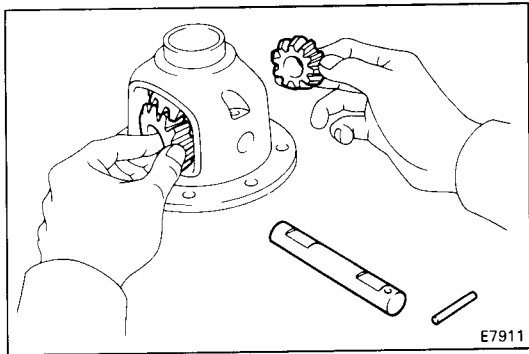
- Pinion shaft
- Two pinion gears
- Two side gears
- Four thrust washers

ASSEMBLY OF DIFFERENTIAL CARRIER



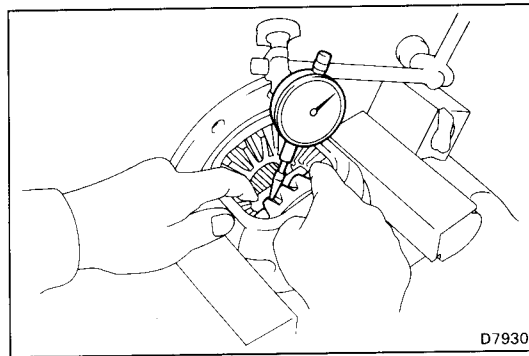
1. (CONVENTIONAL TYPE DIFFERENTIAL)  
ASSEMBLE DIFFERENTIAL CASE

(a) Install the thrust washers to the side gears.



(b) Install the side gears with thrust washers and pinion gears with thrust washers.

(c) Install the pinion shaft.



(d) Check the side gear backlash.

Measure the side gear backlash while holding one pinion gear toward the case.

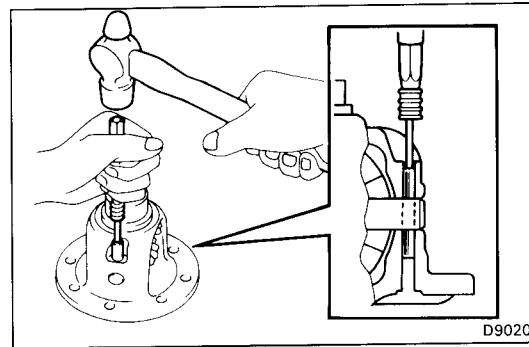
**Backlash: 0.05 – 0.20 mm (0.0020 – 0.0079 in.)**

If the backlash is not within specification, install the side gear thrust washers of different thickness.

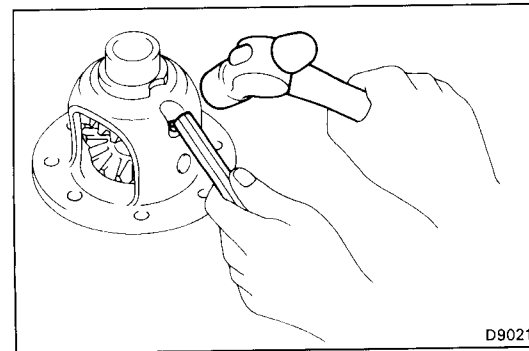
| Thrust washer thickness |               | mm (in.) |
|-------------------------|---------------|----------|
| 0.95 (0.0374)           | 1.10 (0.0433) |          |
| 1.00 (0.0394)           | 1.15 (0.0453) |          |
| 1.05 (0.0413)           | 1.20 (0.0472) |          |

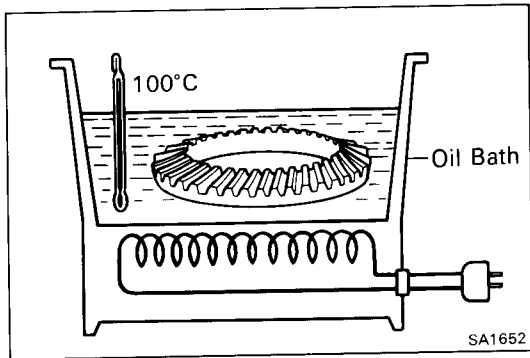
**HINT:** Use washers of same thickness on both the right and left sides.

(e) Using a hammer and punch, drive in the straight pin through the case and hole in the pinion shaft.



(f) Stake the case.





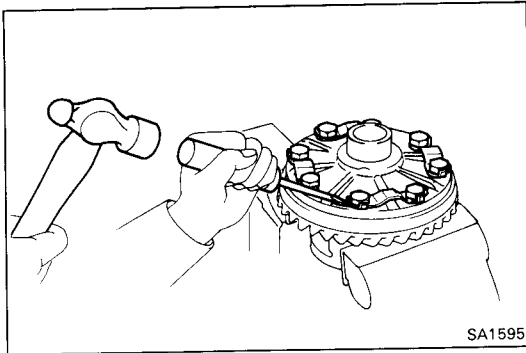
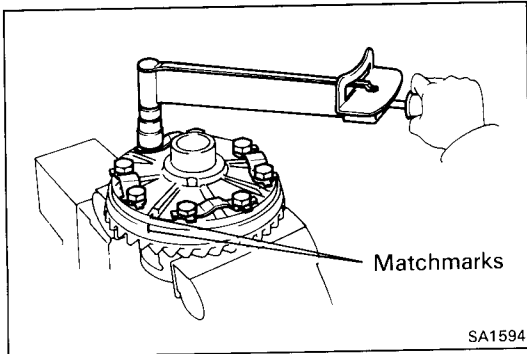
## 2. INSTALL RING GEAR ON DIFFERENTIAL CASE

- Clean the contact surface of the differential case.
- Heat the ring gear to about 100°C (212°F) in an oil bath.

**NOTICE:** Do not heat the ring gear above 110°C (230°F).

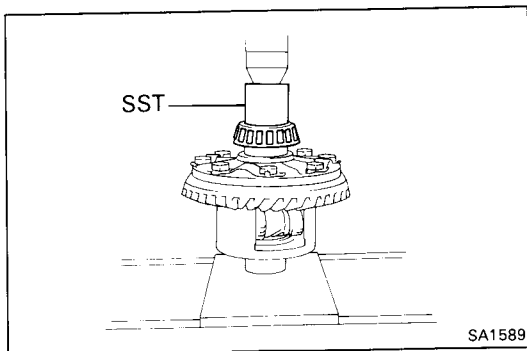
- Clean the contact surface of the ring gear with cleaning solvent.
- Then quickly install the ring gear on the differential case.
- Align the matchmarks on the ring gear and differential case.
- Temporarily install the lock plates and set bolts.
- After the ring gear cools down enough, tighten the set bolts uniformly and a little at a time.

**Torque:** 985 kg-cm (71 ft-lb, 97 N·m)



- Using a hammer and drift punch, stake the lock plates.

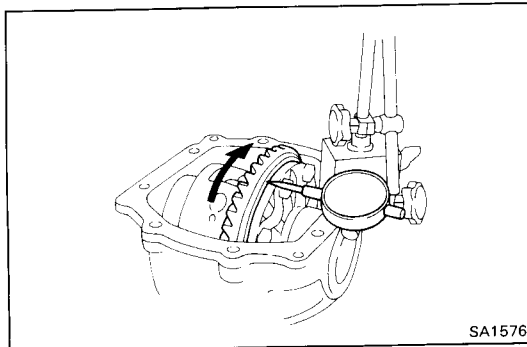
**HINT:** Stake one claw flush with the flat surface of the nut. For the claw contacting the protruding portion of the nut, stake only the half on the tightening side.



## 3. INSTALL SIDE BEARINGS

Using a press and SST, press in the bearings into the differential case.

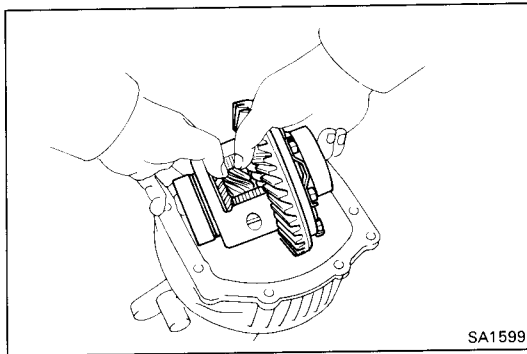
SST 09710-22020 (09710-01030)



## 4. CHECK RING GEAR RUNOUT

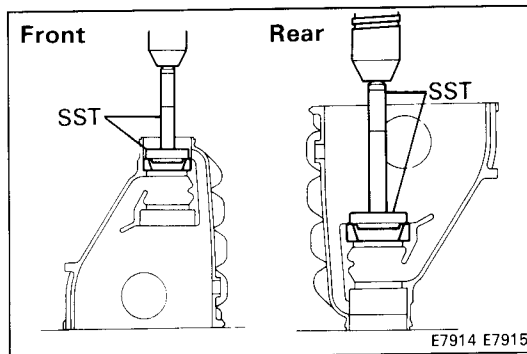
- Install the differential case onto the carrier and install the plate washers to where there is no play in the bearing. (See page SA-55)
- Install bearing caps. (See page SA-57)
- Using a dial indicator, measure the runout of ring gear.

**Maximum runout:** 0.07 mm (0.0028 in.)



## 5. REMOVE DIFFERENTIAL CASE

- (a) Remove the four bolts and two bearing caps.
- (b) Remove the two plate washers.
- (c) Remove the differential case with bearing outer races.



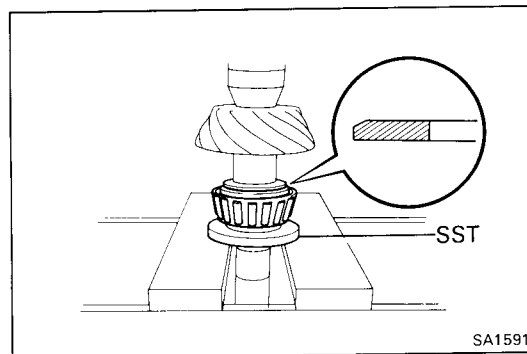
## 6. INSTALL FRONT AND REAR BEARING OUTER RACES

Using a press and SST, press in the front and rear bearing outer races.

SST 09608-30012

Front (09608-04020, 09608-00060)

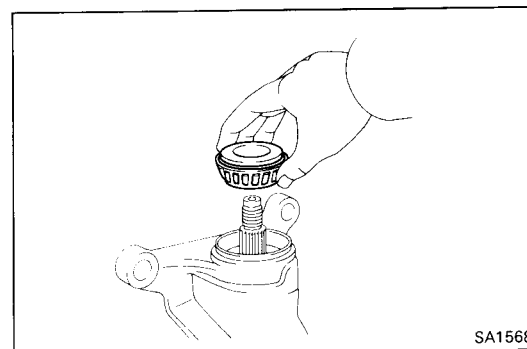
Rear (09608-04020, 09608-04100)



## 7. INSTALL REAR BEARING TO DRIVE PINION

- (a) Install the plate washer on the drive pinion with the chamfered end facing toward the pinion gear.
- (b) Using a press and SST, install the rear bearing onto the drive pinion.

SST 09506-30012

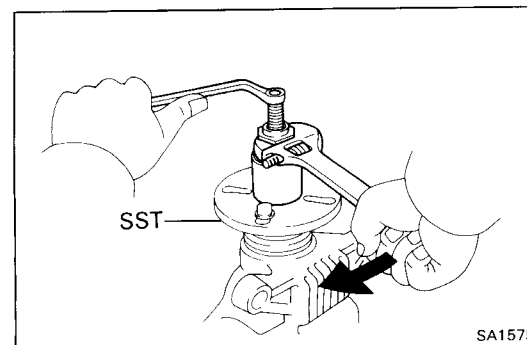


## 8. TEMPORARILY ADJUST DRIVE PINION PRELOAD

- (a) Install the following parts:

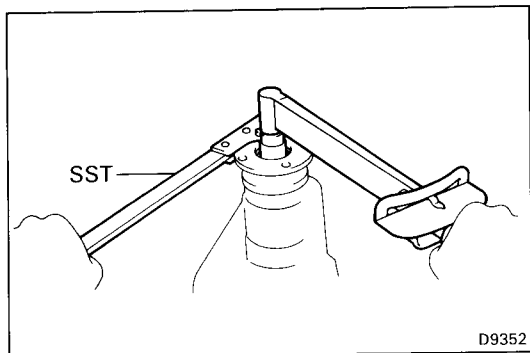
- Drive pinion
- Front bearing

HINT: Assemble the spacer, oil slinger and oil seal after adjusting the gear contact pattern.



- (b) Install the companion flange with SST.

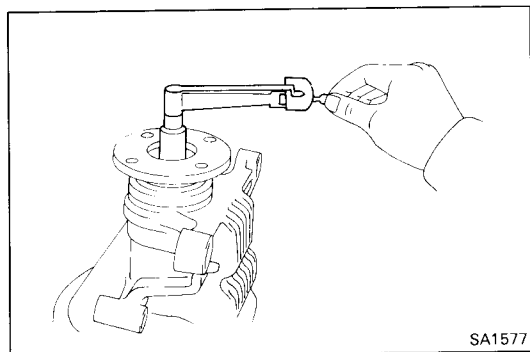
SST 09557-22022



- (c) Adjusting the drive pinion preload by tightening the companion flange nut.

Using SST to hold the flange, tighten the nut.

SST 09330-00021



- (d) Using a torque meter, measure the preload.

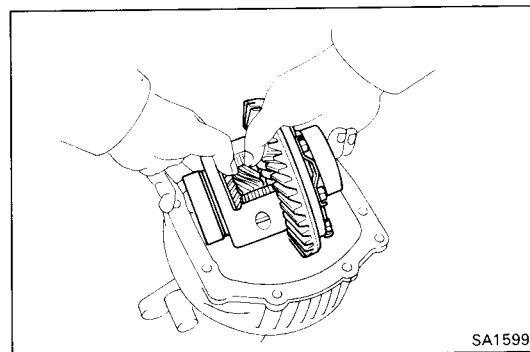
**Preload (at starting)**

**New bearing**

10 – 16 kg-cm (8.7 – 13.9 in.-lb, 1.0 – 1.6 N·m)

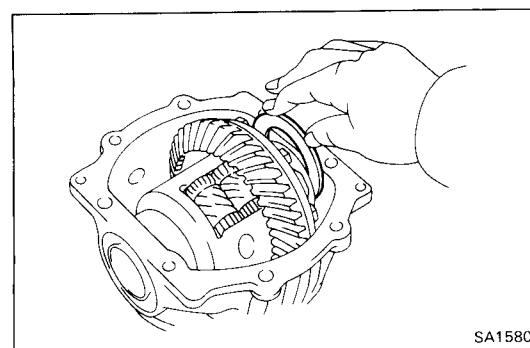
**Reused bearing**

5 – 8 kg-cm (4.3 – 6.9 in.-lb, 0.5 – 0.8 N·m)



## 9. INSTALL DIFFERENTIAL CASE IN CARRIER

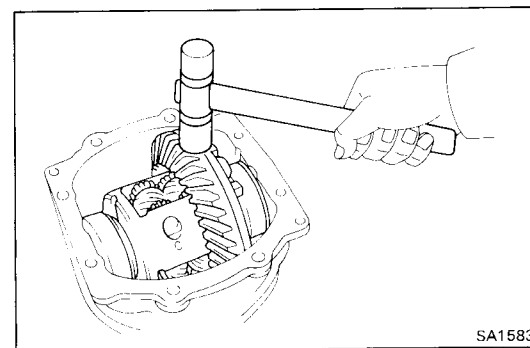
- (a) Place the bearing outer races on their respective bearings. Make sure the left and right outer races are not interchanged.
- (b) Install the differential case in the carrier.



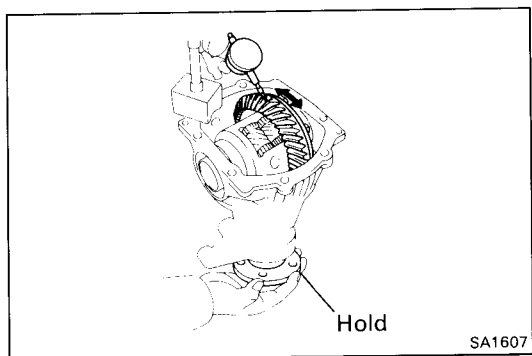
## 10. ADJUST RING GEAR BACKLASH

- (a) Install only the plate washer on the ring gear back side.

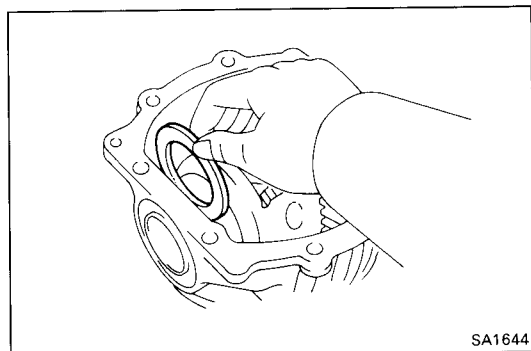
**HINT:** Insure that the ring gear has backlash.



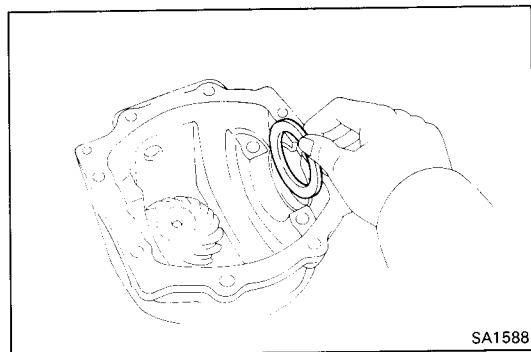
- (b) Snug down the washer and bearing by tapping on the ring gear with a plastic hammer.



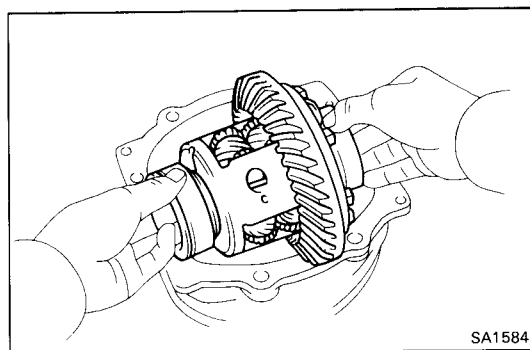
- (c) Using a dial indicator, measure the backlash.
- (d) Select a ring gear back side plate washer so that the backlash is 0.13 mm (0.0051 in.).



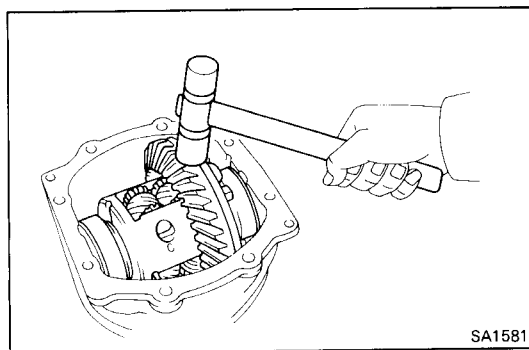
- (e) Select a ring gear teeth side washer with a thickness which eliminates any clearance between the outer race and case.
- (f) Remove the plate washer and differential case.



- (g) Install the plate washer into the ring gear back side.

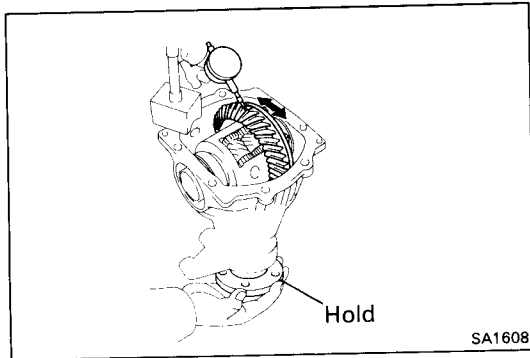


- (h) Place the other plate washer onto the differential case together with the outer race, and install the differential case with the outer race into the carrier.



- (i) Using a plastic hammer, snug down the washer and bearing by tapping the ring gear.





- (j) Using a dial indicator, measure the ring gear backlash.

**Backlash: 0.13 – 0.18 mm (0.0051 – 0.0071 in.)**

- (k) If not within the specification, adjust by either increasing or decreasing the number of washers on both sides by an equal amount.

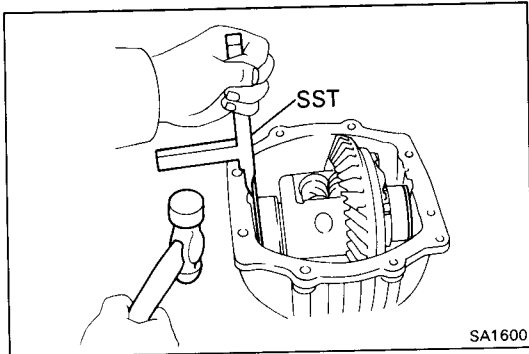
**HINT:** There should be clearance between the plate washer and case.

Insure that there is ring gear backlash.

## 11. ADJUST SIDE BEARING PRELOAD

- (a) Using SST, remove the ring gear teeth side plate washer and measure the thickness.

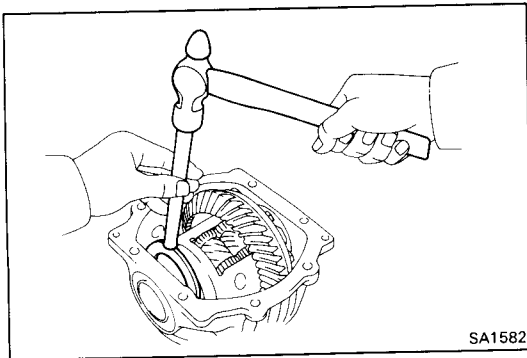
SST 09504-22011



- (b) Install a new washer of 0.06 – 0.09 mm (0.0024 – 0.0035 in.) thicker than the removed washer.

**HINT:** Select a washer which can be pressed in 2/3 of the way by finger.

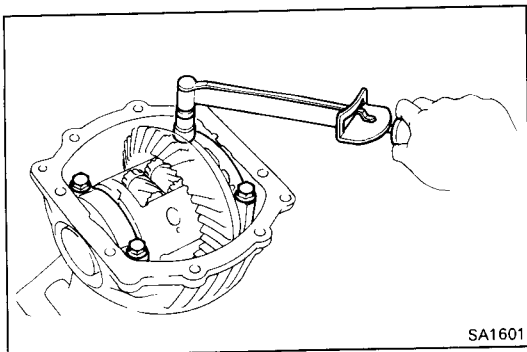
- (c) Using a hammer and brass bar, tap in the side gear thrust washer.



- (d) Install the side bearing caps.

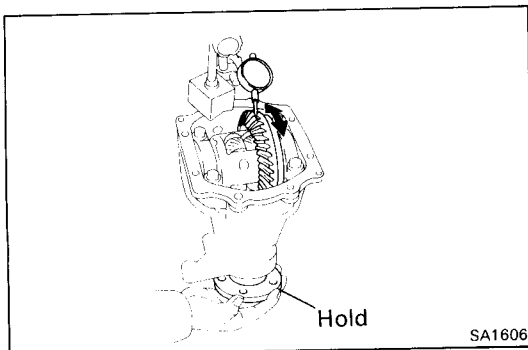
**HINT:** Align the matchmarks on the cap and carrier.

**Torque: 800 kg-cm (58 ft-lb, 78 N·m)**



- (e) Recheck the ring gear backlash.

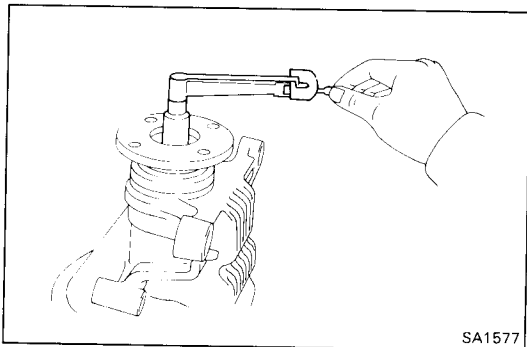
**Backlash: 0.13 – 0.18 mm (0.0051 – 0.0071 in.)**



- (f) If not within the standard, adjust by either increasing or decreasing the washers on both sides by equal amount.

HINT: The backlash will change about 0.02 mm (0.0008 in.) with 0.03 mm (0.0012 in.) alteration of the side gear thrust washer.

| Washer thickness          |                           | mm (in.) |
|---------------------------|---------------------------|----------|
| 2.21-2.23 (0.0870-0.0878) | 2.72-2.74 (0.1071-0.1079) |          |
| 2.24-2.26 (0.0882-0.0890) | 2.75-2.77 (0.1083-0.1091) |          |
| 2.27-2.29 (0.0894-0.0902) | 2.78-2.80 (0.1094-0.1102) |          |
| 2.30-2.32 (0.0906-0.0913) | 2.81-2.83 (0.1106-0.1114) |          |
| 2.33-2.35 (0.0917-0.0925) | 2.84-2.86 (0.1118-0.1126) |          |
| 2.36-2.38 (0.0929-0.0937) | 2.87-2.89 (0.1130-0.1138) |          |
| 2.39-2.41 (0.0941-0.0949) | 2.90-2.92 (0.1142-0.1150) |          |
| 2.42-2.44 (0.0953-0.0961) | 2.93-2.95 (0.1154-0.1161) |          |
| 2.45-2.47 (0.0965-0.0972) | 2.96-2.98 (0.1165-0.1173) |          |
| 2.48-2.50 (0.0976-0.0984) | 2.99-3.01 (0.1171-0.1185) |          |
| 2.51-2.53 (0.0988-0.0996) | 3.02-3.04 (0.1189-0.1197) |          |
| 2.54-2.56 (0.1000-0.1008) | 3.05-3.07 (0.1201-0.1209) |          |
| 2.57-2.59 (0.1012-0.1020) | 3.08-3.10 (0.1213-0.1220) |          |
| 2.60-2.62 (0.1024-0.1031) | 3.11-3.13 (0.1224-0.1232) |          |
| 2.63-2.65 (0.1035-0.1043) | 3.14-3.16 (0.1236-0.1244) |          |
| 2.66-2.68 (0.1047-0.1055) | 3.17-3.19 (0.1248-0.1256) |          |
| 2.69-2.71 (0.1059-0.1067) | 3.20-3.22 (0.1260-0.1268) |          |



SA1577

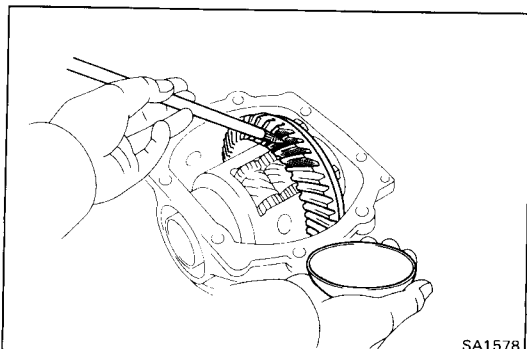
## 12. MEASURE TOTAL PRELOAD

Using a torque meter, measure the total preload.

**Total preload (at starting):**

**Add drive pinion preload**

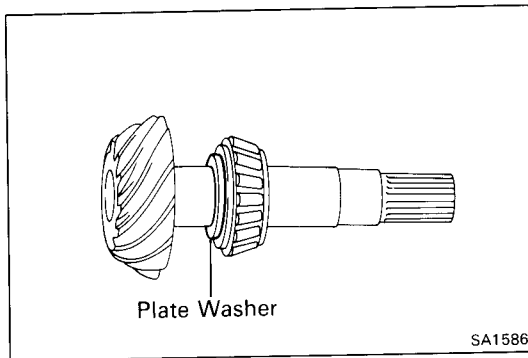
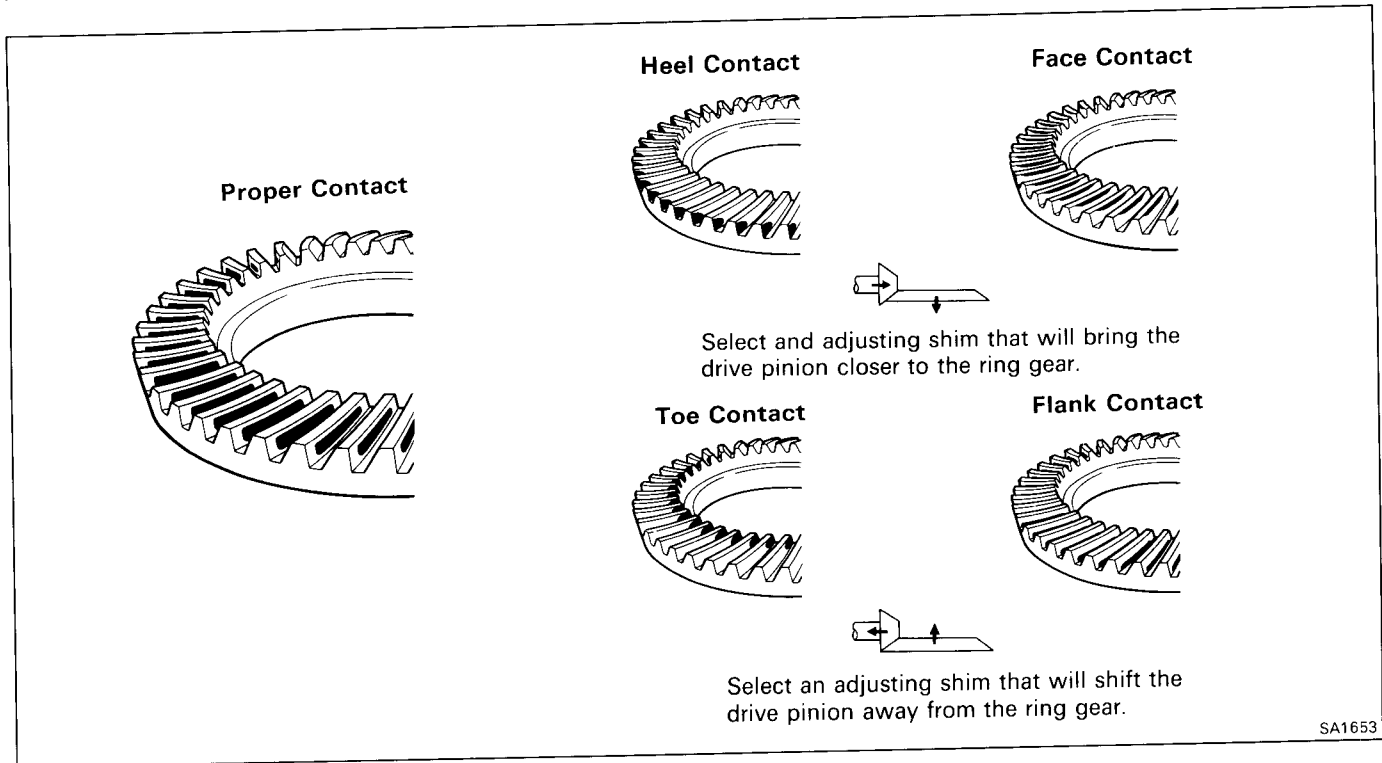
**3 – 5 kg-cm (2.6 – 4.3 in.-lb, 0.3 – 0.5 N·m)**



SA1578

## 13. INSPECT TOOTH CONTACT BETWEEN RING GEAR AND DRIVE PINION

- Coat 3 or 4 teeth at three different positions on the ring gear with red lead.
- Hold the companion flange firmly and rotate the ring gear in both directions.
- Inspect the tooth contact.

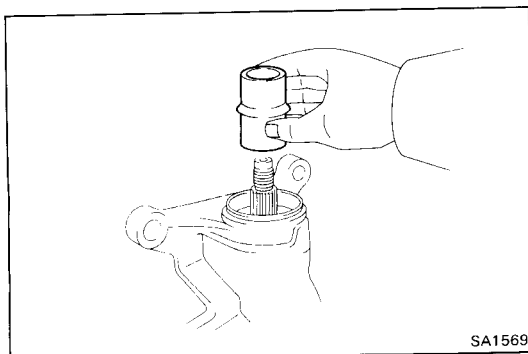


If the teeth are not contacting properly, use the following chart to select a proper washer for correction.

| Thickness                 |                           | mm (in.) |
|---------------------------|---------------------------|----------|
| 2.26-2.28 (0.0890-0.0898) | 2.50-2.52 (0.0984-0.0992) |          |
| 2.29-2.31 (0.0902-0.0909) | 2.53-2.55 (0.0996-0.1004) |          |
| 2.32-2.34 (0.0913-0.0921) | 2.56-2.58 (0.1008-0.1016) |          |
| 2.35-2.37 (0.0925-0.0933) | 2.59-2.61 (0.1020-0.1028) |          |
| 2.38-2.40 (0.0937-0.0945) | 2.62-2.64 (0.1031-0.1039) |          |
| 2.41-2.43 (0.0949-0.0957) | 2.65-2.67 (0.1043-0.1051) |          |
| 2.44-2.46 (0.0961-0.0969) | 2.68-2.70 (0.1055-0.1063) |          |
| 2.47-2.49 (0.0972-0.0980) |                           |          |

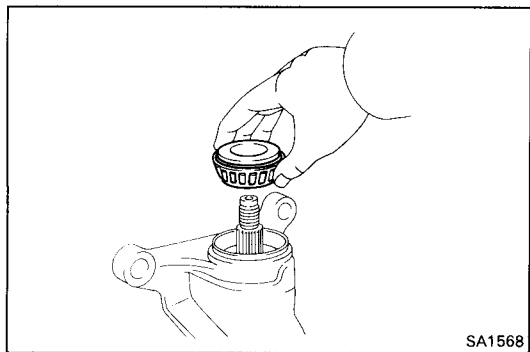
**14. REMOVE COMPANION FLANGE**  
(See step 3 on page SA-47)

**15. REMOVE FRONT BEARING**  
(See step 5 on page SA-48)

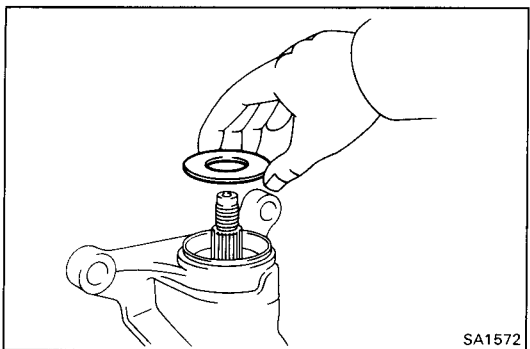


**16. INSTALL NEW BEARING SPACER AND FRONT BEARING**

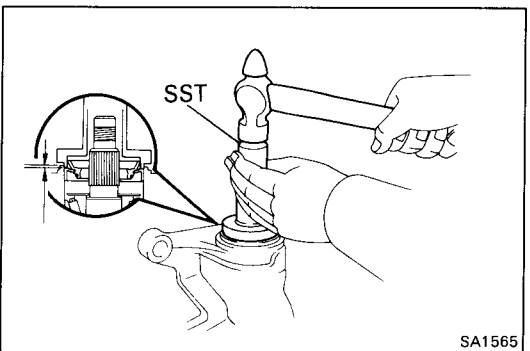
(a) Install a new bearing spacer on the drive pinion.



(b) Install the front bearing on the drive pinion.



#### 17. INSTALL OIL SLINGER



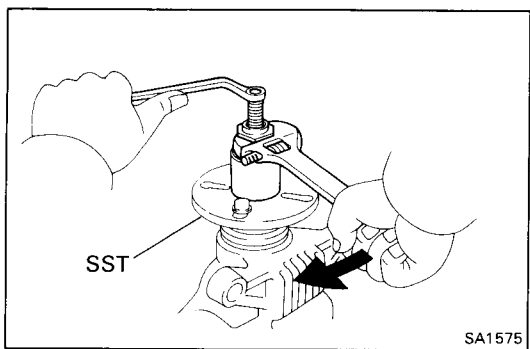
#### 18. INSTALL NEW OIL SEAL

(a) Using SST, drive in a new oil seal.

SST 09554-22010

Oil seal drive in depth: 2.0 mm (0.079 in.)

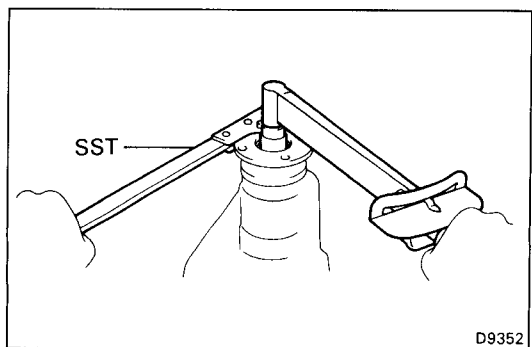
(b) Apply MP grease to oil seal lip.



#### 19. INSTALL COMPANION FLANGE

(a) Using SST, install the companion flange on the shaft.

SST 09557-22022



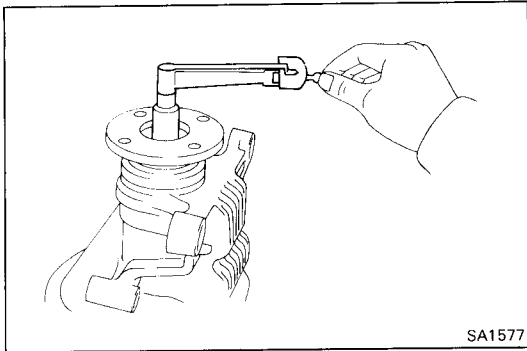
(b) Install the plate washer.

(c) Coat the threads of a new nut with gear oil.

(d) Using SST to hold the flange, tighten the nut.

SST 09330-00021

Torque: 1,100 kg-cm (80 ft-lb, 108 N·m)



## 20. CHECK DRIVE PINION BEARING PRELOAD

Using a torque wrench, measure the preload of the back-lash between the drive pinion and ring gear.

**Preload (at starting):**

|                       |                                                                   |
|-----------------------|-------------------------------------------------------------------|
| <b>New bearing</b>    | <b>10 – 16 kg-cm</b><br><b>(8.7 – 13.9 in.-lb, 1.0 – 1.6 N·m)</b> |
| <b>Reused bearing</b> | <b>5 – 8 kg-cm</b><br><b>(4.3 – 6.9 in.-lb, 0.5 – 0.8 N·m)</b>    |

- If preload is greater than specification, replace the bearing spacer.
- If preload is less than specification, retighten the nut 130 kg-cm (9 ft-lb, 13 N·m) at a time until the specified preload is reached.

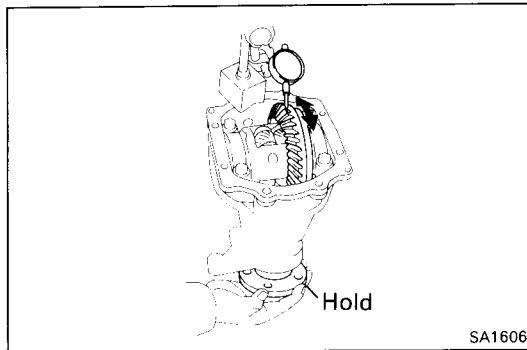
If the maximum torque is exceed while retighten the nut, replace the bearing spacer and repeat the preload procedure. Do not back off the pinion nut to reduce the preload.

**Maximum torque: 2,400 kg-cm (174 ft-lb, 235 N·m)**

## 21. CHECK TOTAL PRELOAD

**Total preload (at starting):**

**Add drive pinion preload**  
**3 – 5 kg-cm (2.6 – 4.3 in.-lb, 0.3 – 0.5 N·m)**



## 22. CHECK RING GEAR BACKLASH

Using a dial indicator, check the backlash of the ring gear.

**Backlash: 0.13 – 0.18 mm (0.0051 – 0.0071 in.)**

If the backlash is not within specification, adjust the side bearing preload.

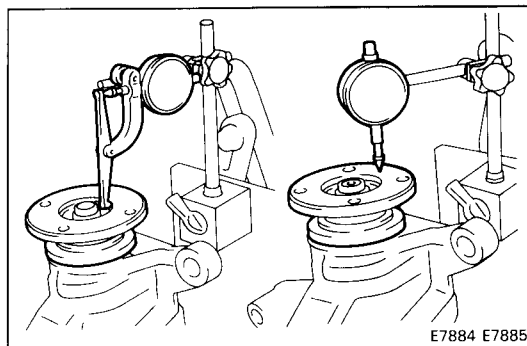
## 23. INSPECT TOOTH CONTACT BETWEEN RING GEAR AND DRIVE PINION (See page SA-58)

## 24. CHECK COMPANION FLANGE RUNOUT

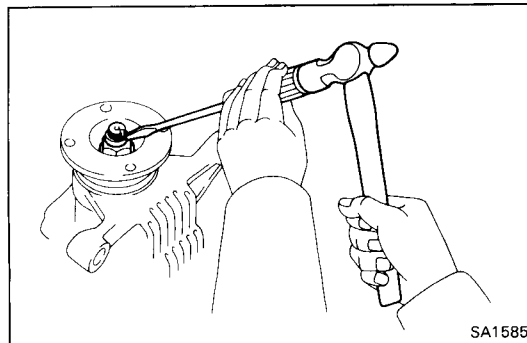
Using a dial indicator, measure the lateral and radial runout of the companion flange.

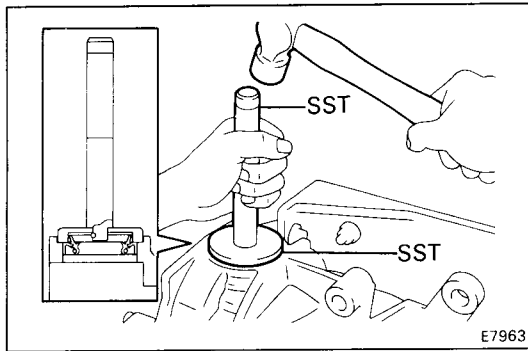
**Maximum lateral runout: 0.10 mm (0.0039 in.)**

**Maximum radial runout: 0.10 mm (0.0039 in.)**



## 25. STAKE DRIVE PINION NUT

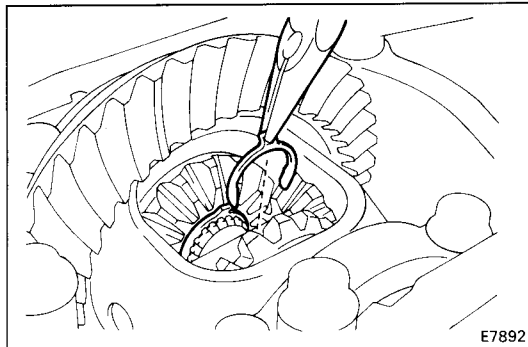


**26. INSTALL NEW SIDE GEAR SHAFT OIL SEALS**

- (a) Using SST, drive in two new oil seals until they are flush with the carrier end surface.

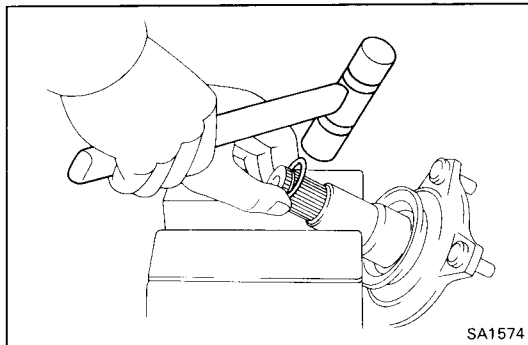
SST 09550-22011 (09550-00020, 09550-00031)

- (b) Coat the oil seal lips with MP grease.

**27. INSTALL SIDE GEAR SHAFTS**

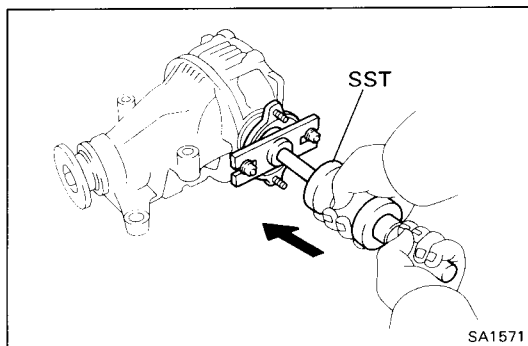
(Conventional Type Differential)

- (a) Install the two side gear shafts to the differential case.
- (b) Install two new shaft snap rings to the side gear shafts.



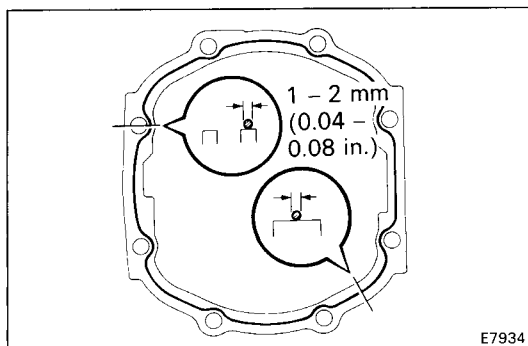
(Torque Sensing Limited Slip Differential)

- (a) Install the new snap ring to the side gear shaft.



- (b) Using SST, drive in the side gear shaft.

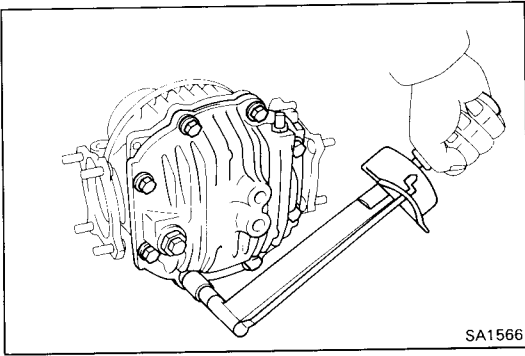
SST 09520-24010

**28. INSTALL DIFFERENTIAL CARRIER COVER**

- (a) Clean contacting surfaces of any residual packing material using gasoline or alcohol.
- (b) Apply seal packing to the carrier.

**Seal packing: Part No.08826-00090, THREE BOND 1281 or equivalent**

**HINT:** Install the carrier cover within 3 minutes after applying seal packing.



(c) Install and torque the eight set bolts.

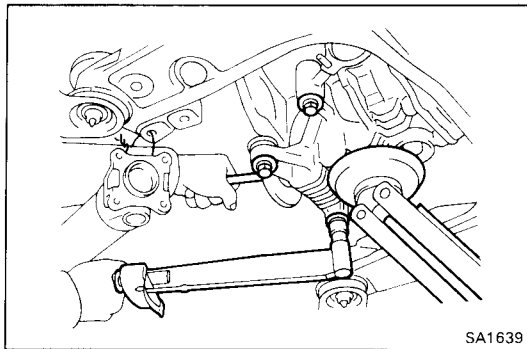
**Torque: 475 kg-cm (34 ft-lb, 47 N·m)**

## INSTALLATION OF DIFFERENTIAL

### 1. INSTALL DIFFERENTIAL

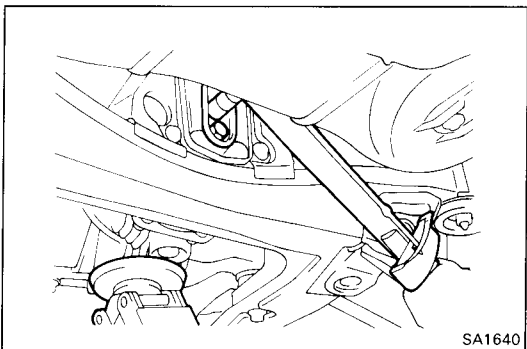
- (a) Position the differential and torque the four bolts and nuts.

**Torque: 970 kg-cm (70 ft-lb, 95 N·m)**



- (b) Install and torque the two bolts.

**Torque: 1,500 kg-cm (108 ft-lb, 147 N·m)**

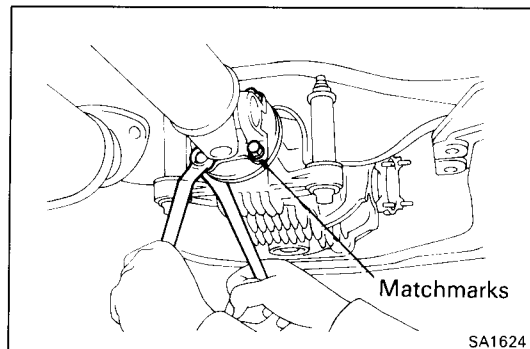


### 2. CONNECT PROPELLER SHAFT

- (a) Align the matchmarks on the flanges and connect the flanges with four bolts, nuts and washers.

- (b) Torque the four bolts and nuts.

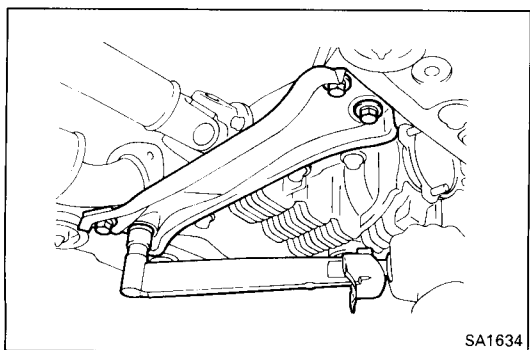
**Torque: 750 kg-cm (54 ft-lb, 74 N·m)**



### 3. INSTALL REAR CROSSMEMBER

Install the rear crossmember with four bolts.

**Torque: 730 kg-cm (53 ft-lb, 72 N·m)**



### 4. CONNECT DRIVE SHAFTS

(See page SA-31)

### 5. FILL DIFFERENTIAL WITH GEAR OIL

- (a) Install the drain plug with new gasket.

**Torque: 500 kg-cm (36 ft-lb, 49 N·m)**

- (b) Fill the differential with gear oil.

**Oil grade: API GL-5 hypoid gear oil**

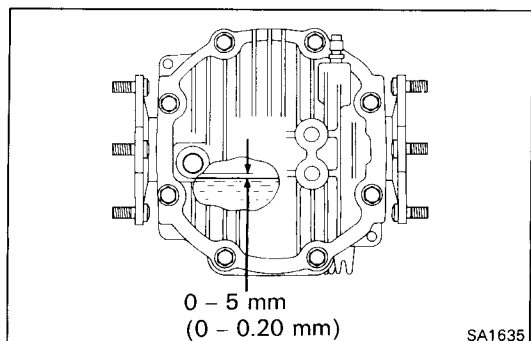
**Viscosity: Above -18°C (0°F) SAE 90**

**Below -18°C (0°F) SAE 80W-90**

**Capacity: 1.1 liters (1.2 US qts, 1.0 Imp.qts)**

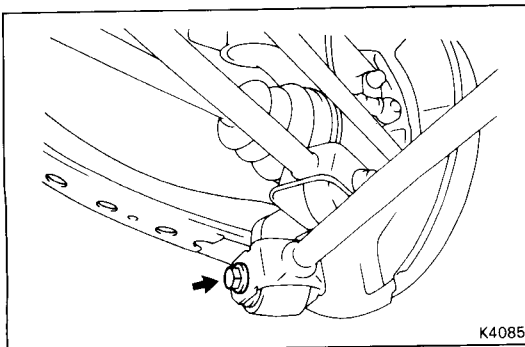
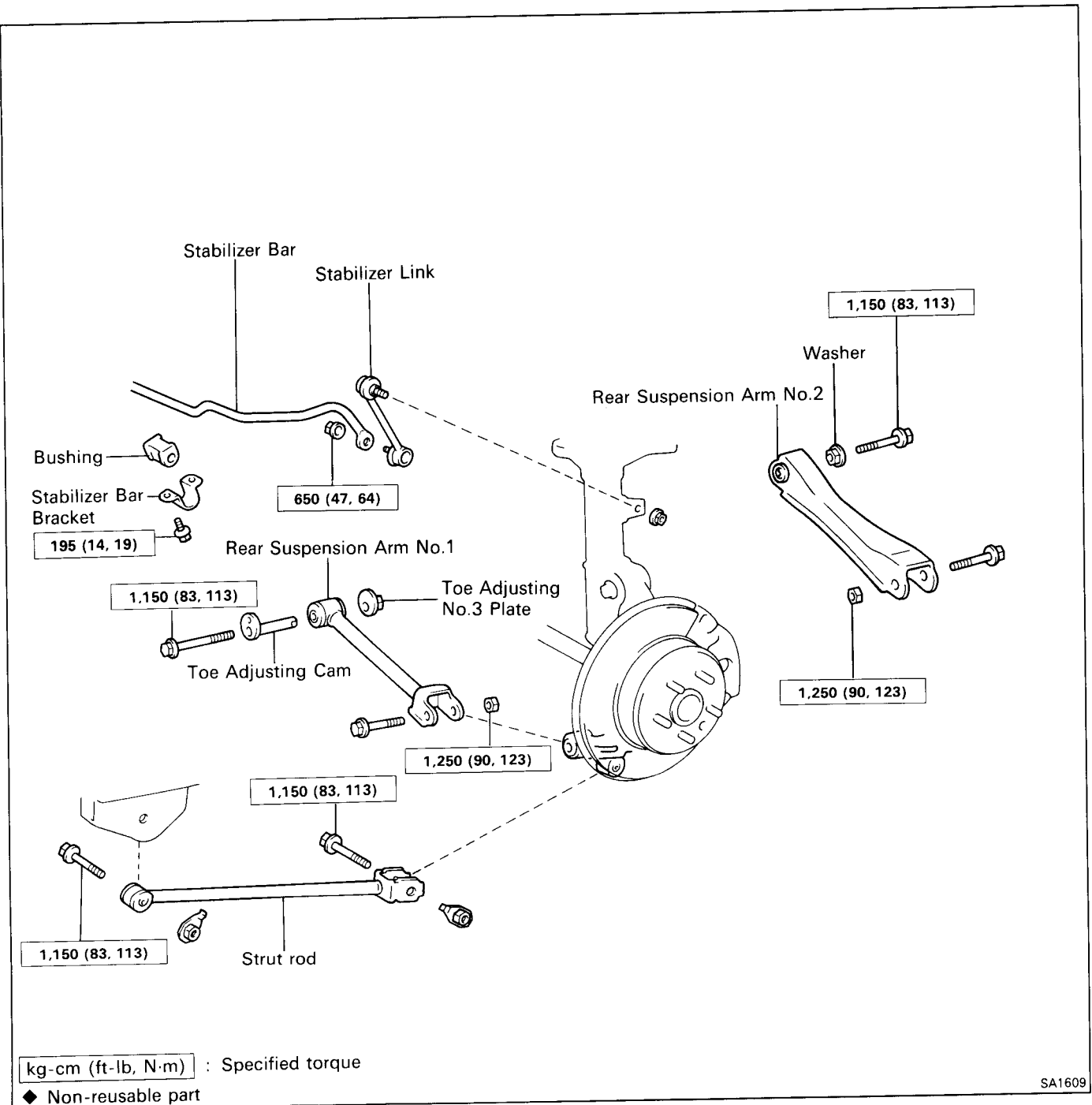
- (c) Install the filler plug with new gasket.

**Torque: 400 kg-cm (29 ft-lb, 39 N·m)**



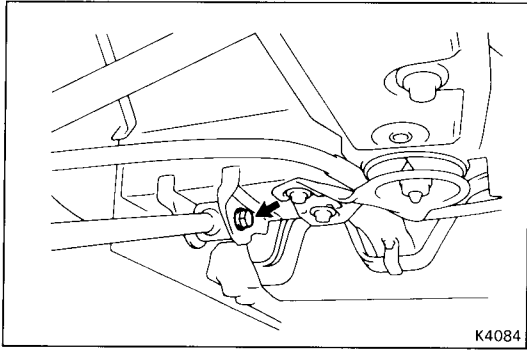


## REAR SUSPENSION COMPONENTS

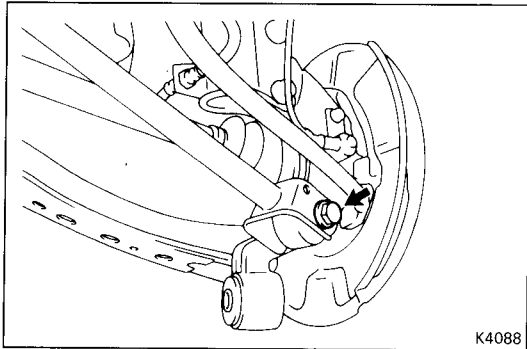


### Strut Rod and Suspension Arm REMOVAL OF STRUT ROD AND SUSPENSION ARM

1. REMOVE REAR WHEELS
2. REMOVE STRUT ROD
  - (a) Remove the bolt and nut from the axle carrier.

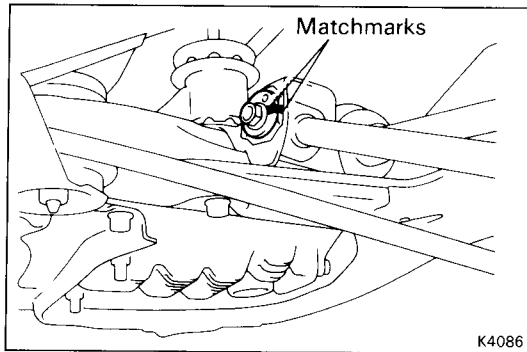


(b) Remove the bolt, nut and strut rod from body.

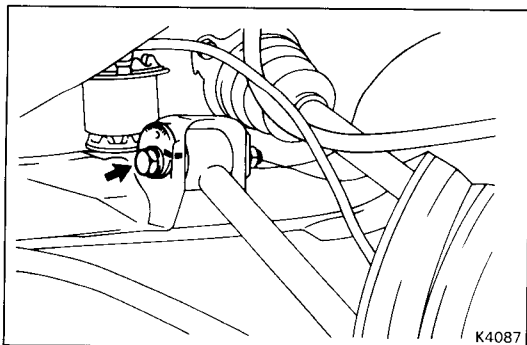


### 3. REMOVE NO.1 SUSPENSION ARM

(a) Remove the bolt and nut from the axle carrier.

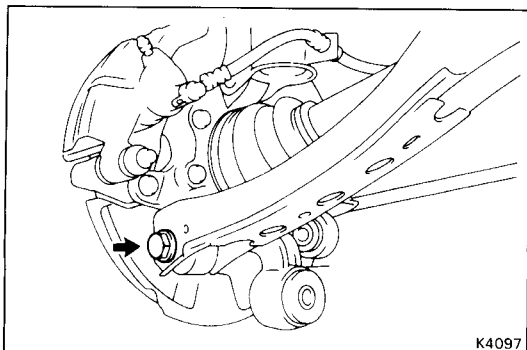


(b) Place matchmarks on the toe adjusting cam and control arm retainer.



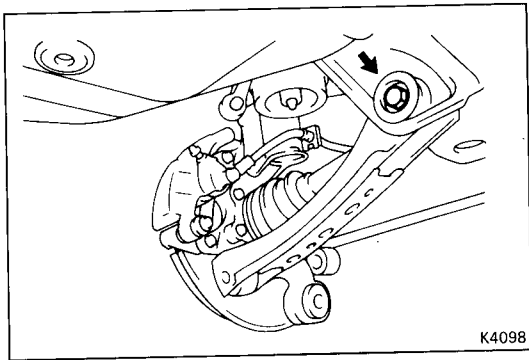
(c) Remove the bolt, toe adjusting cam and toe adjusting plate No.2.

(d) Remove the No.1 suspension arm.

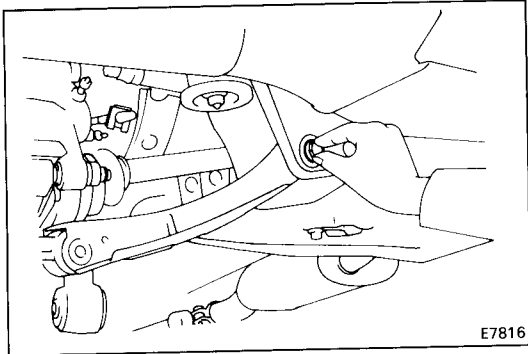


### 4. REMOVE NO.2 SUSPENSION ARM

(a) Remove the bolt and nut from the axle carrier.



- (b) Remove the bolt, washer and No.2 suspension arm.

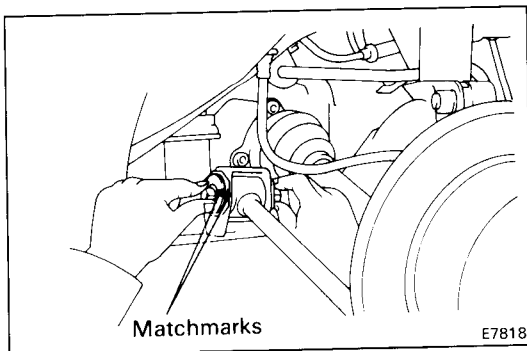
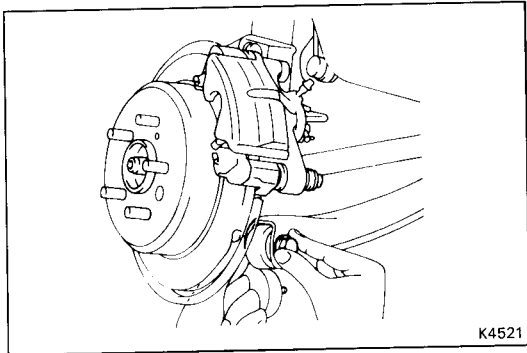


## INSTALLATION OF STRUT ROD AND SUSPENSION ARM

### 1. INSTALL NO.2 SUSPENSION ARM

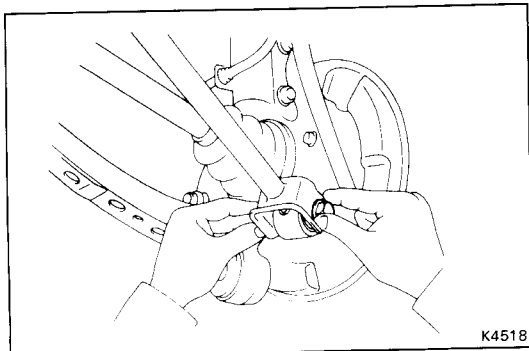
- (a) Install the No.2 suspension arm to the rear suspension member and finger tighten the bolt with washer.

- (b) Install the No.2 suspension arm to the axle carrier and finger tighten the bolt and nut.

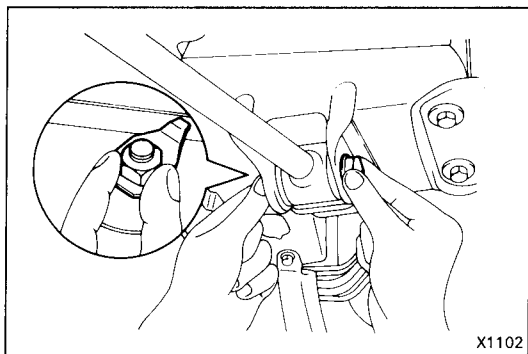


### 2. INSTALL NO.1 SUSPENSION ARM

- (a) Install the No.1 suspension arm to the rear suspension member.
- (b) Align matchmarks on the toe adjusting cam and control arm retainer.
- (c) Install the toe adjusting plate No.2 and finger tighten the bolt.

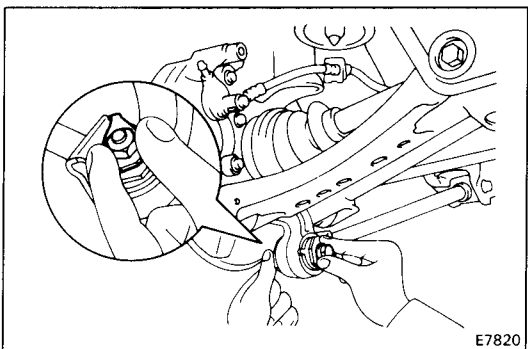


- (d) Install the No.1 suspension arm to the axle carrier and finger tighten the bolt.



### 3. INSTALL STRUT ROD

- (a) Install the strut rod to the body and finger tighten the bolt.



- (b) Install the strut rod to the axle carrier and finger tighten the bolt.

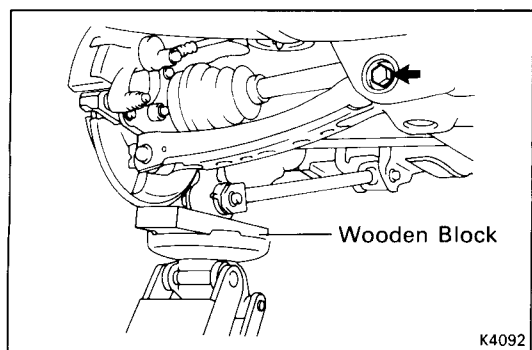
### 4. TEMPORARILY INSTALL REAR WHEELS

### 5. LOWER VEHICLE

Remove the stands and bounce the vehicle up and down to stabilize the suspension.

### 6. REMOVE WHEELS

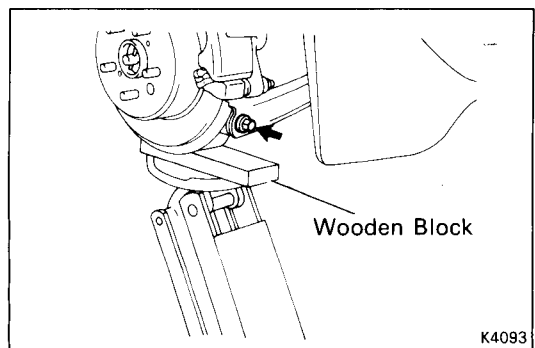
- (a) Jack up vehicle and install stands.  
(b) Remove the rear wheels.



### 7. TORQUE NO.2 SUSPENSION ARM

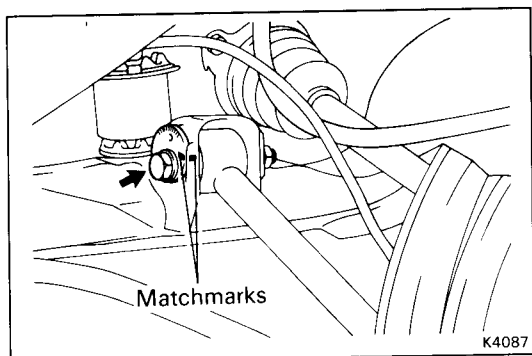
- (a) Jack up the rear axle.  
(b) Torque the No.2 suspension arm bolt of suspension member side.

**Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)**



- (c) Torque the No.2 suspension arm bolt of axle carrier side.

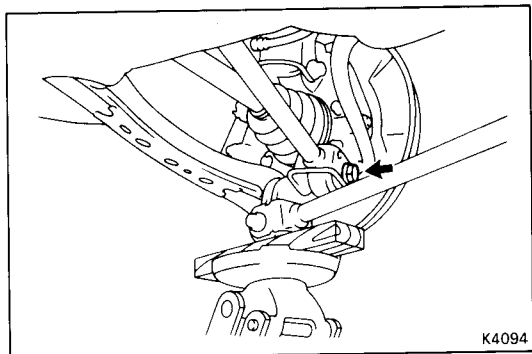
**Torque: 1,250 kg-cm (90 ft-lb, 123 N·m)**



## 8. TORQUE NO.1 SUSPENSION ARM

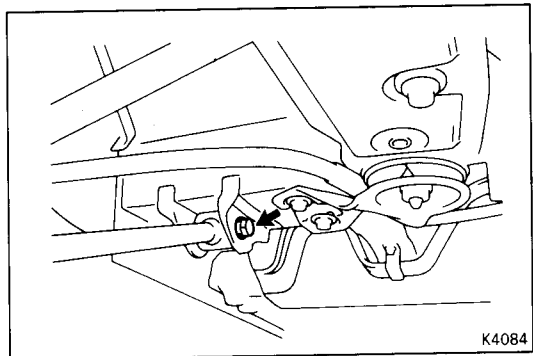
- (a) Align matchmarks on the toe adjusting cam and control arm retainer.
- (b) Torque the No.1 suspension arm bolt of the suspension member side.

**Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)**



- (c) Torque the No.1 suspension arm bolt of the axle carrier side.

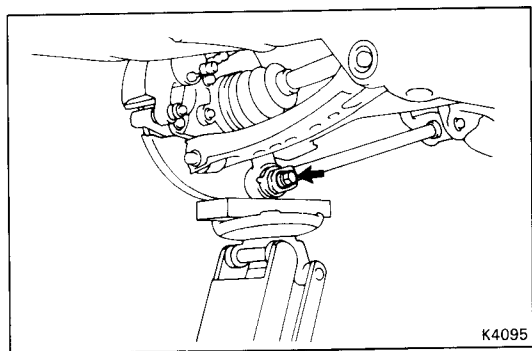
**Torque: 1,250 kg-cm (90 ft-lb, 123 N·m)**



## 9. TORQUE STRUT ROD

- (a) Torque the strut rod bolt of the body side.

**Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)**

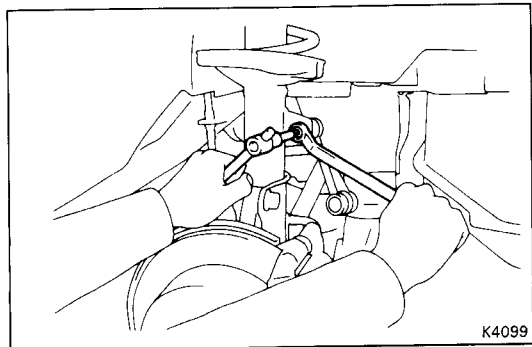


- (b) Torque the strut rod bolt of the axle carrier side.

**Torque: 1,150 kg-cm (83 ft-lb, 113 N·m)**

## 10. INSTALL REAR WHEELS

## 11. CHECK REAR WHEEL ALIGNMENT

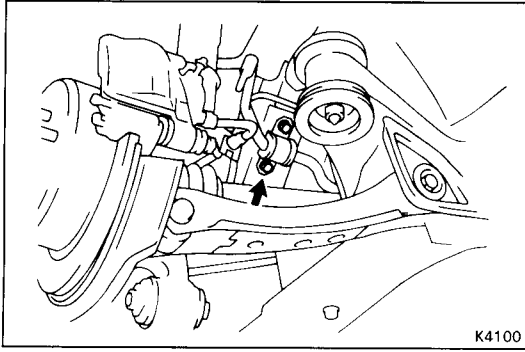


## Stabilizer Bar

### REMOVAL OF STABILIZER BAR

1. REMOVE REAR WHEELS
2. REMOVE STABILIZER LINK

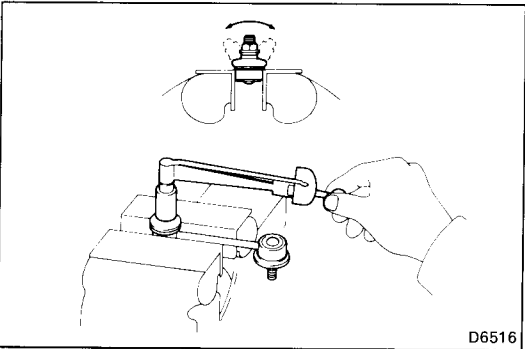
Using a hexagon wrench and offset wrench, remove the two nut and stabilizer link.



### 3. REMOVE STABILIZER BRACKET WITH BUSHING

Remove the four bolts and stabilizer bracket with bushings.

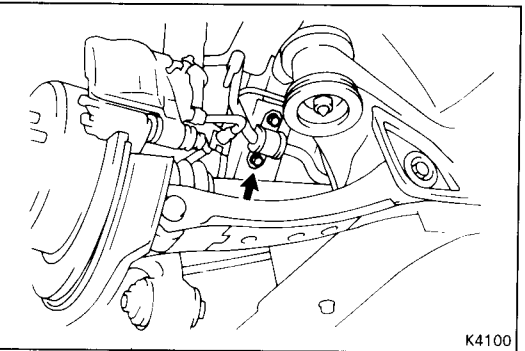
### 4. REMOVE STABILIZER BAR



## INSPECTION OF STABILIZER LINK

### INSPECT STABILIZER LINK

Rotate ball joint arm in all direction. If the movement is not smooth and free, replace the stabilizer link.



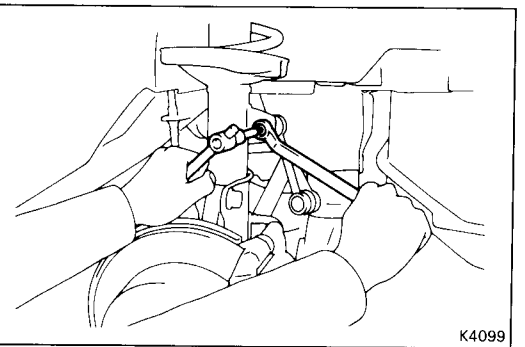
## INSTALLATION OF STABILIZER BAR

### 1. INSTALL STABILIZER BAR

### 2. INSTALL STABILIZER BRACKET WITH BUSHING

- Install the two bushings to the stabilizer bar.
- Install the two brackets with four bolts to the stabilizer bar.

**Torque: 195 kg-cm (14 ft-lb, 19 N·m)**



### 3. INSTALL STABILIZER LINK

- Install the stabilizer link to the stabilizer bar and finger tighten the nut.
- Install the stabilizer link to the shock absorber bracket and finger tighten the nut.
- Using a hexagon wrench, torque the two nuts.

**Torque: 650 kg-cm (47 ft-lb, 64 N·m)**

### 4. INSTALL REAR WHEELS